

SIEMENS

MAGNETOM

Startup System

Start-up

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Upgrades to syngo MR:
from Impact/Expert
from Vision

Answers for life.

Document Version

Siemens reserves the right to change its products and services at any time.

In addition, manuals are subject to change without notice. The hardcopy documents correspond to the version at the time of system delivery and/or printout. Versions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order a current version or refer to our website <http://www.healthcare.siemens.com>.

Disclaimer

Siemens provides this documentation “as is” without the assumption of any liability under any theory of law.

The content described herein requires superior understanding of our equipment and may only be performed by qualified personnel who are specially trained for such installation and/or service.

Copyright

“© Siemens, 1997” refers to the copyright of a Siemens entity such as:

Siemens Healthcare GmbH - Germany

Siemens Aktiengesellschaft - Germany

Siemens Shenzhen Magnetic Resonance Ltd. - China

Siemens Shanghai Medical Equipment Ltd. - China

Siemens Medical Solutions USA Inc. - USA

Siemens Healthcare Diagnostics Inc. - USA and/or

Siemens Healthcare Diagnostics Products GmbH - Germany

1	General	7
1.1	Prerequisites	7
1.2	Auxiliary tools	8
1.2.1	Magnet system	8
1.3	Overall Flowchart	9
2	Prior to switch-on	11
2.1	Safety	11
2.2	Checking the line voltage	12
2.3	Adapting transformer T1 to the line voltage	13
2.4	Adapting the gradient to the line voltage	15
2.4.1	Verification of gradient performance	17
2.5	Prior to switch on the helium compressor	18
2.5.1	Setting the mode selector switch on the F70 helium compressor	18
2.5.2	Line voltage adaption, F70 helium compressor	19
2.5.3	Line voltage adaption, HC10 helium compressor	20
2.6	Additional RCA transformer	22
2.7	Checking the helium gas pressure	26
2.7.1	F70 Helium Compressor	26
2.7.2	HC10 Helium Compressor	27
2.8	Adapting the on-site power to the system	28
2.9	ERDU	30
2.9.1	ERDU Buttons	30
2.9.2	Connecting the ERDU battery	30
2.9.3	Testing the ERDU	31
2.10	Checking the water hoses	33
2.11	Protective conductor test	34
3	Switch-on	35
3.1	Emergency shutdown buttons	35
3.1.1	Function test for emergency shutdown buttons	37
3.2	Starting the refrigerator	39
3.2.1	MAGNETOM Harmony/Symphony: Switch ON Leybold Refrigerator	40
3.2.2	MAGNETOM Sonata: Switch ON APD Refrigerator	42
3.3	Ethernet switches	46
3.4	Switching on the system	47
3.4.1	Circuit breakers	47
3.4.2	The Alarm box	47
3.5	Software configuration	49
3.6	Filling the secondary water circuit	50
3.7	Setting the water flow rates for MAGNETOM Harmony/Symphony	54
3.8	Setting the temperature set point	57

3.9	Cleaning the water filter in case of a local move and Refurbished Systems .	58
3.9.1	CCA Cabinet.	58
3.9.2	RCA / Body Coil	58
3.10	Checks after 12, 24 and 48 hours	63
3.11	Setting the secondary water flow rates for MAGNETOM Sonata	64
3.11.1	Prerequisites	64
3.11.2	Setting the water flow rates	64
3.12	Setting the LHe level meter resistance	67
3.13	Setting the screen reset time	69
3.14	Remote Diagnostics.	70
3.14.1	Introduction.	70
3.14.2	Supported connections	70
3.14.3	Responsibility	70
3.14.4	Support	70
3.14.5	Parameters obtained from the local network manager	71
3.14.6	Parameters obtained from the USC:	71
3.15	TestTools	72
3.16	Warning signs	73
4	Filling the magnet	74
4.1	General Overview	74
4.2	Safety	75
4.3	Auxiliary tooling	77
4.4	LHe filling of Static and Mobile Systems	78
4.4.1	System Preparation	78
4.4.2	Check of magnet pressure.	80
4.4.3	Current lead (HDCL)	81
4.4.4	LHe Dewar Preparation	81
4.4.5	Preparing the dewar syphon	83
4.4.6	Precooling the dewar syphon	83
4.4.7	Helium Transfer with High Efficiency Syphon (HES)	85
4.4.8	Helium Transfer with Smart Syphon	90
4.4.9	Magnet System leak check	101
5	Ramping the magnet	106
5.1	Information	106
5.2	Safety	107
5.2.1	Always	107
5.2.2	Never	107
5.2.3	Protective clothing	107
5.3	Ramping the magnet.	108
5.4	Materials and Tools required.	109
5.5	Magnet Power Supply	110
5.6	HDCL Inspection and Repair	111
5.6.1	Introduction.	111
5.7	Preliminary steps for ramping	119

5.7.1	List of activities which should have been completed	119
5.7.2	Restricting hazardous areas.	119
5.7.3	Removing the covers.	120
5.7.4	Tilting the patient air ventilator	121
5.8	Complete Ramping - flowchart	122
5.9	Preparing the MPS	123
5.10	Depressurization of the Helium Vessel (Magnet)	130
5.11	Inserting the hardwired demountable current lead (HDCL)	134
5.12	Ramping - flowchart	140
5.13	Ramping up the magnet	141
5.13.1	Setting the current and voltage values	141
5.13.2	Switching on the magnet electronics	141
5.13.3	Ramping up the magnet	143
5.13.4	HDCL contact checks	145
5.14	Ramping the magnetic field to the required field strength.	147
5.15	Co-Siting	149
5.16	Checking the magnetic field polarity	150
5.16.1	Checking the field direction with a magnetic compass.	150
5.17	Shimming the magnet	151
5.17.1	Ramping down the magnet.	151
5.17.2	Power failure during magnet ramping	153
5.18	Removing the hardwired demountable current lead (HDCL)	155
5.18.1	Safety checks.	155
5.19	Appendix	160
5.19.1	Energizing the EIS circuit.	160
5.19.2	Ramping down the magnet.	161
5.19.3	Power failure during magnet ramping	162
5.19.4	MPS cabling	164
5.19.5	Footnotes	165
5.19.6	Adding Citric Acid to the Secondary Water Circuit	166
5.19.7	What to do if frequency adjustment failed	170
6	Patient table	172
6.1	Control panel.	172
6.2	Converting the table	173
6.3	Converting the trolley	176
6.4	Testing docking	180
6.5	Adjusting the docking height	181
6.5.1	Deinstalling the front cover.	181
6.5.2	Adjusting the docking height	181
6.5.3	Testing docking after adjustments	184
6.5.4	Fine adjustment of docking height.	184
6.5.5	Reinstalling the front cover.	185
7	TFT color monitor	186
7.1	General	186

8	Intercom	187
8.1	Installation	187
8.2	Start-up	189
8.3	Final Check	191
8.4	Customer Information	192
9	Final work	193
9.1	Tune-up	193
9.2	Gradient Performance Check	194
	9.2.1 Gradient Risettime Check	194
	9.2.2 Possible reasons for this test to fail:	198
	9.2.3 Gradient Burst Check	198
9.3	Gradient Performance Test Protocol	200
9.4	QA (Quality Assurance)	201
9.5	Safety Related Tests	202
9.6	Installation Protocol	203
9.7	Dynamical Backup	204
9.8	Trolley with removable Tabletop (Option)	205
9.9	Cleaning up	206
10	Changes to previous version	207
11	List of Hazard IDs	208

1.1 Prerequisites

- Room lighting must be functioning in every room of the MR-site.
- The system components have been positioned and installed according to the planning guide and installation instructions.
- The control cables have been installed and connected.
- The RF-integrity of the RF-room was measured after the installation of the RF-room and the components.
- The protective conductor connections were measured after completion of system installation.

NOTICE

If the magnet was shipped cold, the refrigerator and magnet monitoring should be put into operation as quickly as possible. To do this, the start-up procedure must be followed beginning with the next chapter.

No warranty will be given if any problem occurs within the system due to work performed by untrained personnel or third party.

- » Only trained personnel (Harmony & Symphony course) are allowed to perform the start-up!

1.2 Auxiliary tools

The auxiliary tools required for starting-up the MR system are:

- Service tool kit cpl.
- Set of standard tools
- ESD service kit
- Digital multimeter, Fluke 8060A
- Protective conductor measurement device
- MAVO monitor

1.2.1 Magnet system

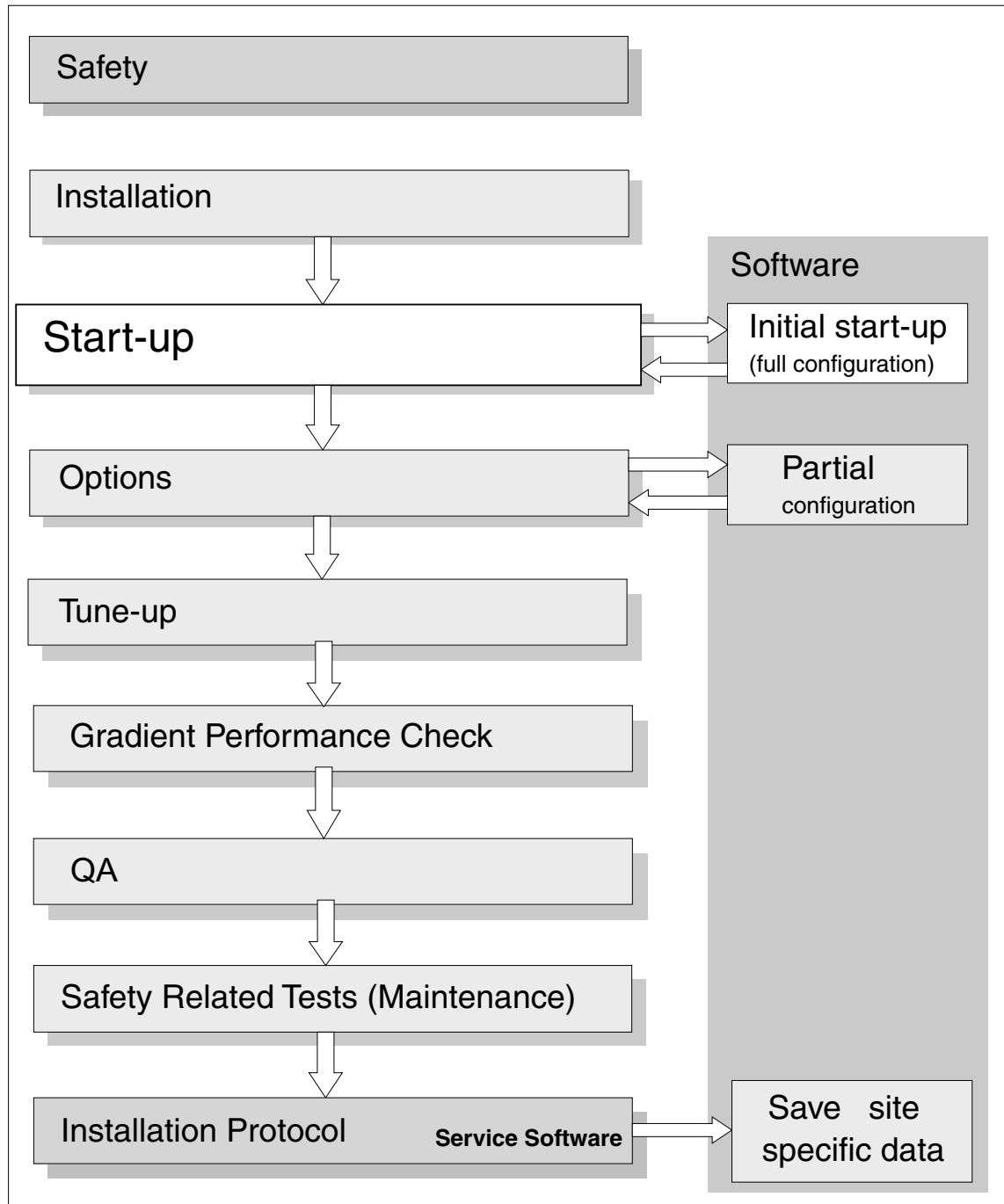
- Array shim device
- Heat gun
- Compass
- Magnet Power Supply
- Protective clothing (face shield, gloves, safety shoes etc.)

1.3 Overall Flowchart



System start-up should be performed in the order of the described steps. Complete each step before you start working on the next one

Fig. 1: Flowchart covering system installation up through the installation protocol





There is no written document for the Tuneup and QA. Use Numaris Online Help for detailed information.

The (→ Gradient Performance Check / Page 194) is described in (→ Final work / Page 193).

2.1 Safety

- For your own safety and the safety of third parties and to protect the MR system and parts, observe the safety information in register (→ General Safety Notes / TD00-000.860.01) and (→ MR-specific safety information / MR-000.860.01) in "Technical Documents" when performing any work on the system.
- Read the special safety information regarding:
 - general safety for service personnel,
 - attaching or checking the "Strong Magnetic Field" warning sign,
 - attaching or checking the "Cardiac Pacemaker" warning sign,
 - bringing ferromagnetic materials into the magnet area,
 - safety information for working when the magnet is under field,
 - hazards involved in cryogen-filling,
 - working with heavy loads,
 - working with lasers,
 - protective conductor test prior to switch-on and after repair work,
 - reduction of the magnetic field to 20 mT when quenching the magnet.

2.2 Checking the line voltage

- Lock all emergency shutdown buttons.
 - Prior to switch-on, several adaptations to the line voltage have to be made.
- Ask your project manager about the on-site line voltage or perform your own measurement.

The on-site line voltage is:	VAC
The on-site line frequency is:	Hz



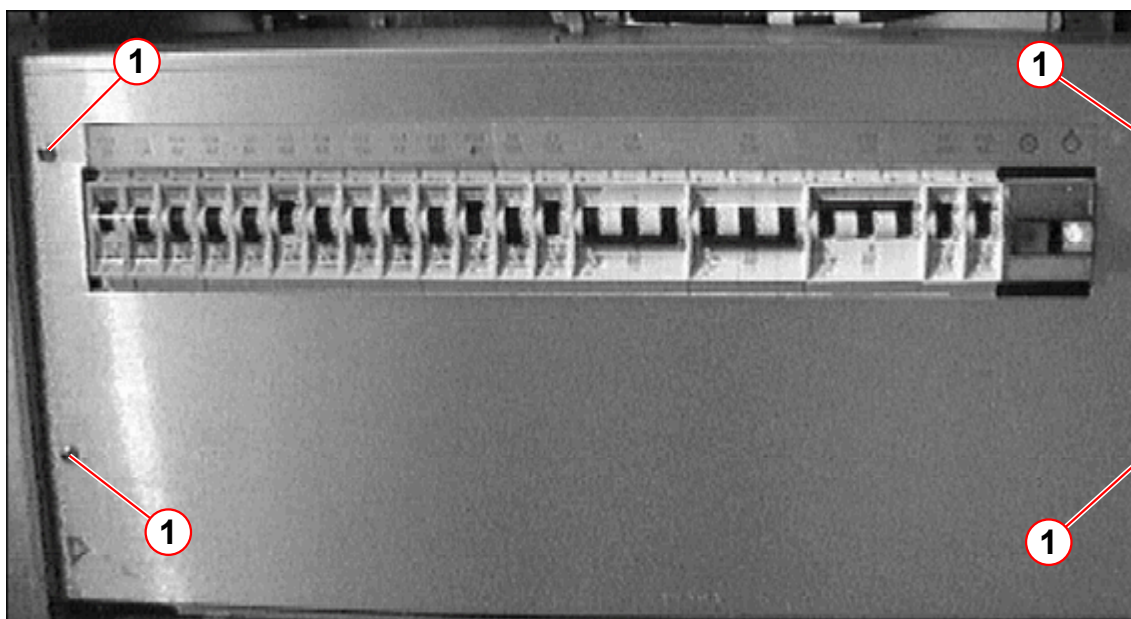
WARNING

Verify that no line voltage is present at the RCA, system transformer T1 in the RCA and at T1 in the GPA.

- » Switch off the on-site power supply (main power distribution to the system) and secure it with a lock to prevent anyone from inadvertently switching it on.

- Open the circuit breaker cover (four screws - refer to (→ Fig. 2 Page 12)).

Fig. 2: Open the cover to get to the adapter for transformer T1



(1) Screw

2.3 Adapting transformer T1 to the line voltage

- Adapt transformer T1 to the line voltage supply according to (→ Fig. 3 Page 13).

Fig. 3: Adapting the transformer T1 to the line voltage

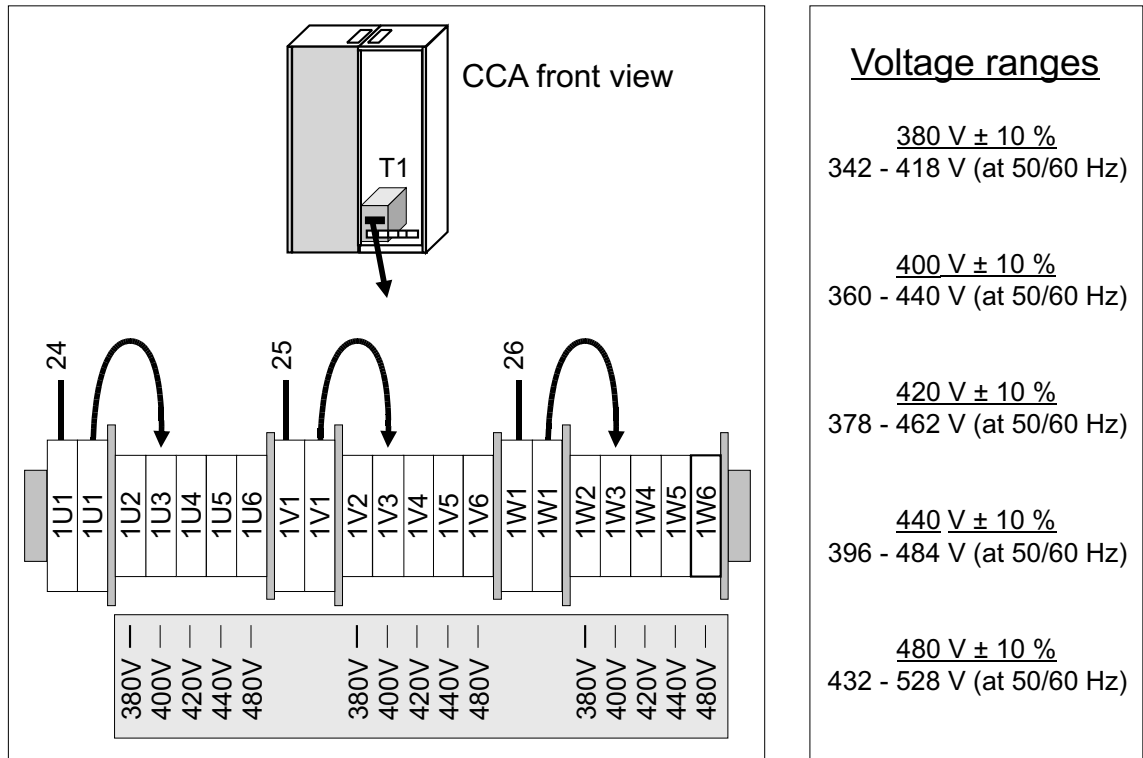
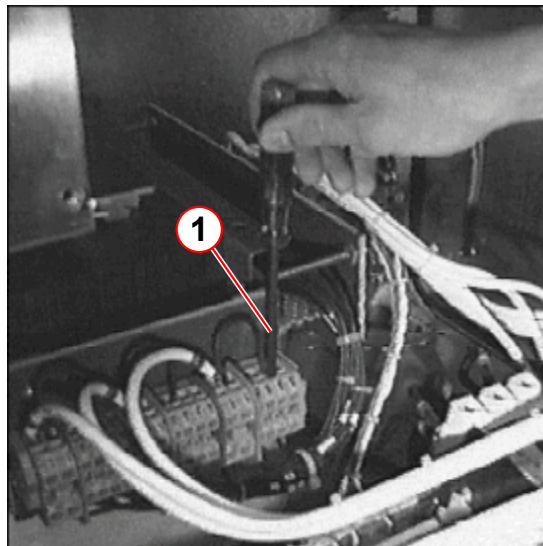
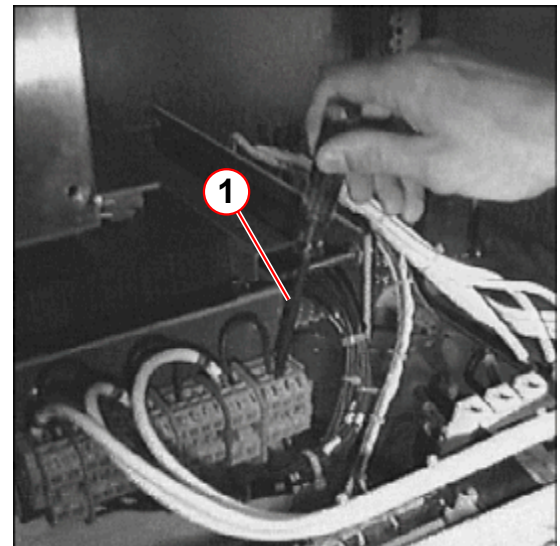


Fig. 4: Slide in screwdriver



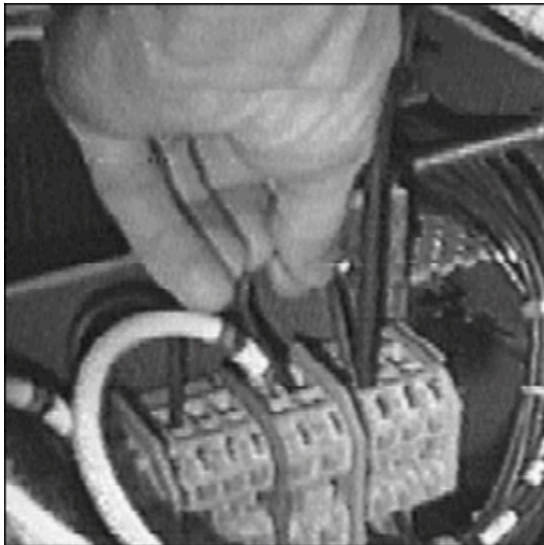
(1) Slide in screwdriver

Fig. 5: Pull forward to open the contact



(1) Pull forward

Fig. 6: Push the wire into the new tap and release the screwdriver



2.4 Adapting the gradient to the line voltage

WARNING

Switch off the on-site power supply (main power distribution to the system) and secure it with a lock to prevent anyone from inadvertently switching it on.

Additionally switch-off the circuit breaker F3 - 80 A for the GPA at top of the CCA (Quantum and Sonata) and F14 in the CCA (GSSU Supply)

- » Verify that no line voltage is present at the RCA, system transformer T1 in the RCA and at T1 in the GPA.

- Locate the unplugged optic fibre cable for the Gradient coil **temperature monitoring** at the top of the GPA labeled with **U4**, refer to (→ Fig. 7 Page 15), (→ Fig. 8 Page 15).

Fig. 7: Connecting the optic fibre cable for Gradient coil temperature monitoring to D17 in the GPA cabinet

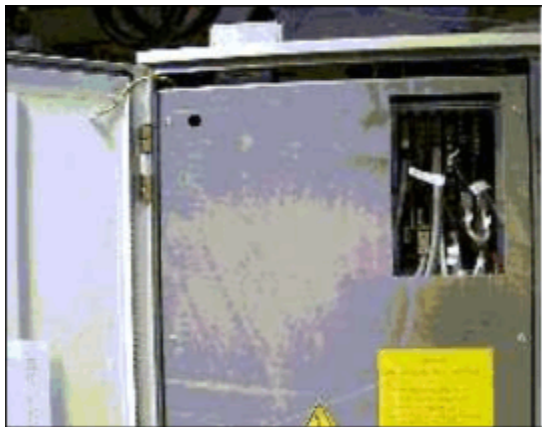
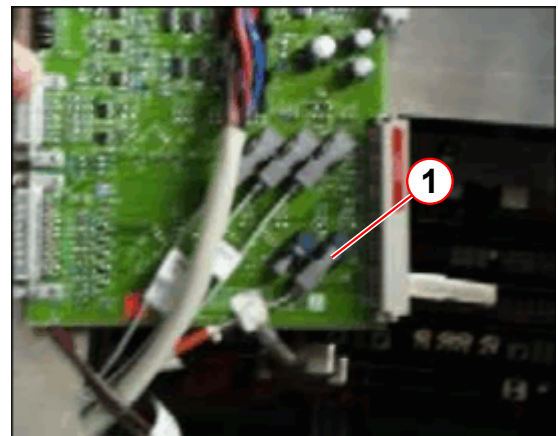


Fig. 8: Connecting the optic fibre cable for Gradient coil temperature monitoring to D17 in the GPA cabinet



(1) Plug U7 at PCB D17

- Pull the PCB D17 located most right at the top of the GPA cabinet (disconnect gently the cables at the front of D17).
- Connect the optic fibre cable labeled with **U4** to the plug labeled with **U7** at the **PCB D17**, refer to (→ Fig. 8 Page 15) (gently slide the board into the slot, reconnect the front plugs).

The factory setting is **480 V** (USA) for the T1 in the gradient cabinet.

- Open the front door of the gradient cabinet.
- Disconnect the water tubes, remove the 4 allen screws, refer to (→ Fig. 9 Page 16), (→ Fig. 10 Page 16).

Fig. 9: Disconnect the water tubes from the rectifier plate, remove the 4 allen screws to release the cover plate



Fig. 10: Disconnect the water tubes from the rectifier plate, remove the 4 allen screws to release the cover plate



- Gently open the cooling plate and the front cover to get access to the terminal block.

Fig. 11: Gently open the cooling plate to access the terminal of the T1 in the GPA

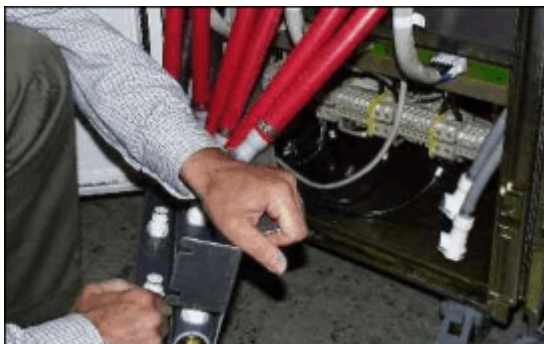
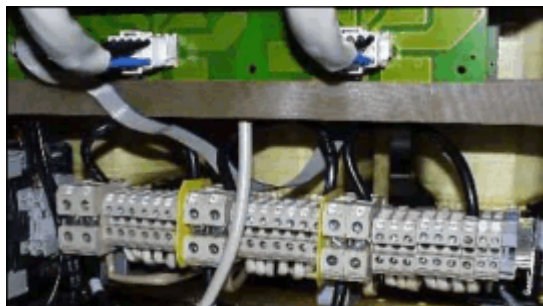
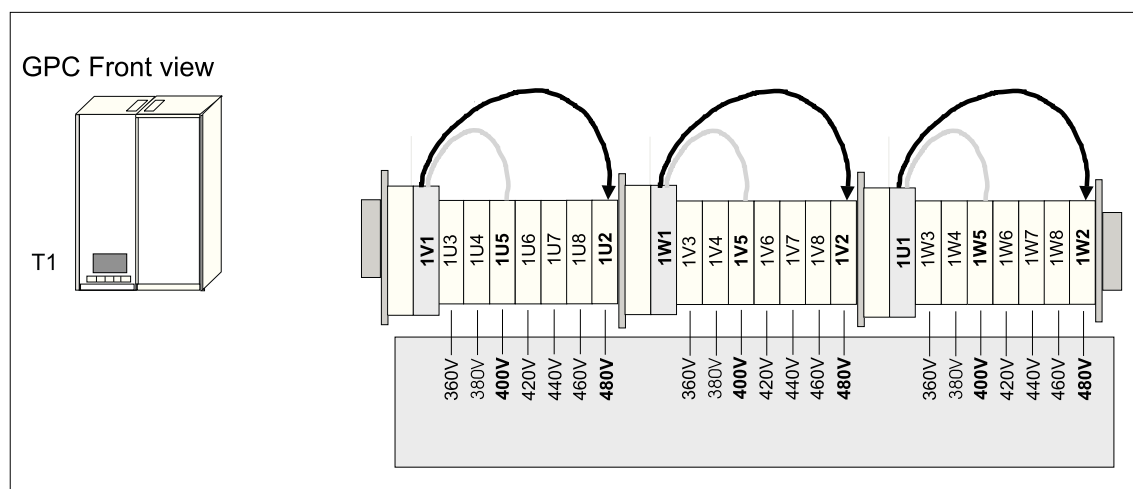


Fig. 12: Gently open the cooling plate to access the terminal of the T1 in the GPA



- Check/ perform wiring for the local line power voltage, refer to (→ Fig. 13 Page 16).

Fig. 13: Adapting the transformer T1 in the GPA to the linevoltage



- When the line voltage adaption has been performed, reassemble the GPA to original state.

2.4.1 Verification of gradient performance

NOTICE

Incorrect adaption to the local line voltage will cause the 400V DC supply voltage for the final stages to be out of specification.

This in turn, may lead to unsatisfactory gradient performance.

- » Please perform the (→ Gradient Performance Check / Page 194) after the Tuneup has been completed. Follow the instructions in (→ Final work / Page 193).

2.5 Prior to switch on the helium compressor

2.5.1 Setting the mode selector switch on the F70 helium compressor



Configured in the factory at mode 1.

Before the compressor's **main power switch** is turned on, the **mode selector switch** must be set to mode **1 (upper position)**.

- » The **ON** and **OFF** buttons are **disabled** in mode **2** position, as the compressor is controlled via the pressure within the magnet (used on 4K systems only).

Fig. 14: F70 mode switch



(1) Mode selector switch



Do not change the mode switch (→ 1/ Fig. 14 Page 18) position after the main power has been switched on.

- Use a flat blade screwdriver to move the switch up.
 - » Switch **up** = mode **1**

2.5.2 Line voltage adaption, F70 helium compressor



The unit is configured in the factory to 400V 3~50Hz.

The F70 has 4 voltage taps, to match the incoming power supply.

The following voltage/frequency combinations are supported:

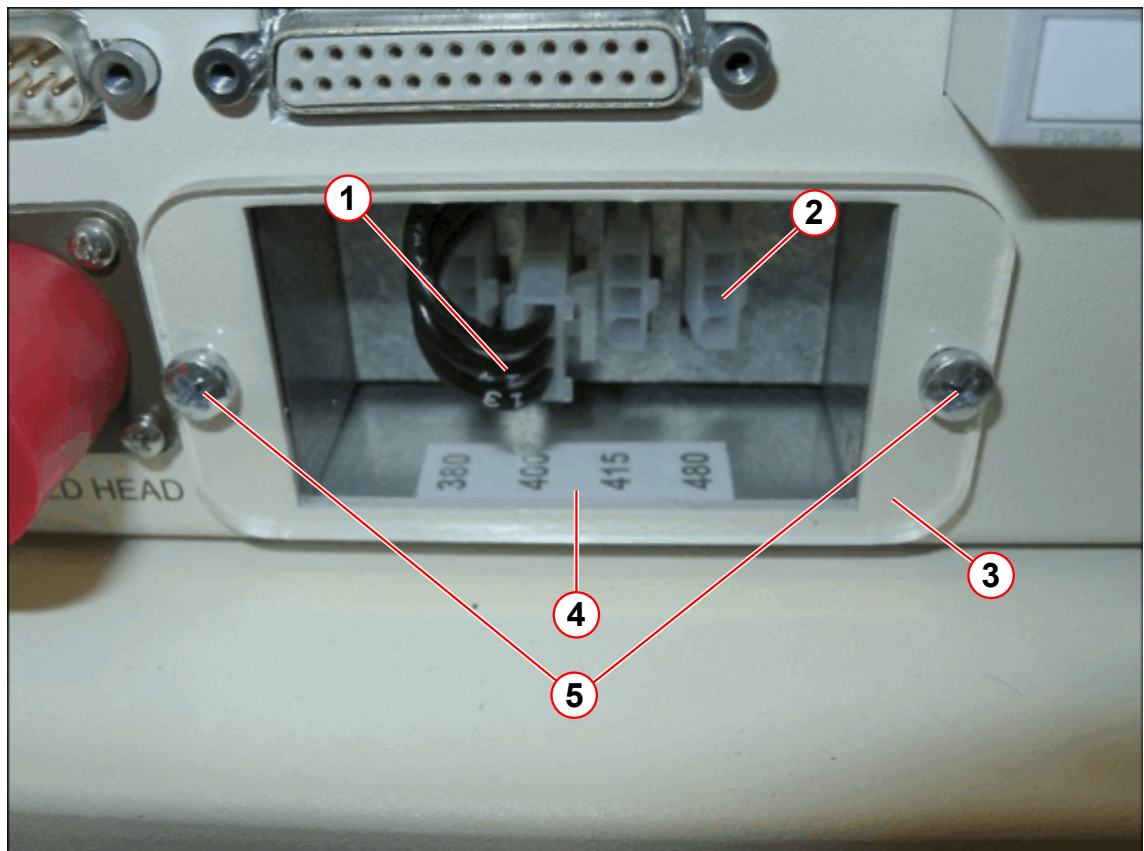
- 480 VAC@60Hz
- 380, 400, & 415 VAC@50Hz



Ensure the compressor is isolated from any power supply.

- Remove the 2 Phillips screws securing the clear plastic cover.
- Plug the voltage connector (1) into the required voltage tap (all 3 phases).

Fig. 15: F70 Compressor voltage selection access panel



- (1) Voltage selection connector
- (2) Voltage taps
- (3) Clear plastic cover
- (4) Voltage label
- (5) Phillips screws (2 off)

- Refit the clear plastic cover using the 2 Phillips screws.

2.5.3 Line voltage adaption, HC10 helium compressor



The unit is configured in the factory to 400V 3~50Hz.

The HC10 has 2 voltage terminals, to match the incoming power supply.

The following voltage/frequency combinations are supported:

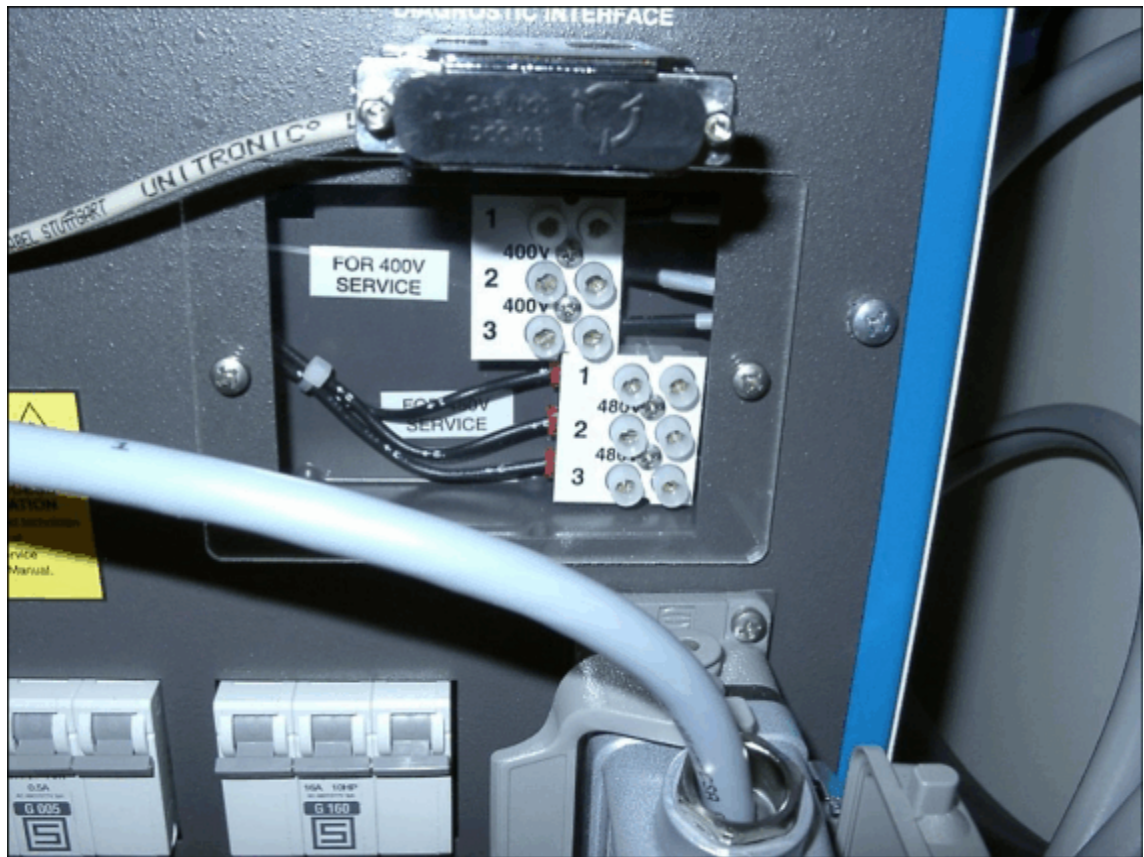
- 400 VAC@50Hz
- 480 VAC@60Hz



Ensure the compressor is isolated from any power supply.

- Check if the compressor is wired to the correct voltage terminal. If necessary, change the wires, using a small blade screwdriver.
- Therefore remove the 2 Phillips screws securing the clear plastic cover.
- Connect the power input wires to the appropriate terminals.
- Replace the cover.

Fig. 16: HC10 Compressor voltage selection terminals



2.6 Additional RCA transformer

An additional transformer as described below needs to be installed for the RCA only if the power conditions at your site are not in accordance with the table below.

Tab. 1 Allowed line voltage/frequency combinations for APD compressors

Frequency	F70 compressor	HC10 compressor
50 Hz	380/415 V ($\pm 10\%$) 3~	380/415 V (+10%-14%) 3~
60 Hz	480 V ($\pm 10\%$) 3~	480 V (+10%-14%) 3~

- » The following 16 kVA transformer can be ordered according to the price book.

Fig. 17: Step-up transformer for RCA

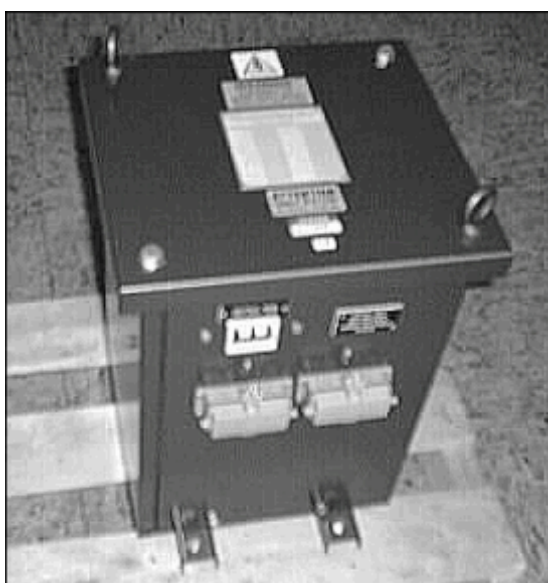
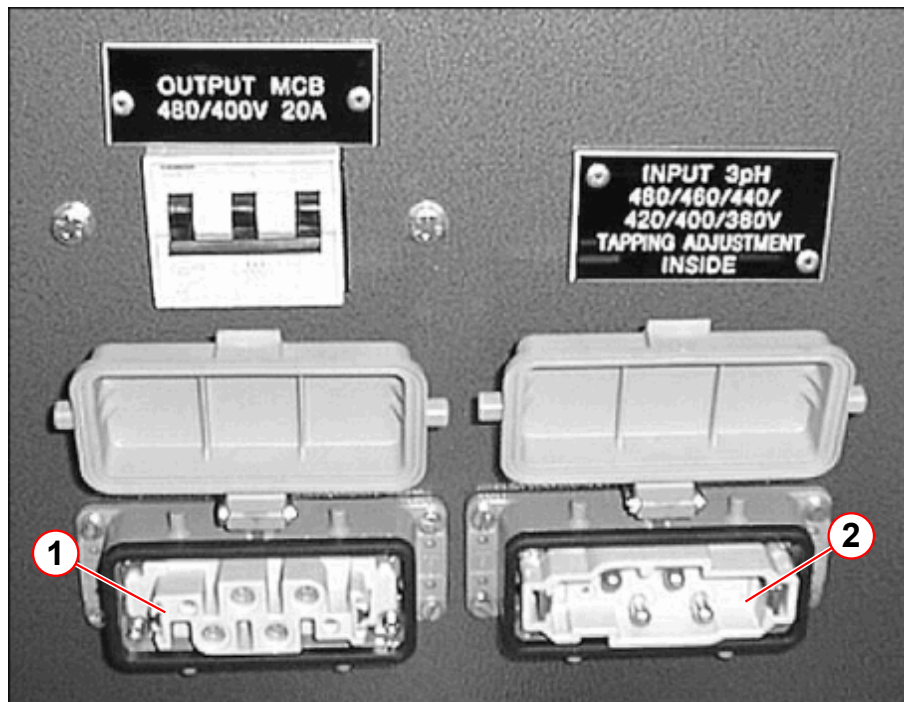


Fig. 18: 20 A circuit breaker in the output circuit with female connector, male input connector



- (1) Output (to RCA)
(2) Input (From CCA)

- Route the power cables from X101.CCA ⇒ transformer input and from transformer output [W63] ⇒ X100.RCA.
- To adapt the transformer to the line voltage it must be opened. Unscrew the two M10 screws and the two M10 transport hooks to remove the top cover.

The input line voltage can be adapted to 380V, 400V, 420V, 440V, 460V or 480V (*1).

- Refer to the tables below to adapt the correct voltages for the refrigerator (check (→ Tab. 1 Page 22)) and for the transformer output and input (*1) according to (→ Fig. 19 Page 25).
- Close the transformer again after adapting.



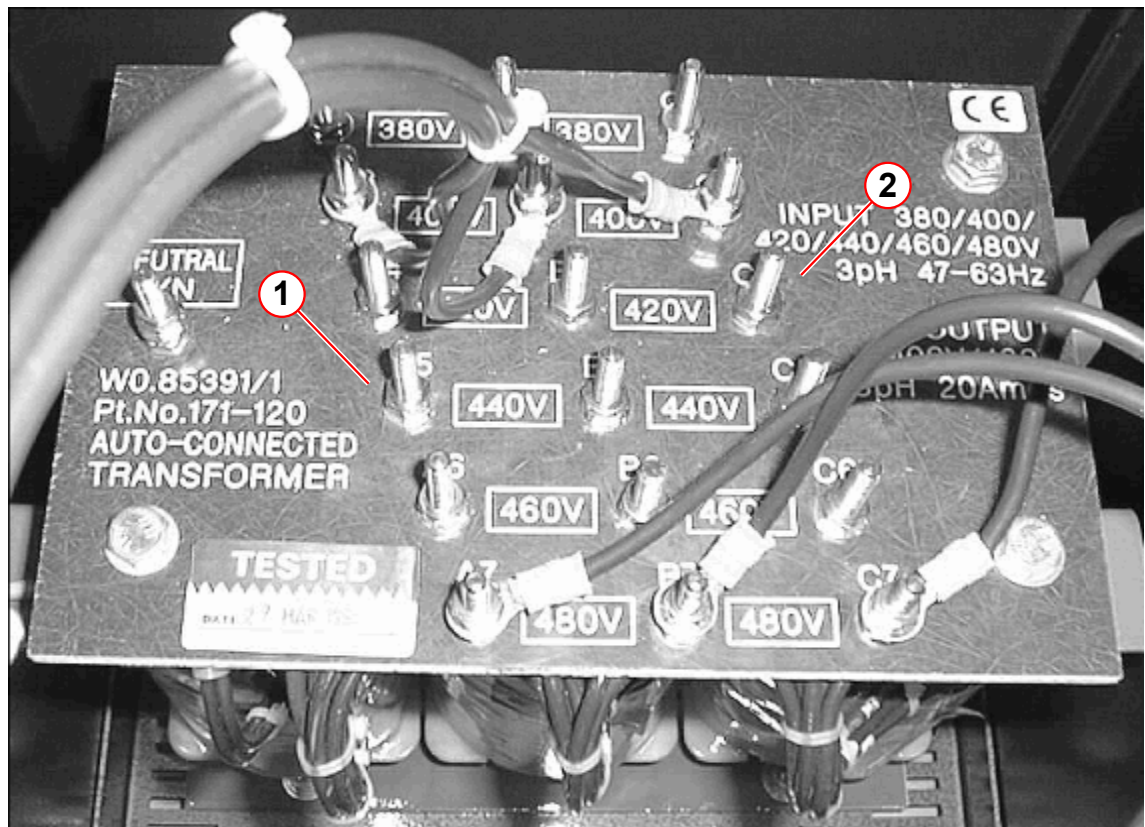
The line voltage used for the compressor depends on the line frequency.

Frequency: 50 Hz			
Main voltage	Refrigerator	Output (to RCA)	Input (from CCA)
380 V	set to 400 V	set to 400 V	*1
400 V	set to 400 V	no transformer necessary	
420 V	set to 400 V	set to 400 V	*1
440 V	set to 400 V	set to 400 V	*1

Frequency: 50 Hz			
Main voltage	Refrigerator	Output (to RCA)	Input (from CCA)
460 V	set to 400 V	set to 400 V	*1
480 V	set to 400 V	set to 400 V	*1

Frequency: 60 Hz			
Main voltage	Refrigerator	Output (to RCA)	Input (from CCA)
380 V	set to 470 V	set to 480 V	*1
400 V	set to 470 V	set to 480 V	*1
420 V	set to 470 V	set to 480 V	*1
440 V	set to 470 V	set to 480 V	*1
460 V	set to 470 V	set to 480 V	*1
480 V	set to 470 V	no transformer necessary	

Fig. 19: Voltage adapter plate inside the step-up transformer housing



- (1) Output (to RCA)
- (2) Input (from CCA)

2.7 Checking the helium gas pressure

2.7.1 F70 Helium Compressor

Fig. 20: Helium compressor



- Check the helium pressure at the compressor. (→ Tab. 2 Page 26)
- Enter the reading in the installation Protocol.

Equalization (static) pressure at 20°C (68°F) for 12 to 20 m long gas lines:

Tab. 2 Helium gas pressures

Cold Head	Static Pressure	Dynamic Pressure
10K - 50Hz	14.3 - 14.6 bar (207 - 212 psig)	21 - 22 bar (304 - 319 psig)
10K - 60Hz	13.6 - 13.9 bar (197 - 202 psig)	21 - 22 bar (304 - 319 psig)

2.7.2 HC10 Helium Compressor

Fig. 21: HC10 Compressor



- Check the He gas pressure on the manometer. The equalization pressure must be:

Tab. 3 Helium gas pressures

Cold Head	Static Pressure	Typical Dynamic Pressure
10K - 50Hz	15.6 - 15.9 bar (226 - 230 psig)	20.0 - 22.8 bar (290 - 330 psig)
10K - 60Hz	14.9 - 15.2 bar (216 - 220 psig)	20.0 - 22.8 bar (290 - 330 psig)

with the helium pressure lines connected to the cold head.

If the pressure is too high vent helium gas from the helium fill port compressor side.

If the helium pressure is too low, refill with helium gas, use only helium of the highest purity (99.999%), refer to TSG.

Typical operating pressure: **20.0 - 22.8 bar (290 - 330 psig) at 50 Hz or 60 Hz**

2.8 Adapting the on-site power to the system

WARNING

Verify that no voltage is present at the line voltage cable.

- » Switch off the on-site power supply (main power distribution to the system) and secure it with a lock to prevent anyone from inadvertently switching it on.

- Make sure the emergency shutdown buttons are still locked and no power is present.
- Open the line power box (one screw) at the front of the top cover.
 - The line voltage cable is ready to connect.
- Remove the isolation tape from the cable ends.
- Connect the 3-phase line voltage and ground.

Fig. 22: Adapting the system to the line voltage, remove the screw at the front to open the top cover



Fig. 23: Adapting the system to the line voltage, remove the screw at the front to open the top cover



(1) Terminal block for line power

- Connect the main power wires to the terminal block (Ground, L1, L2, and L3).

Fig. 24: Connect the ground and main power wires to the terminal block





Verify that the 3 phases are connected in clockwise direction

- at the line power connection jack and
- at the output of the RCA transformer (if available).

- Make sure that the wires are properly fixed.
- Close the line power box.

2.9 ERDU

2.9.1 ERDU Buttons

- Two ERDU buttons are delivered with Magnetom Harmony.
- Check the settings of switches SW2A and SW2B according to (→ Tab. 4 Page 30).

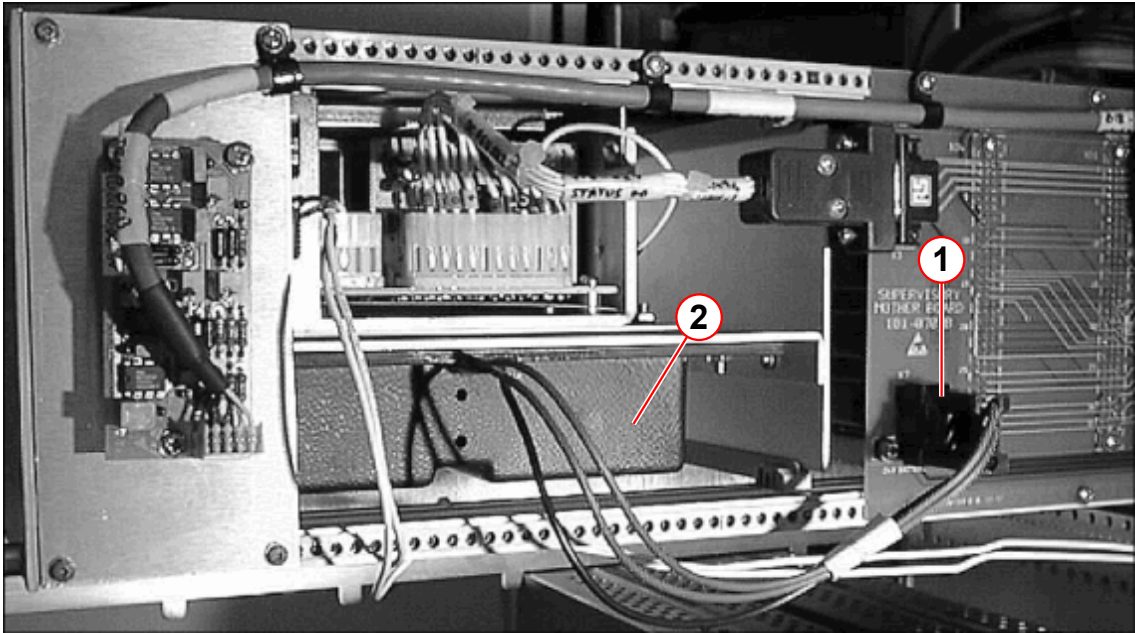
Tab. 4 Switches SW2A and SW2B

Switches on D3 MSUP	Number of ERDU buttons connected to the system	
	2 (standard)	3 (upgrade)
SW2A	closed	open
SW2B	open	open

2.9.2 Connecting the ERDU battery

- Open the MSUP (Magnet SUPERvision) and move the Faston connector of the "Accu PCB" to X7 of the backplane to connect the battery-in circuit.
 - The green "+24V" LED glows on the front plate of D2.

Fig. 25: Connect Pb accu to X7 of the MSUP Backplane



- (1) X7
- (2) Pb-Accu



The battery must be charged for at least 12 hours before
The ERDU can be tested or
It can be used to quench a magnet in case of an emergency.

2.9.3 Testing the ERDU

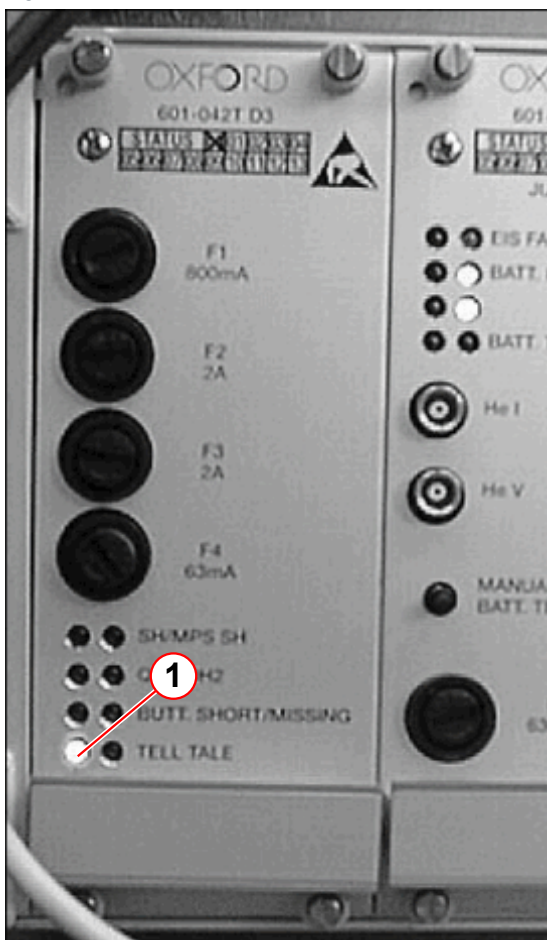
NOTICE

You will need an extra 63 mA fuse for this test.

» **Before the magnet is ramped to field, all ERDU buttons have to be tested to validate that the magnet can be quenched in case of an emergency. Test as follows:**

- Push the ERDU button in the examination room and lock it.
The following LED's are illuminated (with the AC supply OFF):
 - "SH", "QH1", "QH2", and "TELL TALE" on D3 plus "BUTT SUPP" on D2.The following LED's are illuminated (with the AC supply ON):
 - "SH", "QH1", "QH2", "BUTT SHORT" and "TELL TALE" on D3 plus "BUTT SUPP" on D2.
- Release the ERDU button in the examination room and observe the LED's on D3:
 - Only "TELL TALE" should still be illuminated, indicating that fuse F4 / 63 mA (magnet quench-button indicator) has blown (see figure below).

Fig. 26: Fuse F4 has blown if "TELL" LED is illuminated



(1) D3 MSUP

- If the TELL TALE led is not lit when the ERDU button is pushed on battery supply then either the fuse has not blown because of low battery voltage or there is some other problem.
- The led will not light on battery supply unless any ERDU button is pushed.

- Push the ERDU button at the Alarm box and observe the LED's on D3:
 - Again the LED's "SH", "QH1", "QH2", "BUTT SHORT" and "TELL TALE" are illuminated.
- Release the ERDU button at the alarm box.
- Replace fuse F4. Now all LED's on D3 are extinguished. The test has ended successfully.

2.10 Checking the water hoses

- Check all the hose connections prior to switching on the system / pump.



CAUTION

Hose connections tightened at room temperature may not be sufficiently tight after cooling down to the primary water temperature (due to material range contraction).

A leaky connection can cause considerable damage if water should run into one of the cabinets.

- » Please check all hoses for tight seating at the connection. Special care should be taken for the primary circuitry (egs: CCS). Double clamp any connections that could possibly leak.
- » Also the screw connections must be tight.

- Go ahead and check that every time you come to the system (it doesn't take long) - just in case.

2.11 Protective conductor test

- Measure all protective conductors (refer to (→ Component cabling / M1-010.812.01)).

3.1 Emergency shutdown buttons

For safety reasons, all emergency shutdown buttons must be tested prior to starting up the MR system. Refer to the following overview.



The system and UPS emergency circuits must be kept separate.

The UPS emergency circuit is only needed when the UPS option is installed (refer to (→ UPS / MR00-000.814.09)).

Fig. 27: Overview of the emergency shutdown buttons and UPS shut down buttons for Powerware Prestige 3000

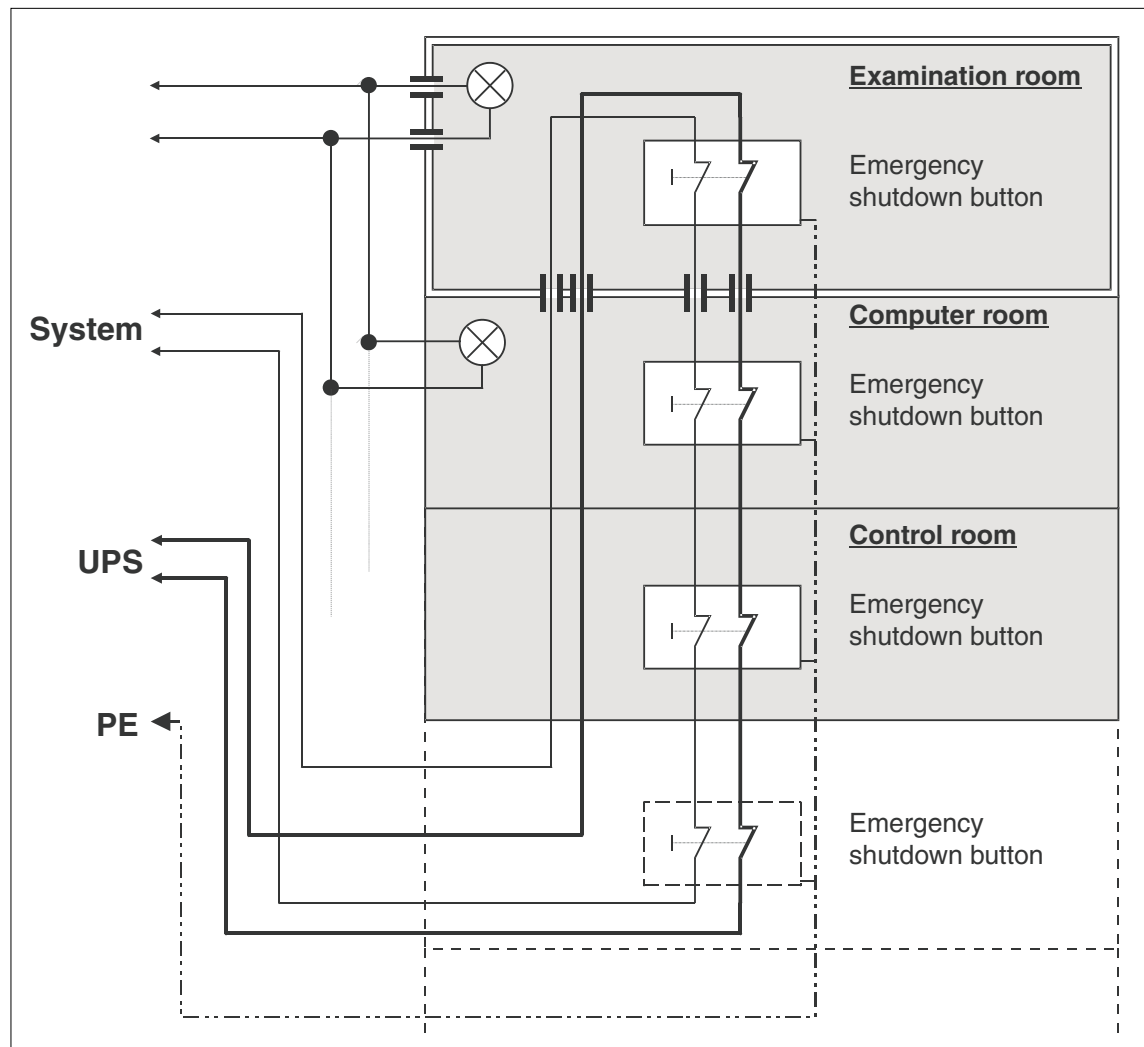
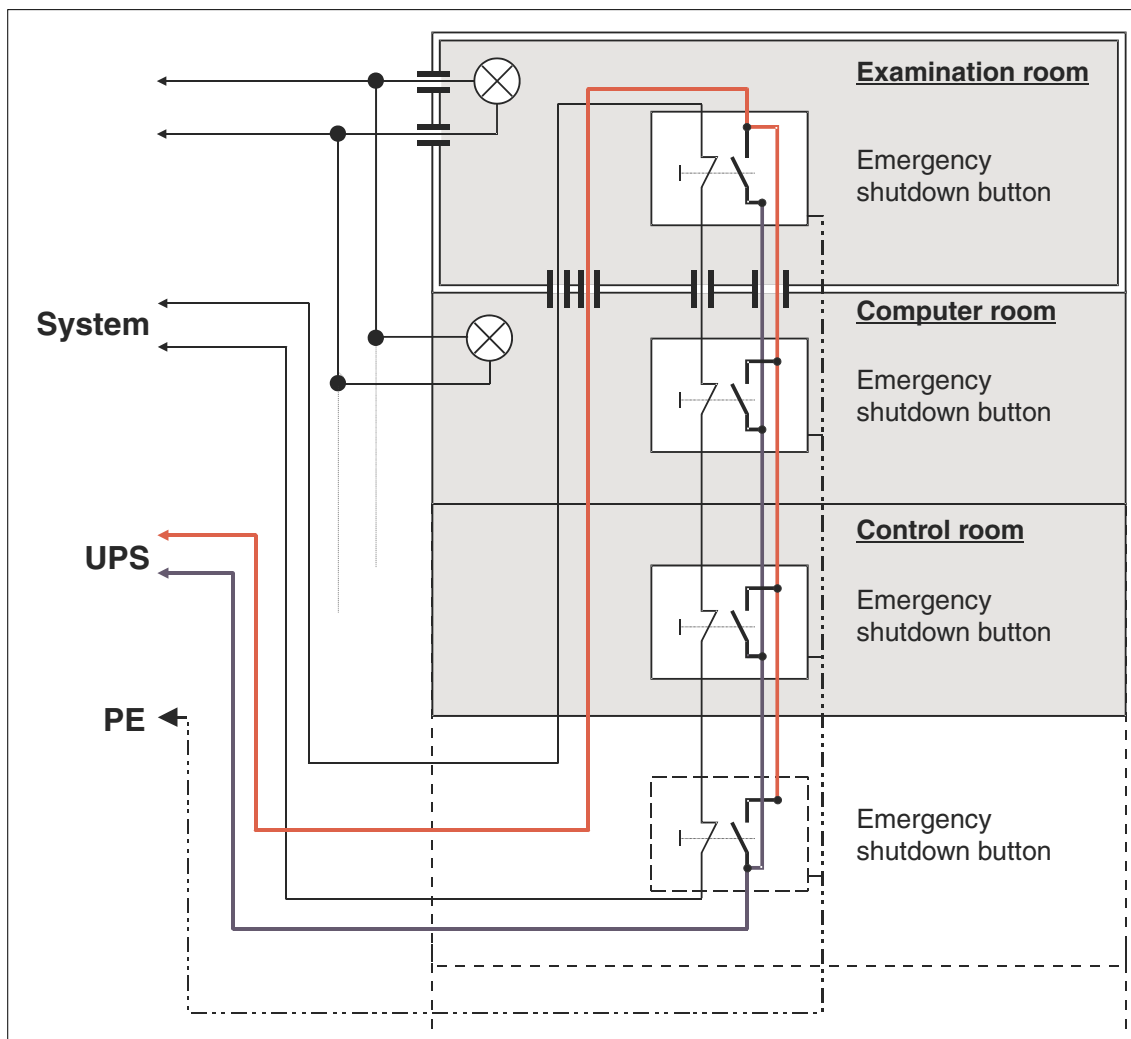


Fig. 28: Overview of the emergency shutdown buttons and UPS shut down buttons for Powerware Prestige 6000



3.1.1 Function test for emergency shutdown buttons



WARNING

High Voltage

If not observed, death or serious physical injury can occur.

- » If the circuit breakers are switched on in the line voltage distribution cabinet (all emergency shutdown buttons released), line voltage is immediately present at the following components:
 - » 1. Line transformer T1 in the CCA.
 - » 2. Line transformer for the Gradient cabinet (via F3 - 80 A, circuit breaker at top of the CCA).
 - » 3. RCA cabinet (via circuit breaker F4 - 32 A)
 - » 4. Magnet Power Supply (via circuit breaker F5 - 20 A)
 - » These components **CANNOT** be switched off via the MR circuit breaker K1. They must be switched off via the line voltage distribution cabinet (with emergency shutdown buttons or circuit breakers).

- Switch off circuit breakers F3, F4, F5 in the mains box/ CCA top and F14 , F23 inside the CCA.
- Switch on the on-site power supply.
- Press and lock the emergency shutdown button.
- Check whether
 - the on-site power supply is switched off and
 - the on-site power supply remains switched off.
- Perform this test for each of the following emergency shutdown buttons:

Emergency shutdown / Computer room

Emergency shutdown / Examination room

Emergency shutdown / Control room

Other room:



If an emergency shutdown button is activated, the system is disconnected from the on-site power supply. Battery-driven components (MSUP) will remain in operation since they cannot be switched off.

- Unlock all emergency shutdown buttons.

**WARNING**

Unlock all emergency shutdown buttons

Otherwise there is the danger of severe damage to power stages and rectifier board D32 of the KAS.

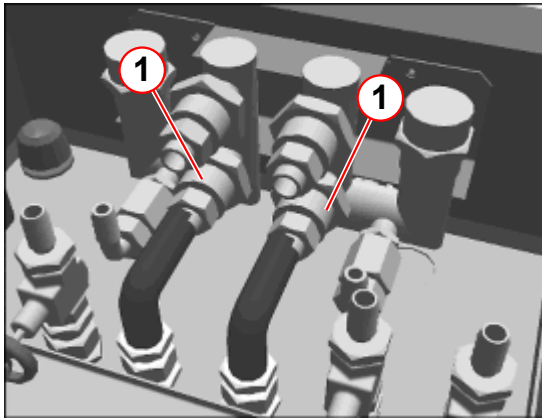
- » Ensure, that the GSSU of the gradient amplifier is switched off (F14/CCA) prior to switching on the 80A circuit breaker F3 in the mainsbox .

- Switch on circuit breakers F3, F4, F5 in the mains box/ CCA top
- Switch on the system with the "System ON" button at the Alarm box or at the CCA.

3.2 Starting the refrigerator

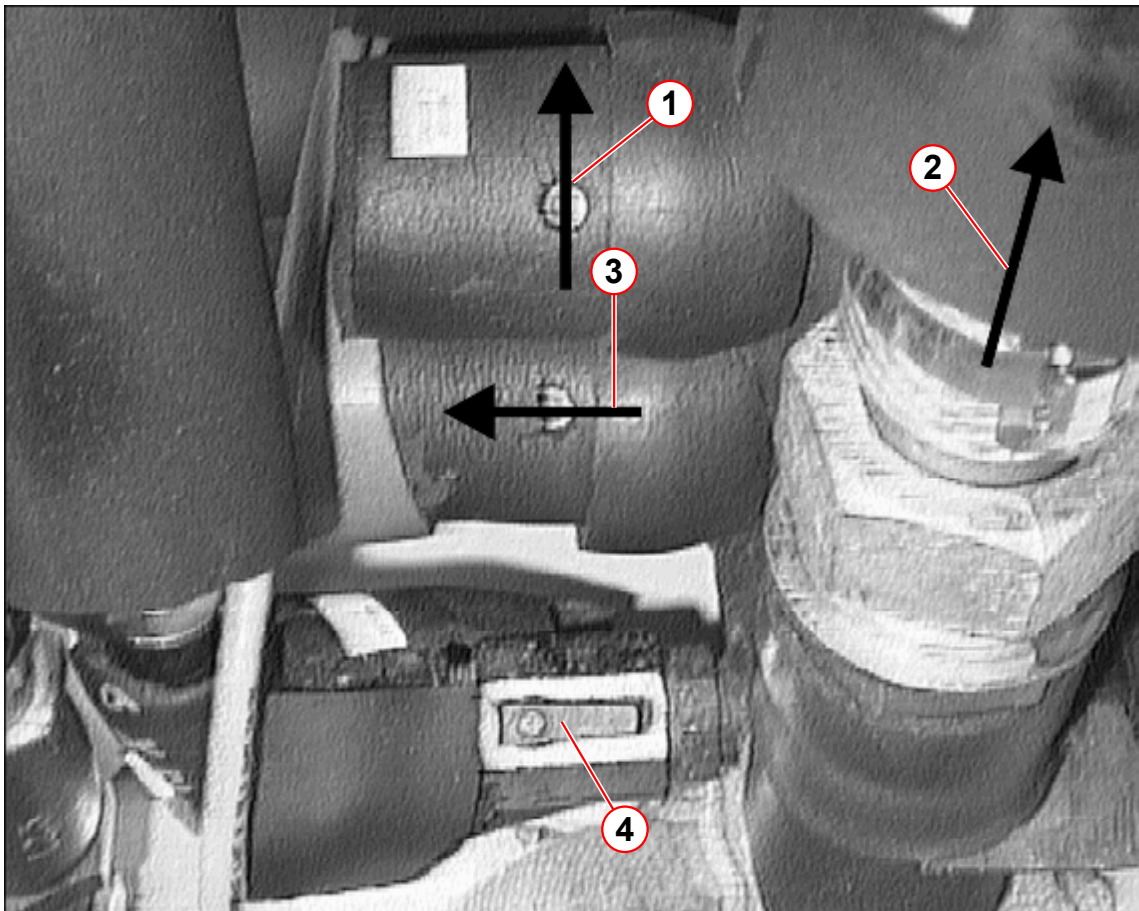
- Open the main system water valve so that the refrigerator (coolpack) begins cooling.
- Make sure that the flow direction of the primary water circuit (Primary water) corresponds to the arrow shown in (→ Fig. 30 Page 40).
- Open the ball valves on both sides with a 5 mm Allen wrench or screwdriver to cool "Cooler 1". The slot must point in the direction of flow.

Fig. 29: Open valves for cooler 1 on both sides



(1) valves

Fig. 30: RCA top view - opening both ball valves (cooler 1) with slot in the direction of flow



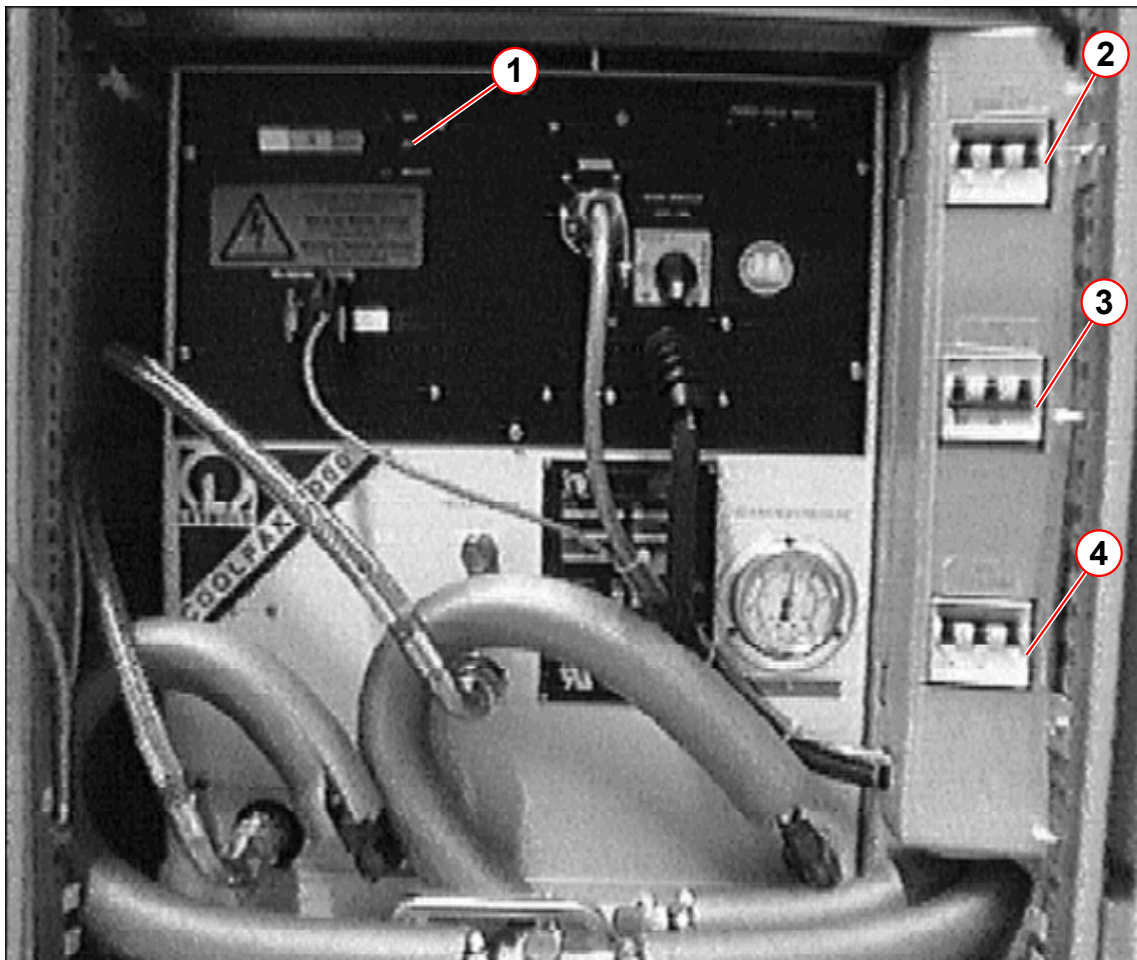
- (1) Cooler 2 closed
- (2) Primary waterflow
- (3) Cooler 1 opened
- (4) open valve only if air conditioning is available

3.2.1 MAGNETOM Harmony/Symphony: Switch ON Leybold Refrigerator

- Switch fuse F3 (→ 4/Fig. 31 Page 41) on.

- Switch the refrigerator on (coolpack) with the "ON" switch (refer to (→ 1/ Fig. 31 Page 41)).
 - The LCD display on the refrigerator displays "System ON".
 - The operating pressure on the manometer increases at first to approximately 23 bar and then decreases slowly within the next 24 hours to approximately 20 bar (due to the He gas cooling).
Check the pressure after the shield temperature has stabilized.
Typical operating pressure:
50 Hz: 20 - 21 bar (290 - 305 psig)
60 Hz: 19 - 20 bar (276 - 290 psig)

Fig. 31: Refrigerator front view



- (1) ON
- (2) F1
- (3) F2
- (4) F3

- Check again, to ensure that the primary hose connections do not have any leaks.

NOTICE

Check the hoses again for tight seating at the connections. Double clamp any connections that could possibly leak.

A leaky connection can cause considerable damage if water should run into one of the cabinets.

- » Refer to subtitle (→ Checking the water hoses / Page 33) in chapter (→ Prior to switch-on / Page 11) for more information.

3.2.2 MAGNETOM Sonata: Switch ON APD Refrigerator

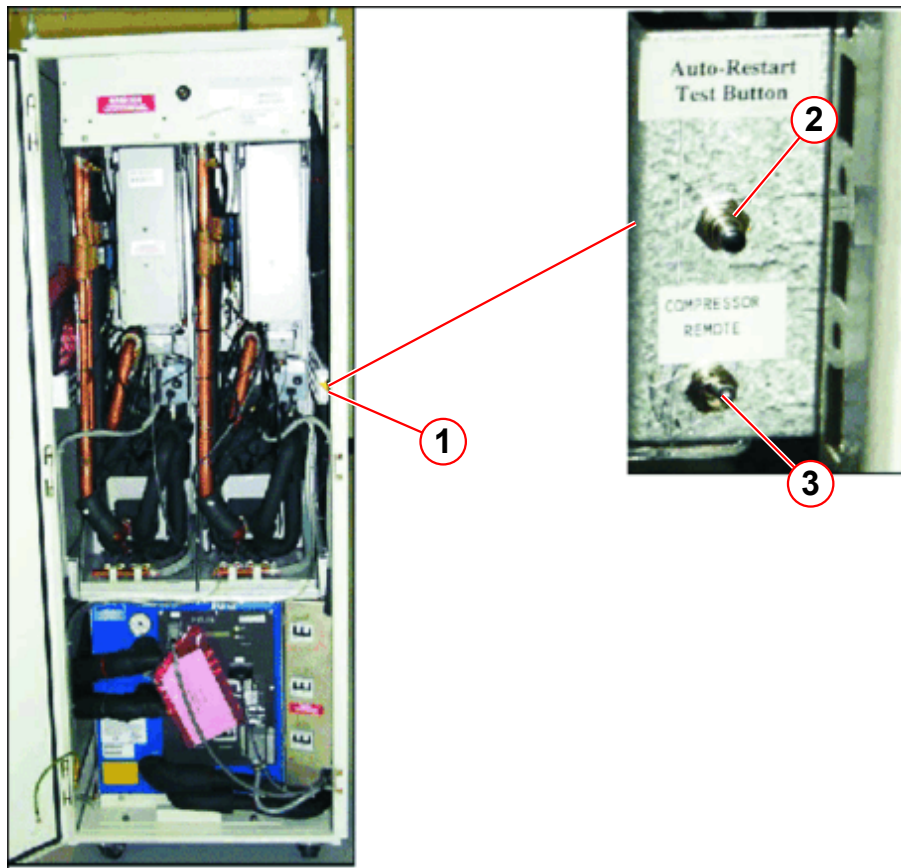
Additional functionality has been added to the RCA cabinet. An auto-restart command is sent to the helium compressor every 3 hours. The purpose of this is to ensure that in case of a failure (e.g. loss of primary water) the compressor will try to restart on a periodic basis.

NOTICE

In order that the compressor can run with the auto restart module, the compressor must be started from the remote switch.

- » DO NOT switch the compressor ON using the on/off button on the front panel of the compressor (→ 4/Fig. 33 Page 44)

Fig. 32: Compressor remote switch



- (1) Position of test button and remote switch for the APD compressor
- (2) Test button (30 second duration)
- (3) Remote switch for the compressor, UP = Off, Down = On

- Check if the remote switch is in OFF position (→ 3/Fig. 32 Page 43)
- Switch fuse F3 (→ 3/Fig. 33 Page 44) ON.
- Close the MAIN POWER circuit breaker (→ 5/Fig. 33 Page 44)
- Close the CONTROL SYSTEM circuit breaker (→ 4/Fig. 33 Page 44)
- Check that the display is ON but **DO NOT** switch the compressor **ON** using the on/off button on the front panel of the compressor
- Switch the compressor **ON** using the **Remote on/off** switch (→ 3/Fig. 32 Page 43)



If the compressor starts but does not build pressure, turn it off immediately. It could be running in reverse.

How does the compressor remote switch work?

- When the switch is in the **OFF** position (up) and power is applied to the compressor the display will come on and the compressor can be started in the usual manner (using on/off switch on the front panel of the compressor). With the switch in this posi-

tion, the display can be read without the compressor running. If the compressor is started in this mode, then the restart command will not work.

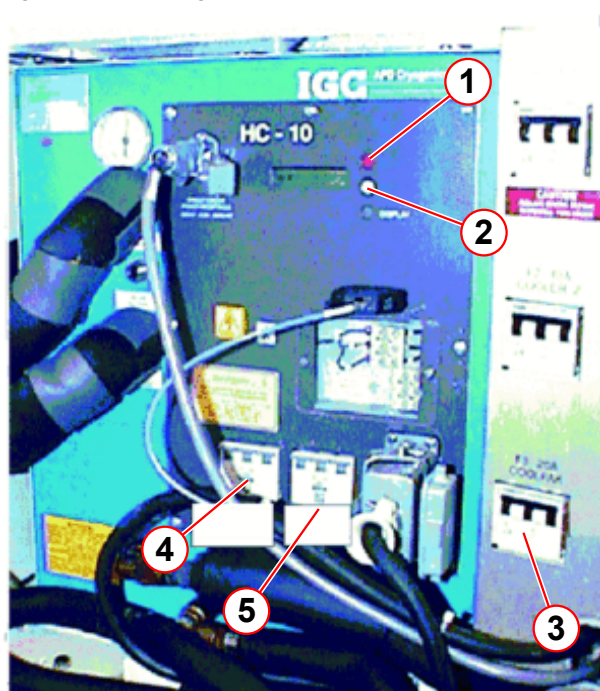
When the switch is in the **ON** position (down), and the power is applied to the compressor, the compressor will start automatically. This will enable the auto-restart of the compressor to function correctly. A restart pulse is sent to the compressor every 3 hours.

What is the test button for?

- The test button (when pressed) will send a restart pulse every 30 seconds to the compressor. This can be used to send a restart command manually.

It is also recommended that the test button be pressed during service visits to check that the circuit is operating correctly.

Fig. 33: APD Refrigerator front view



- (1) OFF
- (2) ON
- (3) F3
- (4) Control System
- (5) Main Power

- Check again, to ensure that the primary hose connections do not have any leaks.

NOTICE

Check the hoses again for tight seating at the connections. Double clamp any connections that could possibly leak.

A leaky connection can cause considerable damage if water should run into one of the cabinets.

- » Refer to subtitle (→ Checking the water hoses / Page 33) in chapter (→ Prior to switch-on / Page 11) for more information.

- Check the helium gas pressure
Typical operating pressure: 20.0 - 22.8 bar (290 - 330 psig) at 50 Hz or 60 Hz

3.3 Ethernet switches

A System with MRC console only needs one Ethernet switch for the communication between MRC host and MMC; the communication to the external Ethernet is done from the MRC host.

A System with MRC and MRSC console needs one additional MRSC Ethernet switch for high speed communication between MRC and MRSC; the communication to the external Ethernet is done by the MRSC Ethernet switch port 5. A twisted pair cable offered as an option connects the port 5 to the external Ethernet.

If a MRSC Ethernet switch is present:

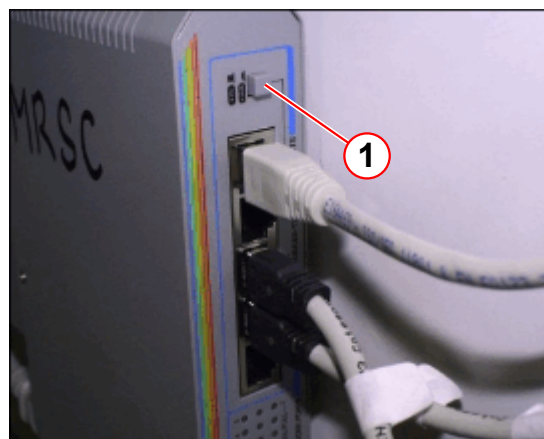
- Switch the MRSC Ethernet switch to "HUB" (button pressed).

For the MRC Ethernet switch the button position doesn't matter, because the HUB/XPC button only refers to port 5, which is not connected at the MRC switch

Fig. 34: MRSC Ethernet switch



Fig. 35: MRSC Ethernet switch: button in "HUB" position



(1) to "HUB" position

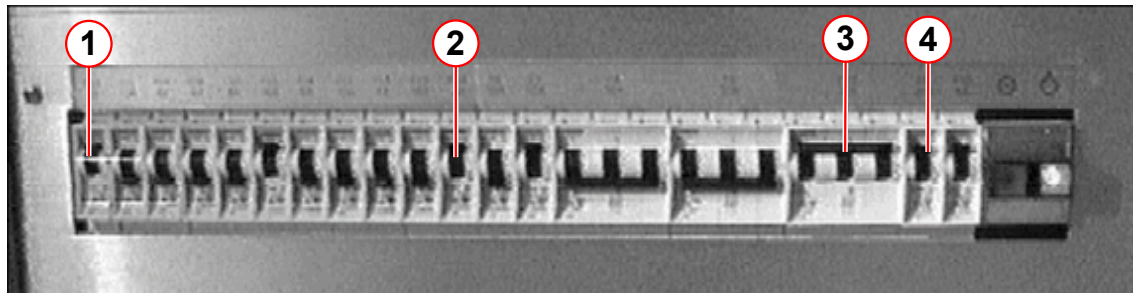
3.4 Switching on the system

3.4.1 Circuit breakers

- Switch on the following circuit breakers in the CCA:
 - F22 (RCA Control)
 - F24 (Msup)
 - F23 (CCA)
 - F2 (UPS)

None of the other circuit breakers should be switched off at this time.

Fig. 36: Switch these circuit breakers on



- (1) F22
- (2) F24
- (3) F23
- (4) F2

3.4.2 The Alarm box

The Alarm box indicates the system status via LED's:

- Line lights up, if line voltage is present.
- System ON lights up if the system (not the MRC) is switched on.
- LHe Level lights up, if the helium level drops below the minimum level specified.
- Magnet Stop lights up in case of an ERDU error or if a Magnet STOP switch was activated.

The Alarm box contains the following switch functions:

Previous Status	Push button ... to change status to...		
	Host Standby	System ON	System OFF
MRC OFF System OFF	MRC ON System remains OFF	MRC ON System ON	no action

Previous Status	Push button ... to change status to...		
	Host Standby	System ON	System OFF
MRC ON System OFF	no action	MRC remains ON System ON	MRC OFF System OFF
MRC ON System ON	MRC remains ON System OFF	no action	MRC OFF System OFF

Fig. 37: Alarm box



3.5 Software configuration

- Check the installed software version; refer to register Installation Software, chapter Configuration

3.6 Filling the secondary water circuit

Approximately 50 to **70 liters** of **distilled water** are required to fill the secondary water circuit. An additional 6 to 10 liters of distilled water are required to fill the magnet power supply.

- With the hoses at maximum hose length of 4 x 15 m between the CCA and RCA, approximately 13 more liters of water are needed to fill the circuit than at the minimum hose length of 4 x 2.5m (you need approximately 0.285 liters of water per meter of 3/4" hose used).

NOTICE

Do not use deionized water; use distilled water only.

After initial shimming procedure has been completed, citric acid must be added to the water circuit.

- » **Refer to (→ Ramping the magnet / Page 106), (→ Adding Citric Acid to the Secondary Water Circuit / Page 166).**

NOTICE

For MAGNETOM Sonata Cooler 1 and Cooler 2 has to be filled with additional 20l of distilled water.

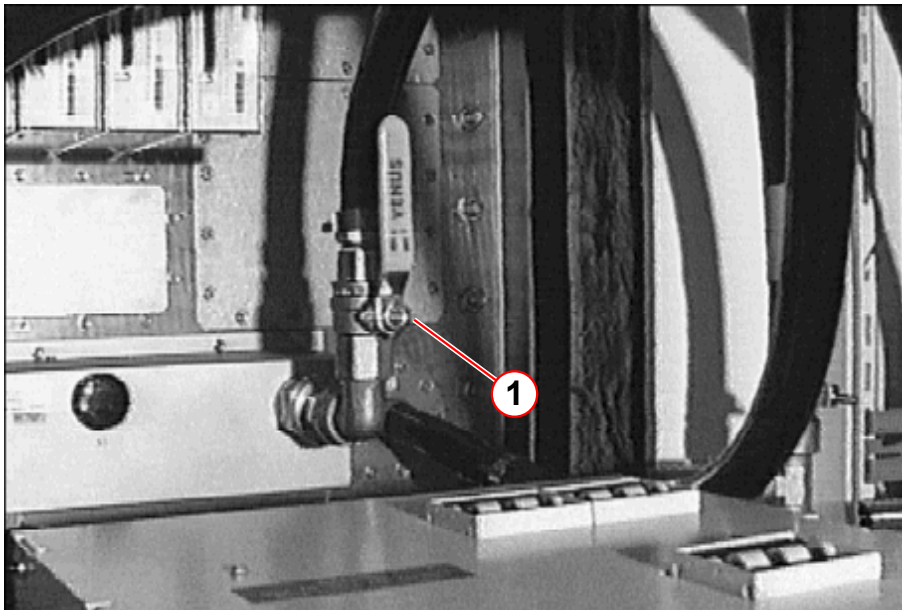
- » **Also refer to (→ Setting the secondary water flow rates for MAGNETOM Sonata / Page 64).**

- Remove the cover of the "cooler".
- Open the following valves in the secondary water circuit:
 - At the filter plate (quick disconnect (→ Fig. 38 Page 51))
 - Return for gradient coil and GPS at the RCA (refer to (→ Fig. 39 Page 51)).
- Close the water flow valves (→ Fig. 44 Page 55) and then open them a maximum of two to three turns. (at 50 Hz line frequency) or open them a maximum of one turn (at 60 Hz).



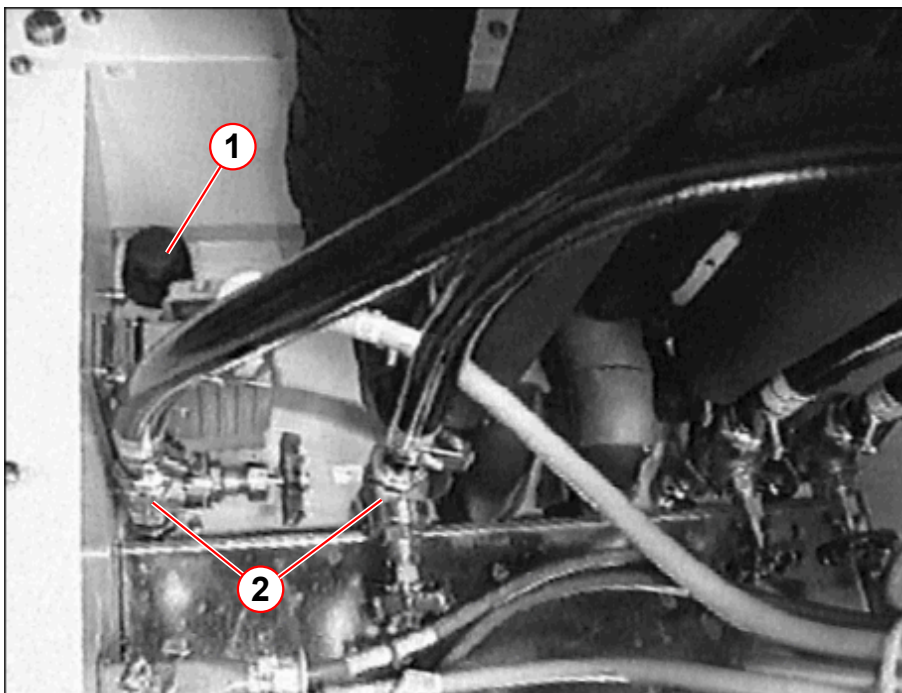
If the water flow valves are open too far, the overpressure monitor will switch the pump off shortly after it starts.

Fig. 38: Water valve opened at the filter plate



- Open the valve cap of the water tank and insert the funnel extension.

Fig. 39: Open the valve cover of the water tank and insert the extension hose of the funnel



- (1) Water tank sealing cap
(2) Return valve open max.

- Switch on fuse F1 "Cooler 1" (refer to (→ Fig. 36 Page 47)).

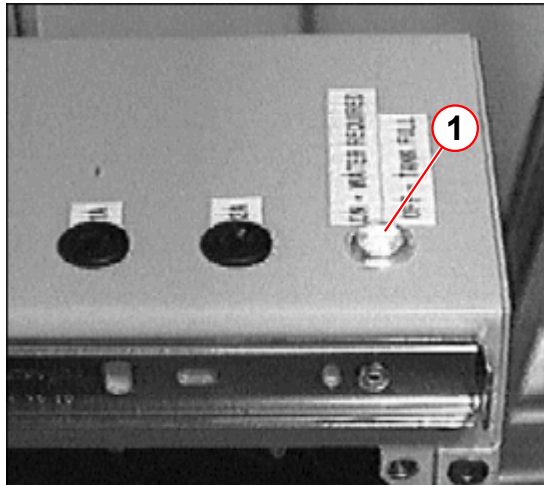
- Fill the water tank to its maximum level with distilled water (monitor the yellow lamp - refer to (→ Fig. 41 Page 53)).

Fig. 40: Filling procedure at RCA cabinet (front)



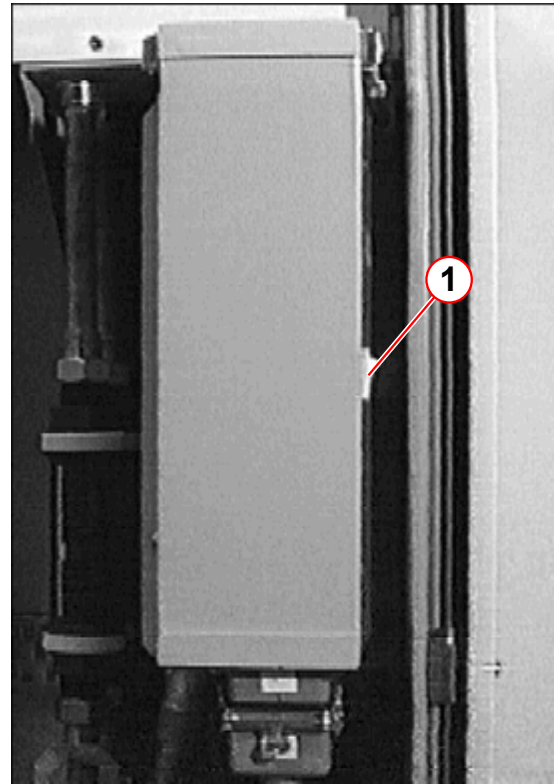
- Set the pump switch to "MAN".
 - The pump gradually starts and fills the water circuits of the gradient coil and the electronics cabinets. When the minimum acceptable water level is reached, the pump switches off automatically. No air is pumped into the circuits. The yellow lamp is illuminated.

Fig. 41: Water indicator - yellow Lamp LP1



(1) LP1

Fig. 42: Auto / Man switch



(1) Auto / Man switch

- Refill the water tank, until the yellow lamp goes out.
 - At that point, the pump will gradually begin to run again and refill the water circuit.
- The cycle begins again; refill the tank with distilled water until the yellow lamp goes out.

NOTICE

Refill the water only when lamp LP1 lights up.

Refill only enough water until lamp LP1 goes out.

» **A residual air pocket must remain in the tank.**

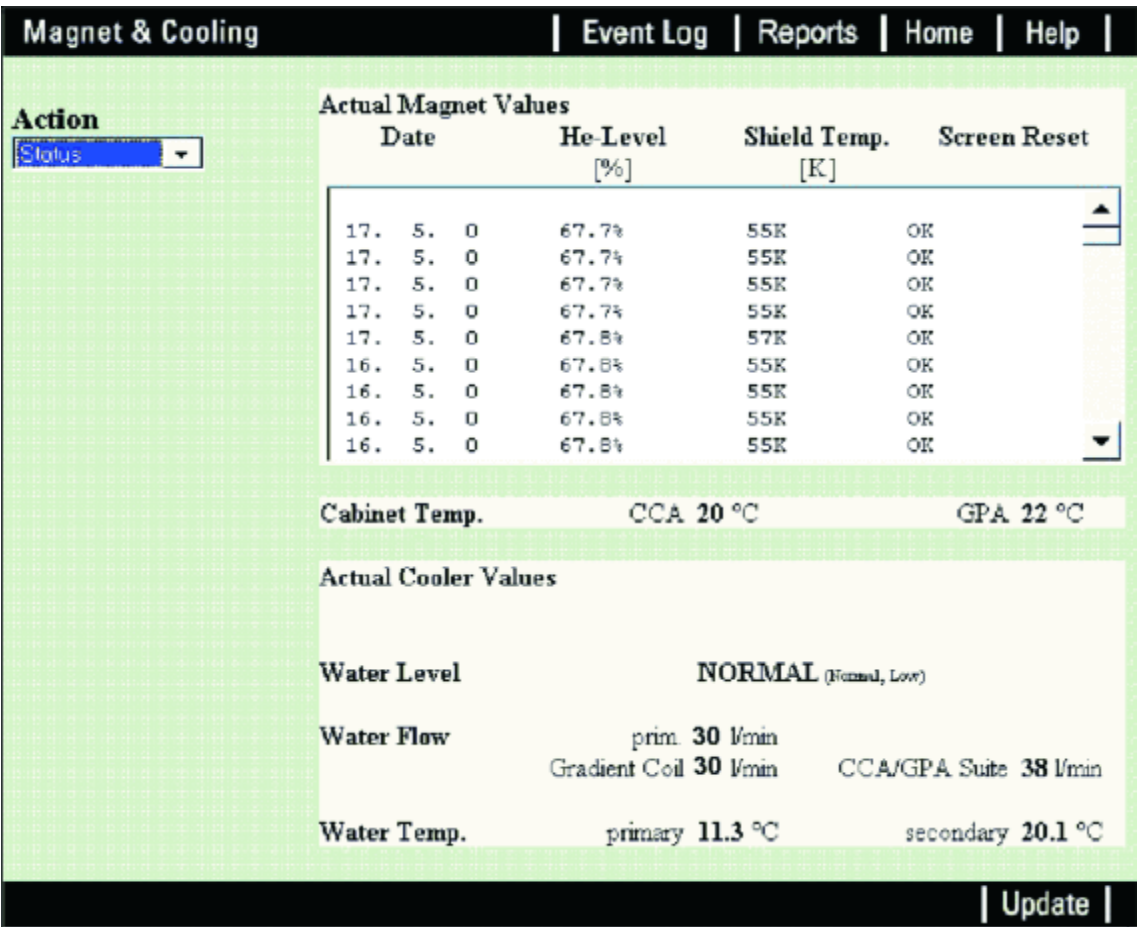
- If lamp LP1 remains off when the pump is running, the refill procedure is ended. Close the cap of the water tank.

3.7 Setting the water flow rates for MAGNETOM Harmony/Symphony

The flow rates of the secondary water circuit can be checked using the service software ("SeSo"). All following steps should be executed "on-line" at the MRC. Online Help explains the individual steps.

- Access the service menu in the SeSo (Service Software) as follows at the MRC:
- Enter your password and select menu item **"Magnet & Cooling"**.

Fig. 43: SeSo Menu Magnet & Cooling - Action "Status"



- Select **"Status"** in the "Action" menu (takes up to 20 sec in time).
- The current flow rates are listed in the field **"Actual Cooler Values"** under **"Water-flow"**

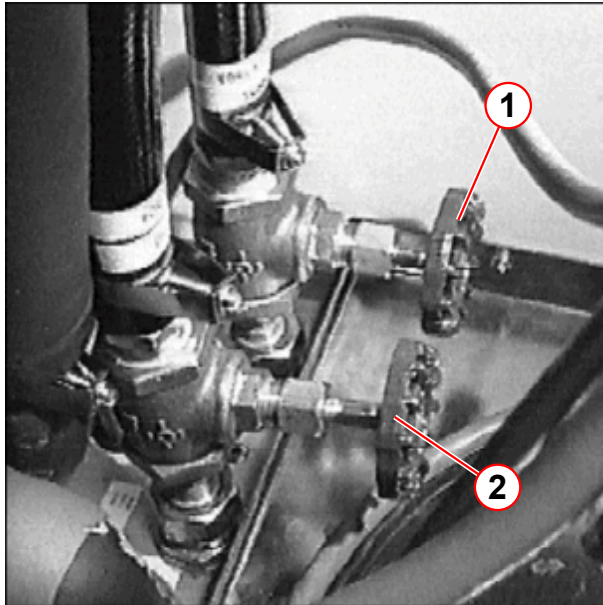
i The status menu shows only the primary water flow rate for the cooler (30 l/min). The flow rate for the helium compressor (10 l/min) and for the water cooled cabinet cooling system (20 l/min) is not shown.

- Adjust the flow valves (refer to (→ Fig. 44 Page 55)), make sure that the flow rates are within specification, refer to (→ Tab. 5 Page 55).

i

Adjust the flow from the closed position of the valve in small step to the required value. If the water flow valves are open too far, even if the flow rates are within spec, the over-pressure monitor will switch the pump off shortly after it starts. It takes some time for the software to indicate the change in flow.

Fig. 44: Slowly adjust the water flow valves for flow control



(1) Gradient cabinet

Tab. 5 Flow rates and warning levels for MAGNETOM Symphony: Turbo/Ultra and Quantum/Q

Harmony / Symphony	Turbo/Ultra		Quantum / (Q Japan only)	
	Gradient coil	CCA/GPA Suite	Gradient coil	CCA/GPA Suite
Flow rates	30 l/min	38 l/min	29 l/min	38 l/min
Warning level	25 l/min	33 l/min	24 l/min	33 l/min
Primary water Flow rate	30 l/min		30 l/min	
Warning level	25 l/min		25 l/min	

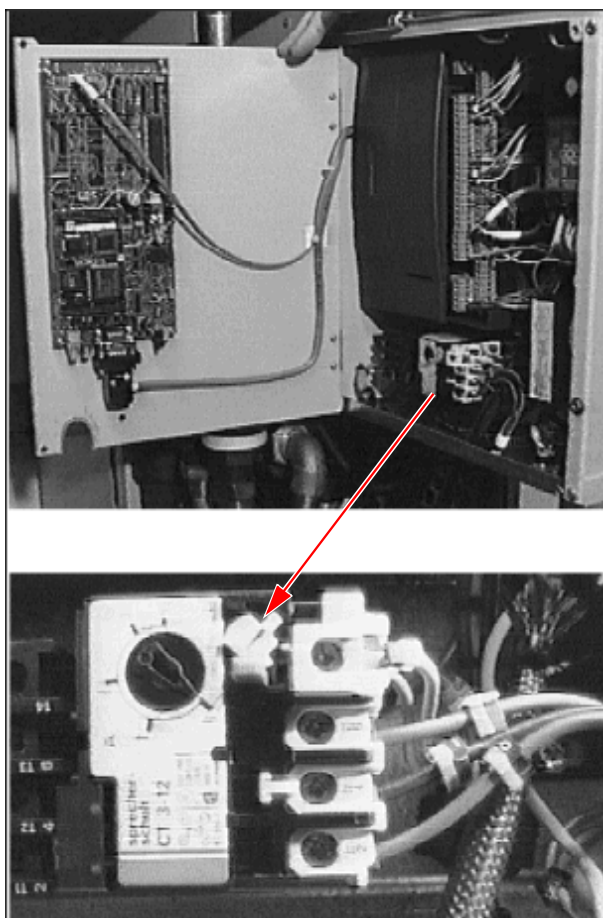
NOTICE

If the indicated flow rates cannot be reached (e.g. only 2/3 of the indicated flow), the pump rotation is not correct.

- » In this case, switch two of the connection wires (50, 51, 52 - arrows in (→ Fig. 45 Page 56)) at the motor protection circuit in the pump control. Then perform the flow adjustment again.

- Recheck the secondary hose connections for leaks.

Fig. 45: Motor protection circuit on the water pump - change the pump rotation by switching the wires



- Adjust the pump switch to "Auto" (→ Fig. 42 Page 53).
- Switch on the entire MR system at the Alarm box with the "System ON" switch.

3.8 Setting the temperature set point

- Select **"Initialization"** in the "Action" menu.
- Check the standard temperature value under **"Cooler"** in the "Temperature set point" field (→ Fig. 57 Page 67). The value must be **20°C** to avoid water condensation.
- Save if settings have been changed.

3.9 Cleaning the water filter in case of a local move and Re-furbished Systems

General: Clean the filter after cooling was running for at least 1 hour

Clean following filters:

- Filter of GPA / CCA Cabinet
- Filter of RCA
- Filter of watercooled Bodycoil (if available)

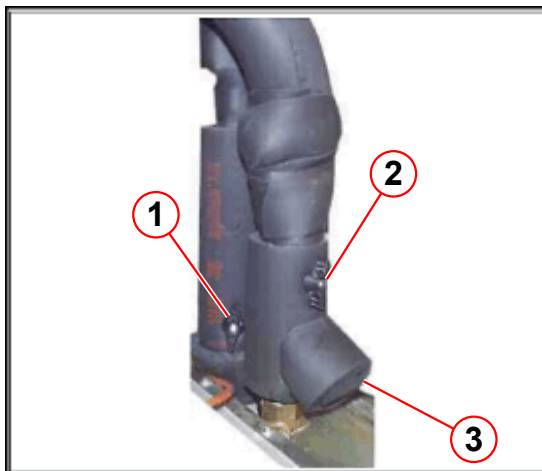
3.9.1 CCA Cabinet

- Switch off the CCA cabinet
- Carefully swivel out the electronic section of the CCA

Checking the water-cooled CCS

Cleaning the filter

Fig. 46: Top of the CCS



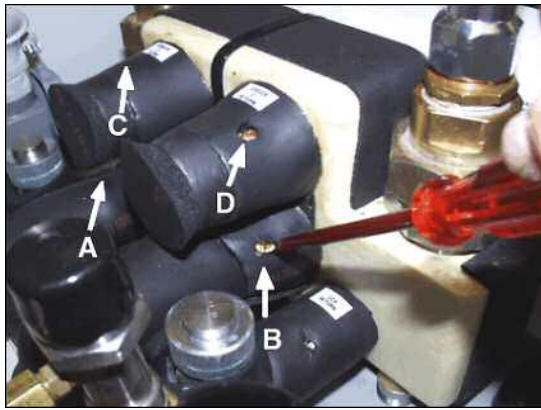
- (1) Return valve
- (2) Flow valve
- (3) Filter

- Close the shut off flow and return valves on the top of the CCS, (→ Fig. 46 Page 58) .
- Remove the filter using the open-end wrench, (→ Fig. 46 Page 58).
- The filter can be cleaned by rinsing it under tap water.
- Mount the filter and open the valves.

3.9.2 RCA / Body Coil

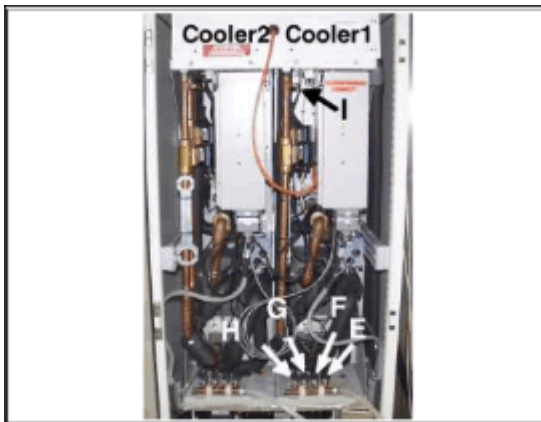
- Switch off compressor and F1, F2, F3 in the RCA [! Unresolved GRAPHICLINK !].
- Switch off F22 in the CCA
- Switch off F4 on top of CCA
- Remove the electronic box

Fig. 47: Intake and outlet valves



- **Turn off the intake (A) and outlet (B) valves** to isolate the primary water circuit, Cooler 1 (→ Fig. 47 Page 59).
- Sonata is also connected to Cooler 2. The strainer has to be cleaned here as well. **Turn off the intake (C) and outlet (D) valves** and isolate the primary water circuit, Cooler 2.

Fig. 48: Drain valves



- Open the primary circuit drain valves (E, F). Open the bleeding valves (I) or remove the filter cap to speed up the draining process (→ Fig. 48 Page 59).
- With Sonata, also open the corresponding drain valves on Cooler 2.



Cooler 2 applies only with MAGNETOM Sonata.

Fig. 49: Flowmeter



- Loosen screw (E), and pull flow meter (F) slightly to the right (→ Fig. 49 Page 60).

Fig. 50: Primary strainer



- Open the strainer on Cooler 1 and, if applicable, Cooler 2. Using a 27 mm wrench, refer to "Inspect the strainer for contamination" and carry out cleaning procedures, if necessary (→ Fig. 50 Page 60).
- Reassemble strainer.

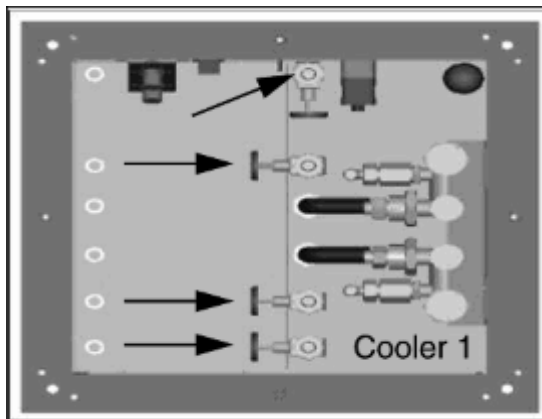
- Close drain valves.
- Assemble flow meter.
- Assemble electronic box.
- Turn on intake (A) and outlet valves (B). If Cooler 2 is connected, open intake (C) and outlet (D) valves (→ Fig. 47 Page 59).
- Switch on F22 in the CCA.
- Switch on F4 on top of CCA.
- Switch on F1, F2, F3 in the RCA and compressor (COOLPACK) **[! Unresolved GRAPHICLINK !]**.

Checking the secondary water circuit

Cleaning of the strainer

- Disconnect all power from machine
- Switch off compressor and F1, F2, F3 in the RCA **[! Unresolved GRAPHICLINK !]**.
- Switch off F22 in the CCA
- Switch off F4 on top of CCA

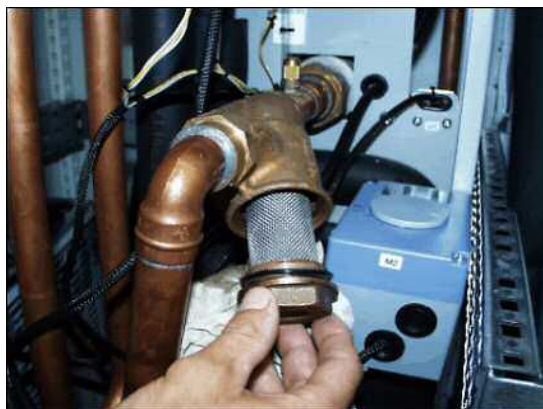
Fig. 51: Secondary valves



- Turn off the inlet and outlet valves of Cooler 1 on top of RCA (→ Fig. 51 Page 61) to isolate the secondary water circuit.

- Remove the electronic box.
- Open the secondary circuit drain valves (the two left valves) (→ G, H/Fig. 48 Page 59).
- Open the bleeding valves or remove the filter cap to speed up the draining process.

Fig. 52: Secondary strainer



- Open the strainer on Cooler 1 by using a 37 mm wrench. Inspect the strainer for contamination and clean it by rinsing under tap water. (→ Fig. 52 Page 61)
- Install the filter.

- Close the drain valves (→ Fig. 48 Page 59).
- Open the inlet and outlet valves.



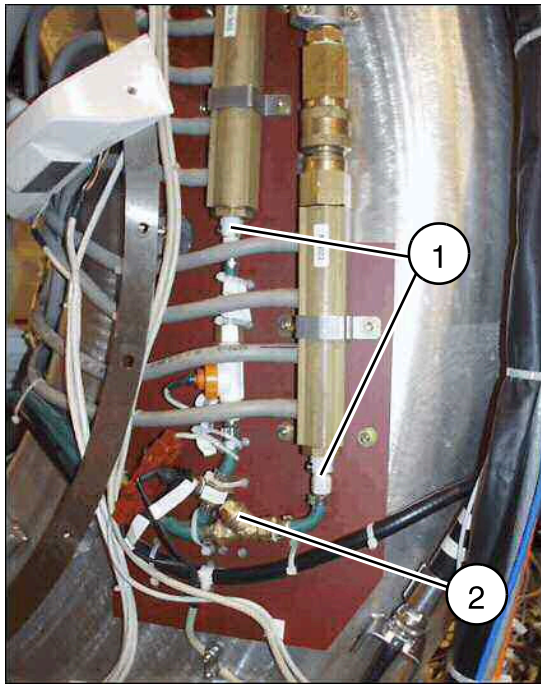
With Sonata, the work procedures for Cooler 2 are the same as for Cooler 1.

Cleaning the body coil water filter (water cooled body coil only)

The previous versions of the Symphony / Sonata body coils are water cooled. The newer **Body Coil-2** is not water cooled. Check the system configuration to identify your type of body coil. Newer type body coils have a BTB-2 type, e.g. BTB047-2.

Harmony body coils are always water cooled.

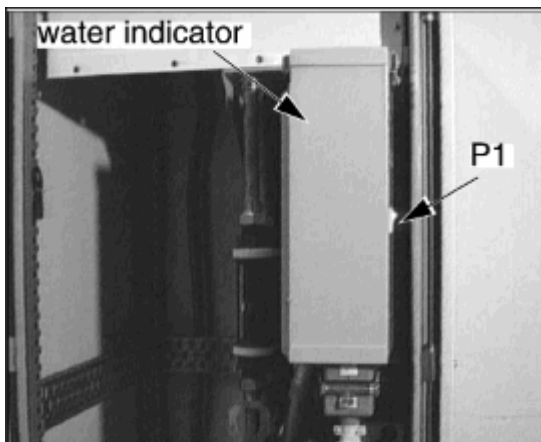
Fig. 53: Water cooled body coil



- Remove the rear magnet cover.
- Disconnect the two quick connectors (1) of the body coil cooling circuitry at the gradient coil water distributors (→ Fig. 53 Page 62).
- Open and clean the small water filter (2).
- Close the filter again.
- Reconnect the two quick connectors.
- Check for leaks
- Close the magnet cover.

- Switch on F22 in the CCA,
- Switch on F4 on top of CCA,
- Switch on F1, F2, F3 and the compressor in the RCA [! **Unresolved GRAPHICLINK !**].

Fig. 54: RCA cabinet



- Check the water level (use service software: refer to "Magnet&Cooling"). If necessary, refill the water tank with distilled water. The yellow light on the control box indicates the water level (→ Fig. 54 Page 62).
- Adjust the flow rates for gradient coil and CCA/GPA cabinet.
- Check the water temperature (20 degrees centigrade +/- 3degrees) using the service software.

3.10 Checks after 12, 24 and 48 hours

- Check the level in the water tank. If the yellow lamp LP1 (→ Fig. 41 Page 53) glows, refill with distilled water until the lamp goes out.
- Check the "flow rates" again in the service software.

3.11 Setting the secondary water flow rates for MAGNETOM Sonata

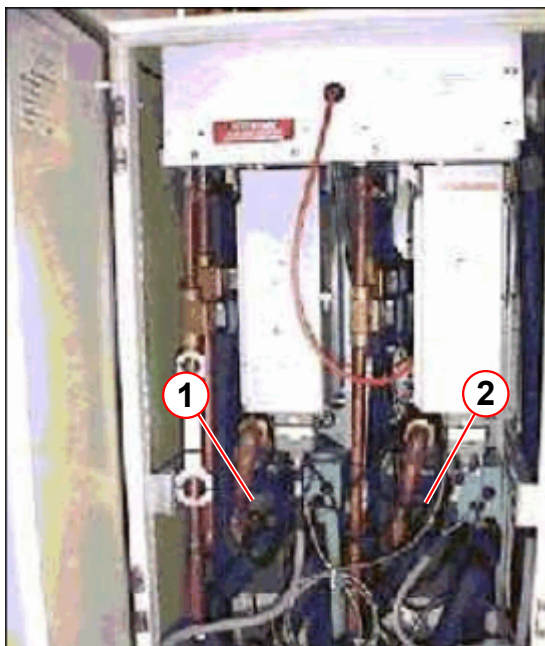
3.11.1 Prerequisites

- Cooler 1 and cooler 2 has been filled according to the procedure (→ Filling the secondary water circuit / Page 50).
- The primary water valves are open for both coolers. See (→ Starting the refrigerator / Page 39)
- The fuses F1 and F2 are switched on. See (→ Fig. 31 Page 41)



Cooler 1 supplies cooling water for the CCA/GPA Suite.
Cooler 2 supplies cooling water for the Gradient Coil.

Fig. 55: RCA with 2 coolers

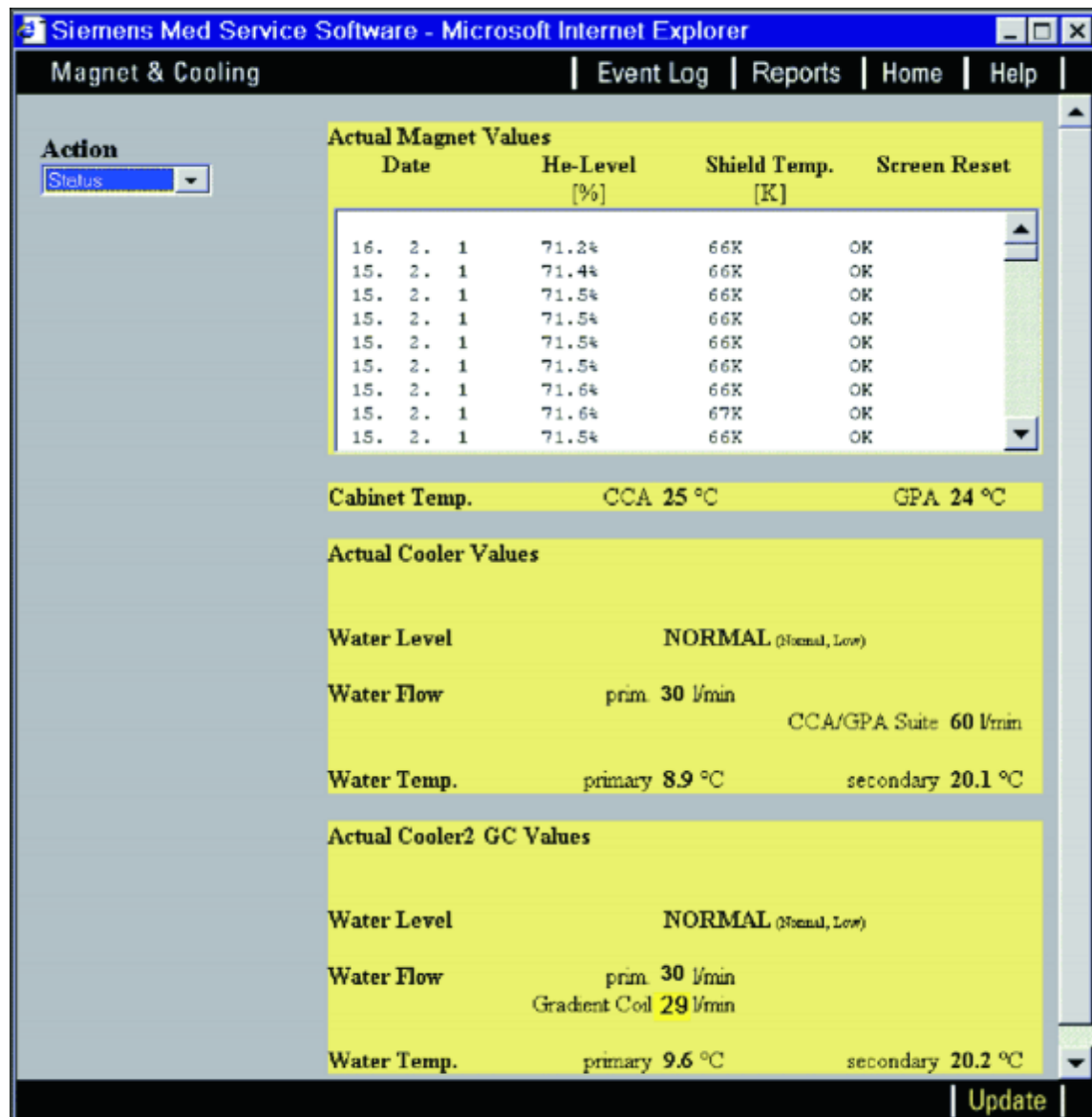


- (1) Cooler 2
- (2) Cooler 1

3.11.2 Setting the water flow rates

- Start Service Software
- Select "**Magnet & Cooling**"

Fig. 56: Sonata SeSo Menu Magnet & Cooling - Action "Status"



The status menu shows only the primary water flow rate for the coolers (cooler 1: 30 l/min; cooler 2: 30 l/min). The flow rate for the helium compressor (10 l/min) and for the water cooled cabinet cooling system (20 l/min) is not shown.

- Adjust the flow valves make sure that the flow rates are within specification, refer to (→ Tab. 6 Page 66).



Adjust the flow from the closed position of the valve in small step to the required value. If the water flow valves are open too far, even if the flow rates are within spec, the over-pressure monitor will switch the pump off shortly after it starts. It takes some time for the software to indicate the change in flow.

Tab. 6 Flow rates and warning levels for MAGNETOM Sonata

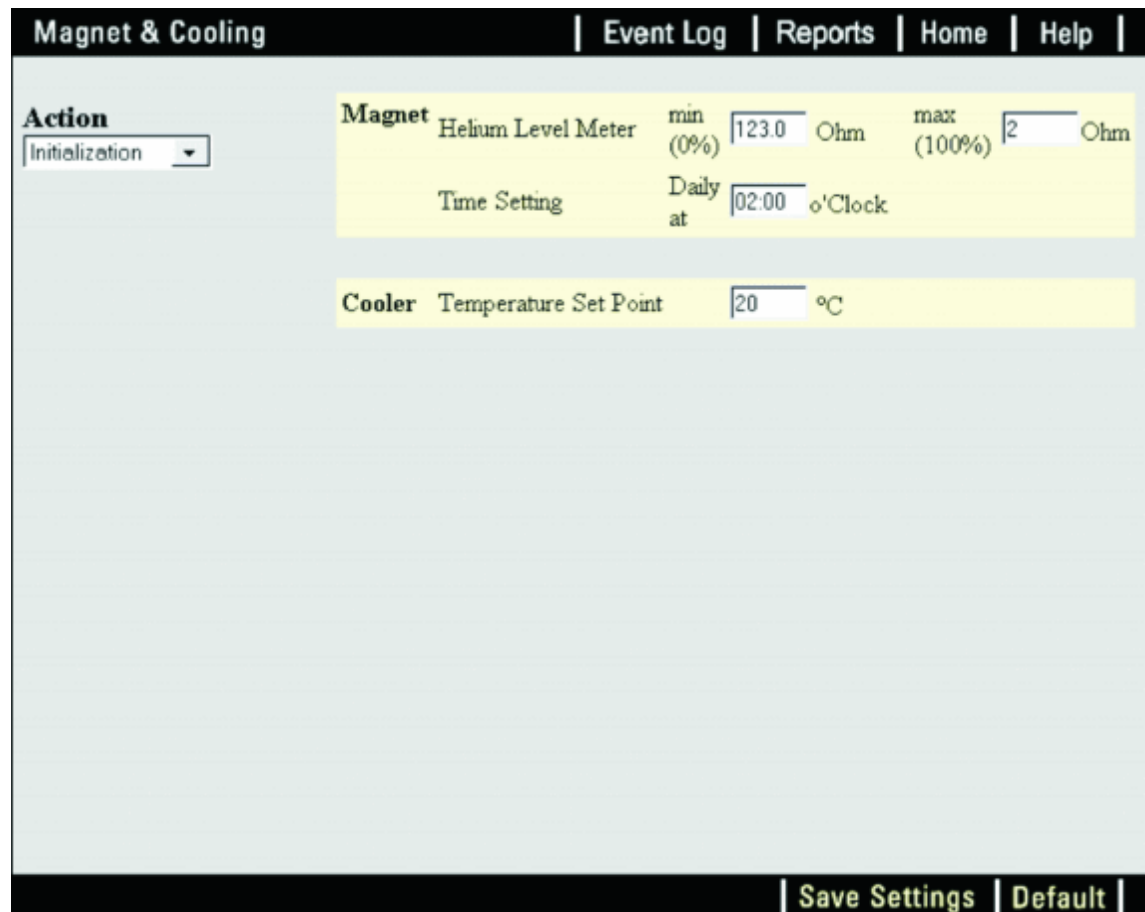
	Cooler 1 Service software: "Cooler"		Cooler 2 Service software: "Cooler 2 GC"	
	Gradient coil	CCA/GPA Suite	Gradient coil	CCA/GPA Suite
Secondary water Flow rates	n.a.	60 l/min	29 l/min	n.a.
Warning level	n.a.	55 l/min	24 l/min	n.a.
Primary water Flow rate	30 l/min		30 l/min	
Warning level	25 l/min		25 l/min	

3.12 Setting the LHe level meter resistance

- Select **"Initialization"** in the "Action" menu.
- Enter the resistance of the LHe Probe under "Helium Level Meter" in the following Se-So menu (refer to the "Oxford - Factory System Acceptance" document, subtitle **"Helium level probe resistances** when helium level is zero (%)"):

Magnet		min (0%)	max (100%)
Helium Level Meter		123.0 Ohm	2 Ohm
Time Setting	Daily at	02:00 o'Clock	

Fig. 57: SeSo Menu Magnet & Cooling - Action "Initialization" (with example of probe resistance)



Magnet & Cooling | Event Log | Reports | Home | Help |

Action
Initialization ▾

Magnet

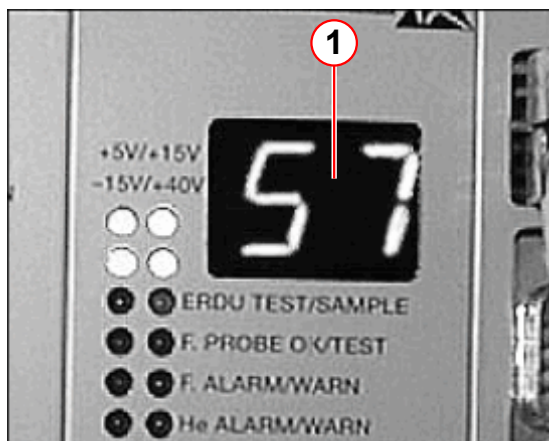
Magnet		min (0%)	max (100%)
Helium Level Meter		123.0 Ohm	2 Ohm
Time Setting	Daily at	02:00 o'Clock	

Cooler Temperature Set Point 20 °C

| Save Settings | Default |

- Save the settings.

Fig. 58: MSUP D1 - LHe level meter



(1) actual LHe level

3.13 Setting the screen reset time

- Select **"Initialization"** in the "Action" menu.
- Enter the time for reset under "Time Setting" (→ Fig. 57 Page 67) in the **"Magnet"** field.
 - The reset must occur outside the normal time frame for patient examinations. Select e.g., '2:00 a.m.'
- Save the settings.

3.14 Remote Diagnostics

3.14.1 Introduction

Remote Diagnostics allows to establish a connection from a remote location in order to provide remote diagnostics of the system, on-line technical assistance, automated reporting of system quality assurance data (QDB), magnet monitoring data, and system utilization data (customer option).

For new installations with the syngo-MR software VA21A and beyond only Router connections are supported.

3.14.2 Supported connections

- ISDN Router (Router provided by SIEMENS)
- Customer ISDN Router
- Direct connection via Internet Security EndPoint (**VPN tunneling**)
Authentication: SHA1 or MD5
Supported encryption: DES or 3DES
- Router with analog modem interface via modem (Router provided by SIEMENS)
- Customer Router with analog modem interface via modem

3.14.3 Responsibility

The project manager is obligated to plan for a remote connection in corporation with the customer.

3.14.4 Support

The most recent documentation regarding Remote diagnostics is available at the SIEMENS Intranet (→ Siemens Remote Service, <http://cs.med.siemens.de/cms/productinfo/default.asp?startfolder=e3371dfd0ace1e5181c7d36777ef3b4b>) or contact your USC/TSC/HSC

NOTICE

Regarding the remote configuration of the syngo-MR system, Refer to the on-line help text "configuration".

- » For detailed information regarding rdiag configuration, refer to (→ Software Installation syngo MR A30 (NUMARIS/4VA30A) / MR-000.816.26).

3.14.5 Parameters obtained from the local network manager

Use following form to plan SRS router configuration located at siemens Intranet: (→ Siemens Remote Service, <http://cs.med.siemens.de/cms/productinfo/default.asp?startfolder=e3371dfd0ace1e5181c7d36777ef3b4b>)

After filling out the network checklist please send it by **email**: srs_final.med@siemens.com

Systems

Tab. 7

IP address and netmask for the syngo MR console:	
IP Number MRC	
IP Number MRSC	
Netmask	

Systems connected via **router**:

Tab. 8

Login information / Communication information syngo MR (ROUTER)	
Router ISDN Number	
Router Name	
Router IP Number	
Router Callback	(Yes /No)
Router Chap Password	

3.14.6 Parameters obtained from the USC:

- FTP account and password for your account on the RDIAG server.
- Recipient Mail Address (USC account)

Tab. 9

For your notes:		
Rdiag username / password	Pmr _____	
FTP account / Passwd	rdiagftp	
Recipient Mail Address		

3.15 TestTools

Mandatory testing of the TestTools is required at the end of the installation.

Instead of running all tests at one sitting, it is recommended to perform them individually over a period of time (e.g., during wait times, lunch break, etc.).

To get an idea of how much time is needed for the tests, please refer to the following table. The given times are for a fully equipped system. Multi-nucleus tests are not mandatory but recommended, the time will increase with every measured nucleus. Please plan to perform the tests accordingly:

Test time in hh:mm:ss	functional	communication/status	interactive	multi nucleus	complete
MMC	00:08:31	00:01:00	00:06:04		00:15:35
RF	00:48:28	00:00:16	00:45:09	01:03:17	02:53:07
LCCS plug x-int			00:33:00	00:33:00	00:01:06
GPA	00:25:02	00:00:07			00:25:09
IMAGER	00:03:57				00:03:57
HOST	00:00:02		00:10:00		00:10:02
All comm. tests		00:01:21			

Normally it is not necessary to perform the cable-tests in the RF-section. Only in case of problems while trouble-shooting they should be used to isolate the defective component. The following tests can be performed prior to Tune-Up/QA as they do not require a tuned system:

- MMC complete
- Imager complete
- Host complete
- Communication tests complete



Before handing over system, please ensure that all tests of the TestTools have been performed and are within specification.



When the patient table was removed for moving the system into the site it is highly recommended to run the LCCS-int tests prior to shimming of the magnet.



The LCCS-int test requires no interaction, only a pop-up has to be confirmed at the start of the test (not at the start of the test session!).

3.16 Warning signs



Prior starting up the magnet system make sure that the warning signs (→ Fig. 59 Page 73) to (→ Fig. 62 Page 73)) are posted!

The warning signs are available in different languages. Pick the correct language. The following pictures are examples.

Fig. 59: Sign_MAGNETOM



Fig. 61: Sign_Filling_up

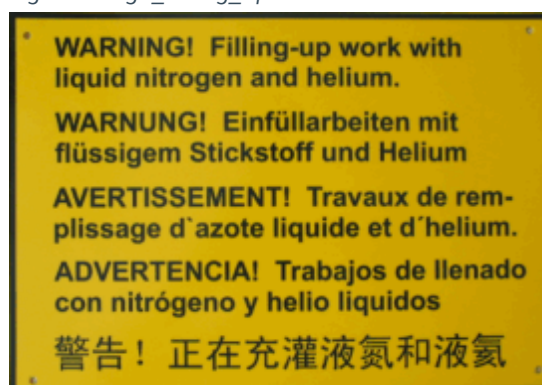
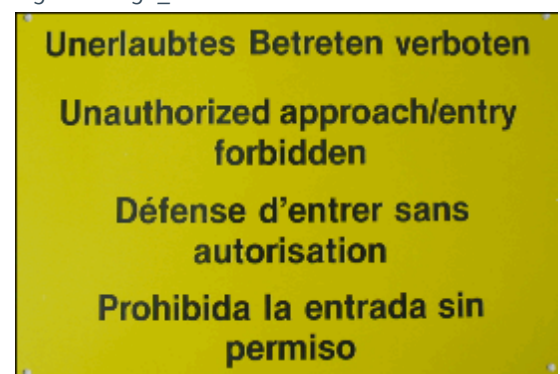


Fig. 60: Sign_Pacemaker



Fig. 62: Sign_Unauthorized



4.1 General Overview

This procedure describes the filling of OR24 and OR70 magnet systems when using the standard Oxford dewar syphon with the High Efficiency Syphon (HES) or Smart syphon when performing the fill with the magnet at its full operating field.

The procedure also explains the different filling conditions for Static systems compared to those used for Mobile systems.

It is extremely important that the procedures are read carefully paying particular attention to the 'Notices and Warnings' as well as all the Health and Safety requirements when working with cryogenic liquids and magnet systems that are at field.

The "Filling the Magnet" procedure is in Sections which are in the logical process steps.

The type of LHe syphon fitted to the system will need to be established before commencing the fill so that the correct syphon fill procedure is followed.

The main process steps are:

- Safety & auxiliary tooling
- System, syphon & dewar preparation
- Precooling the dewar syphon
- Helium transfer procedure with HES syphon
- Helium transfer procedure with Smart syphon
- Stopping the transfer & disconnecting the syphons
- Changing the LHe supply dewars
- After the transfer & leak check

NOTICE

Establish type of syphon fitted and read appropriate procedures.

- » The filling procedures for the HES and SMART syphon types is completely different!!! Please read, be familiar with and carefully follow the appropriate procedure for your specific syphon in order to prevent unintentional quenching of the magnet.

WARNING

Procedures and warnings not followed.

Personnel injury or damage to equipment or the magnet system may occur.

- » Read all the Safety instructions, Important Notes and Aux. Tools described. They are valid for all system and syphon types.

4.2 Safety

This section on safety is valid for the different filling procedures and syphon types used.

- For your safety and the safety of third parties and to protect the MR system and its parts from damage, adhere to the safety information contained in:
 - General safety notes (→ General Safety Notes / TD00-000.860.01)
 - MR-specific safety information (→ MR-specific safety information / MR-000.860.01)
 - Register (→ Safety guidelines / MR-000.860.01) of this manual.
- Prior to beginning work, read the special information in the manual regarding:
 - the general safety information for service personnel,
 - the restrictions regarding ferro-magnetic materials,
 - the safety information for working with the magnet at field,
 - the information on working with heavy loads,
 - the information on cryogen handling.
 - The information on handling pressurized gas cylinders.



WARNING

Protective clothing, gloves and eye protection must be worn during the following procedure.

Personnel injury may occur.

- » Trained personnel only should perform the following.



WARNING

Check ferrous content of equipment (including LHe dewars).

Personnel injury may occur or damage to equipment.

- » Only non-magnetic equipment may be taken into the magnet room.



WARNING

Do not vent Helium gas into confined areas.

Possible asphyxiation to personnel.

- » Follow correct HES procedures to minimize gas venting.
- » Depressurize all dewars outside the imaging suite or by connecting to the system's vent line.

**WARNING**

Restrict access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

- » Unauthorized personnel must be excluded from the imaging suite.

**WARNING**

Use appropriate warning signs and fence area to limit access.

- » "Warning: authorized personnel only"
- » "Warning: filling area for liquid Helium"
- » "Warning: danger of cold burns and asphyxiation"
- » "Warning: Strong magnetic fields"

**WARNING**

Dewar valve positions.

If the access or vent valves on the dewar are left open, air will freeze in the neck of the dewar with the strong possibility of causing an explosion.

- » See LHe supplier's manual for safe operation.

4.3 Auxiliary tooling

- Sufficient Helium supplied in the largest possible supply dewars for the site.



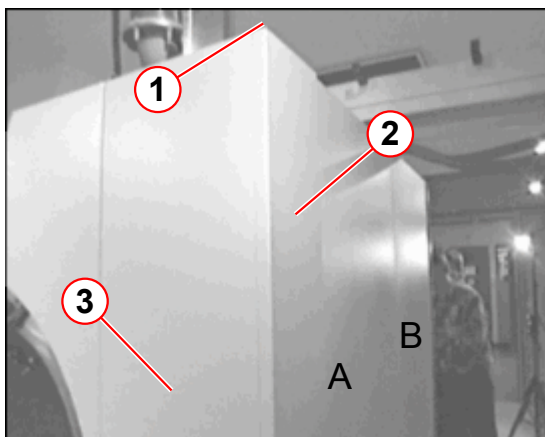
Larger dewars will increase efficiency and minimize the risk of air ingress into the syphons.

- Helium gas cylinder (at least 99.996%, equals 4.6 quality in some regions).
- Pressure regulator 200bar/ 0 - 1.5 bar
- Helium syphon (dewar end).
- Gas tubing for pressurizing the dewar.
- Syphon/dewar adaptor for the LHe dewar (syphon diameter = 16 mm).
- Replacement bursting disk and poppet seal "O" ring for quench valve (if needed).
- Heat gun and paper towelling.
- Safety step ladder.
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes and personnel oxygen monitor.
- Non magnetic torch (beam type) with HDCL inspection sight glass.
- Leak detection fluid or suitable electronic leak detector.
- Read descriptions and operating manuals for the cryogen dewars and gas transfer equipment used.
- Appropriate warning signs.

4.4 LHe filling of Static and Mobile Systems

4.4.1 System Preparation

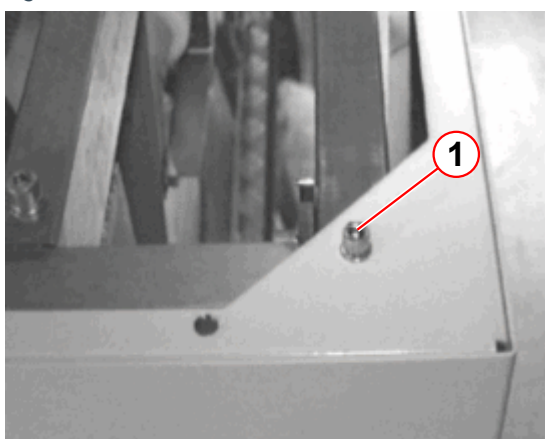
Fig. 63: Rear corner cover



- (1) 6 mm Allen screw
- (2) Side panels
- (3) Rear corner cover

Removing the Cover

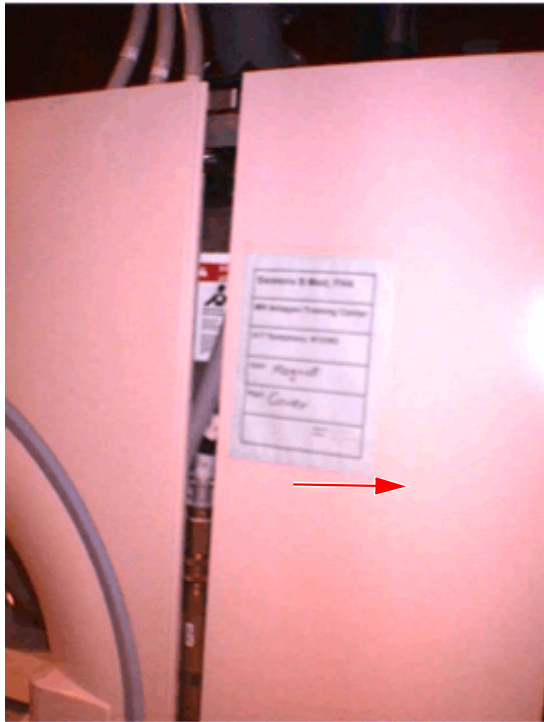
Fig. 64: Rear corner cover screw



- (1) 6 mm Allen screw

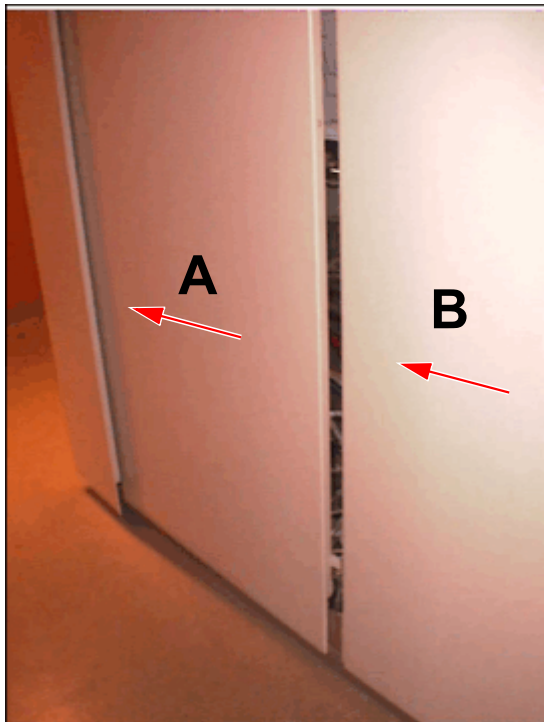
- Remove the 6 mm Allen screw on top of the rear corner cover.

Fig. 65: Rear corner cover



- Move the rear corner cover to the right approximately 5 cm (2 inches)

Fig. 66: Side panels



- Move the side panel A to the left
Move it behind the rear corner cover
- Move the side panel B a little to left
until you have access to both sides
of the side panel B
Lift the side panel with both hands

- The magnet system syphons are designed to be permanently fitted to the system and should remain in position at all times. If the syphon needs to be replaced or there is a requirement to fill the system from empty or from Nitrogen temperature then contact Oxford Service Department.

- Static and Mobile systems have different vent valve ratings. Before commencing LHe fill check the system type and operating pressure expected.

NOTICE

Mobile and Static systems are fitted with different magnet venting assemblies.

STATIC SYSTEMS

OR24/70 are fitted with a vent assembly with two 1/3rd psi valves in series with a bypass valve in parallel. When gas freely vents through the valves a pressure of 0.6 psi [0.041 bar] will be seen on the system pressure gauge.

OR41/42/35/32 systems are fitted with only one 1/3rd psi valve. When gas freely vents through the valve a pressure of 0.3 psi [0.02 bar] will be seen on the system pressure gauge.

MOBILE SYSTEMS

OR24/70 are fitted with a 1/3rd psi valve and 16 psia absolute valve which will give a higher differential pressure which can be as high as 4.2 psi [0.28 bar] on the system gauge at high altitudes. Standard atmospheric pressure will see the system gauge reading about 1.6 psi [0.11 bar].

OR41/42 are fitted with a single 16 psia valve. When gas vents freely through the valve a pressure of [1104 mBar absolute]

» Refer to (→ Fig. 84 Page 105) for pressure readings at different altitudes.)

4.4.2 Check of magnet pressure

- Check that the system pressure gauge is reading a POSITIVE pressure. If not, the transfer must not proceed. It is essential for the system to be at a positive pressure to ensure that there is no possible air ingress or ice within the turret which will cause inefficient LHe transfers.

NOTICE

A positive system pressure is required for correct cooling of the HES syphon prior to connecting with the dewar syphon.

» Follow procedure (→ Connecting to the HES syphon / Page 88).

- If the system is at atmospheric (zero) pressure, there is a leak. This will result in air ingress which will cause icing within the system.
- Before continuing the filling procedure the system must be checked for leaks.
- Check the turret, venting and system valves.
- Check the syphon end cap.
- Check the quench valve burst disc and "O" ring.
- See section: (→ Magnet System leak check / Page 101).

- If no leaks can be found contact Uptime Service Centre.

NOTICE

Ice will prevent an efficient transfer.

- » Using an HDCL sight glass check for large amounts of ice looking down the current lead (HDCL) port. If ice is visible contact Uptime Service Centre.

NOTICE

Pressure can be raised in the system by operating the EIS switch heater when the magnet is at field.

- » (refer to the (→ Pressurisation of the helium vessel / MR-000.840.17).

- Check that the connections to the system's Helium exhaust port are intact, and that the Helium exhaust is venting freely.

4.4.3 Current lead (HDCL)

- The demountable current lead (HDCL) must be removed from the system if the magnet is at field, before commencing the Helium transfer.



WARNING

Remove HDCL before starting LHe fill.

If the demountable current lead is left in the system when a Helium transfer is started, there is a significant risk of the magnet quenching.

- » Remove HDCL and replace with baffle assembly.

4.4.4 LHe Dewar Preparation

Certain operations need to be done without delay to ensure efficient Helium transfer.

- Ensure that there is open and clear access to the magnet for the Helium dewar.
- Ensure that there is open and clear access for the syphon (particularly height restrictions)
- If the Helium storage dewar is highly pressurized, release the pressure slowly to minimize the loss of Helium.

! WARNING

Venting of Helium gas in a confined area.

Possible asphyxiation.

- » Depressurisation of the dewar must be done outside the imaging suite or by connecting to the magnet system's vent line.

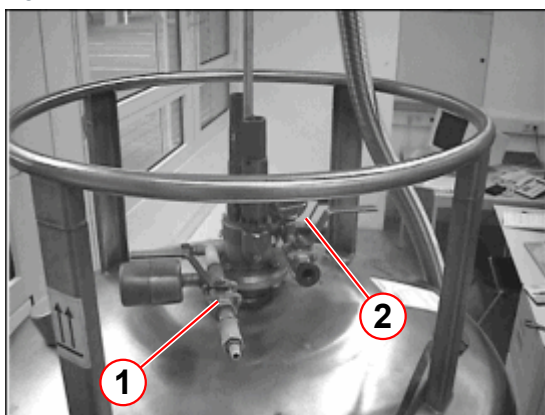
- Check the quantity by weighing, or by other means, as available.
- Fit the syphon dewar (neck) adaptor to the top of the dewar.
- Check the pressure gauge on the LHe dewar is accurate and displays zero bar after being depressurized. (or connect an electronic pressure meter if available).
- Connect the pressurizing gas supply hose to the dewar, after purging it of air, ensuring that the hose is long enough to allow the dewar to be moved near to the magnet but with the gas cylinder remaining outside the imaging suite.

! CAUTION

Use only pure [at least 99.996%] Helium to pressurize the dewar.

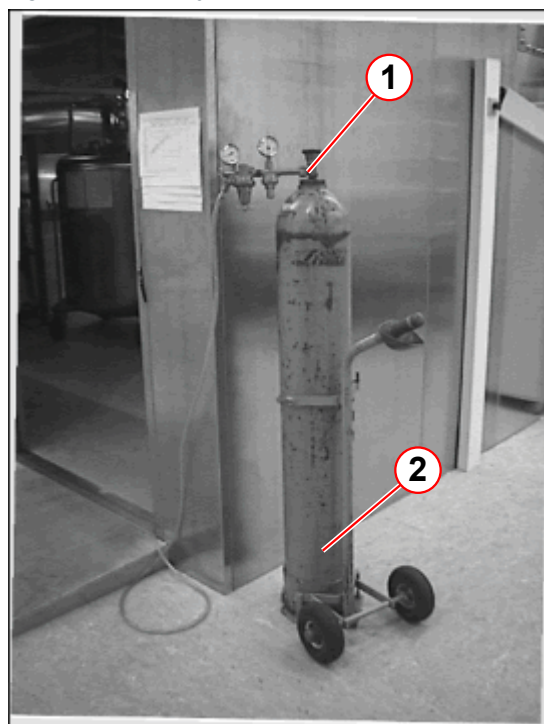
- » Impure Helium will cause build up of sludge and ice.

Fig. 67: He-dewar



- (1) Pressure relief valve
(2) Vent valve

Fig. 68: He-Gas cylinder



- (1) Pressure regulator
(2) He-Gas cylinder

4.4.5 Preparing the dewar syphon

- Fit extensions as necessary to suit the size and type of LHe dewar being used, including the sludge inhibitor, to the dewar syphon.
- Check the physical condition of the syphon. The flow valve opens freely and that the bayonet leg, threads and seals of the coupling are serviceable and not damaged.



CAUTION

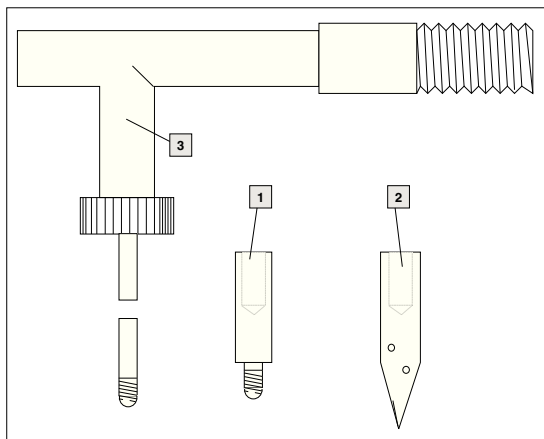
When using the LHe syphon without the extension and sludge inhibitor ((→ 2/ Fig. 69 Page 83)).

This will prevent ice sludge from blocking the syphon.

- » Check the syphon end is at least one inch above the bottom of the dewar.

Fig. 69: Component parts of syphon

component parts of syphon.



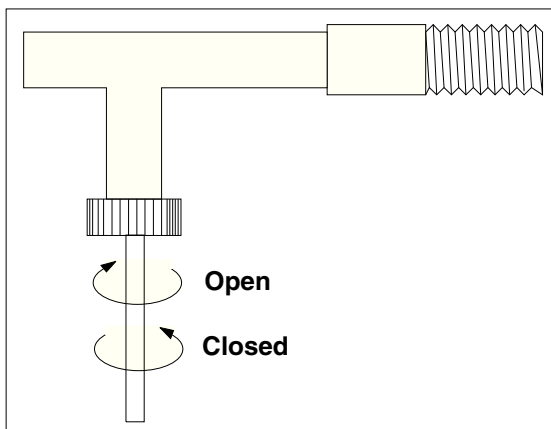
- (1) Syphon extension leg
- (2) Sludge inhibitor
- (3) Dewar syphon

4.4.6 Precooling the dewar syphon

- Close the vent and low pressure relief valves on the dewar.

- Open the dewar syphon flow valve full (approximately 4 turns. (→ Fig. 70 Page 84)).

Fig. 70:



NOTICE

About four turns clockwise will fully open the valve. [viewed from the top]

- » The flow valve is fully open when the two guide pins at the top of the knurled brass nut are flush with the top of the stainless steel body.

- Ensure the sludge inhibitor is fitted to the dewar syphon down leg.
- Insert the syphon down leg into the dewar adaptor, and tighten the adaptor "O" ring clamp so as to make a gas seal through which the syphon can slide. The seal does not have to be completely leak tight at this stage.
- Open the dewar access valve, and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.
- Push the syphon fully down into the dewar until the sludge inhibitor touches the bottom.
- Now tighten the syphon neck adaptor "O" ring clamp.
- Open the gas supply on the regulator from the gas cylinder
- Allow the syphon to cool until a cold "cascading white plume" appears (refer to (→ Fig. 71 Page 85). Do not allow a large pressure to build in the dewar, only enough to cool the syphon.
- Regulate the dewar pressure by opening the dewar pressure release valve if required.
- Set the gas supply pressure to 4 psi [0.28 bar].
- When the syphon is cold, close the flow valve and move the dewar close to the magnet so that the syphons can be coupled without excessive bending.

Fig. 71: "Plume" during syphon cool down



NOTICE

Do not over bend the syphons.

- » Tight bends in syphons reduce the efficiency of Helium transfer.

NOTICE

Do not open the vacuum valve of the dewar syphon near the bayonet.

- » This valve is the syphon vacuum valve, only to be used when re-pumping the syphon. Contact Uptime Service Centre if damaged.

4.4.7 Helium Transfer with High Efficiency Syphon (HES)

You can easily recognize whether you have a High Efficiency or a Smart Syphon fitted. Fig. 9 shows the configuration with a High Efficiency Syphon (from August 2002 a new revised HES is now fitted to the systems).

The HES syphon will have a second vent tube fitted to the syphon bracket for venting away gas during the syphon cooldown.

The magnet system is designed to have the HES syphon permanently fitted at all times. If the magnet needs to be filled from empty (Nitrogen temperature), or the syphon needs to be removed for any reason, contact Uptime Service Center.



CAUTION

Damaged HES syphon

Damage to ring nut, syphon leg or end cap could cause air ingress.

- » Contact Uptime Service Centre for advice.

**CAUTION**

Do not connect the two syphons immediately.

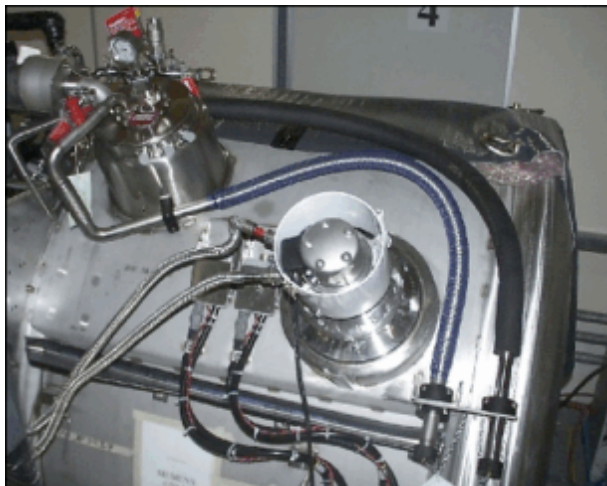
The HES syphon must be fully pre-cooled before connecting the dewar syphon.

» Warm gas pushed into the system will cause the magnet to quench.

4.4.7.1 Important Notes

- Monitor the filling procedure continuously by checking:
 - the transfer pressure and fill level
 - the service turret and venting (check gas flow through valves and venting).
 - that all connections are properly sealed and leak tight.

Fig. 72: New High Efficiency Syphon on the magnet OR70 or OR24



4.4.7.2 Cooling the HES system syphon

- ONLY when the system is pressurized to its operating pressure can you begin the cool-down of the HES system syphon (gauge needle in the green). System pressure will vary depending on atmospheric pressure but minimum for any system must 0.5 psi [0.34 bar].
- Remove the magnet syphon and the vent tube end caps.
- Fix the short "U" tube in place (refer to (→ Fig. 73 Page 87)) so that the initial gas is vented into the exhaust pipe. When the U-tube starts getting frosty, remove it to get a better visual control of the gas flow.
- Allow the syphon to vent until it is cold (2-4 minutes, looks like "pluming", see (→ Fig. 74 Page 87)).

NOTICE

The magnet should not be completely depressurized!

- » The system pressure is required to precool the HES syphon.

Fig. 73: "U" tube fitted for venting down



- Open the dewar syphon flow valve again FULLY and cool down until the 'long white cascading plume' can be seen.

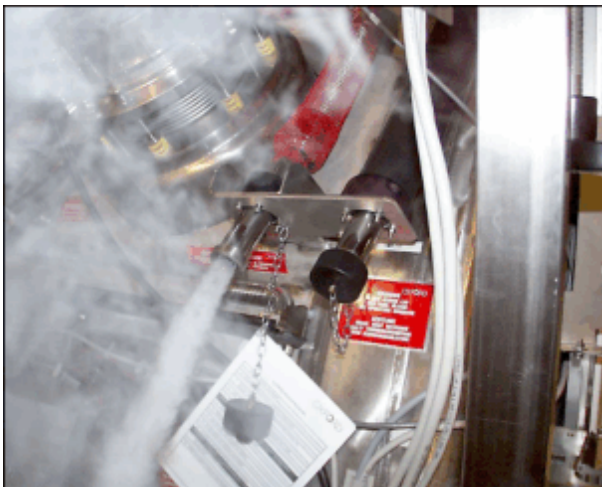
NOTICE

OPEN the system bypass valve and leave it open for the entire refill!!!

This procedure is now common for Mobiles and Static systems.

- » Refer to (→ Fig. 81 Page 99) and (→ Fig. 82 Page 99).

Fig. 74: "U" tube removed from the syphon end, and cap fitted to the vent-down tube



4.4.7.3 Connecting to the HES syphon

! WARNING

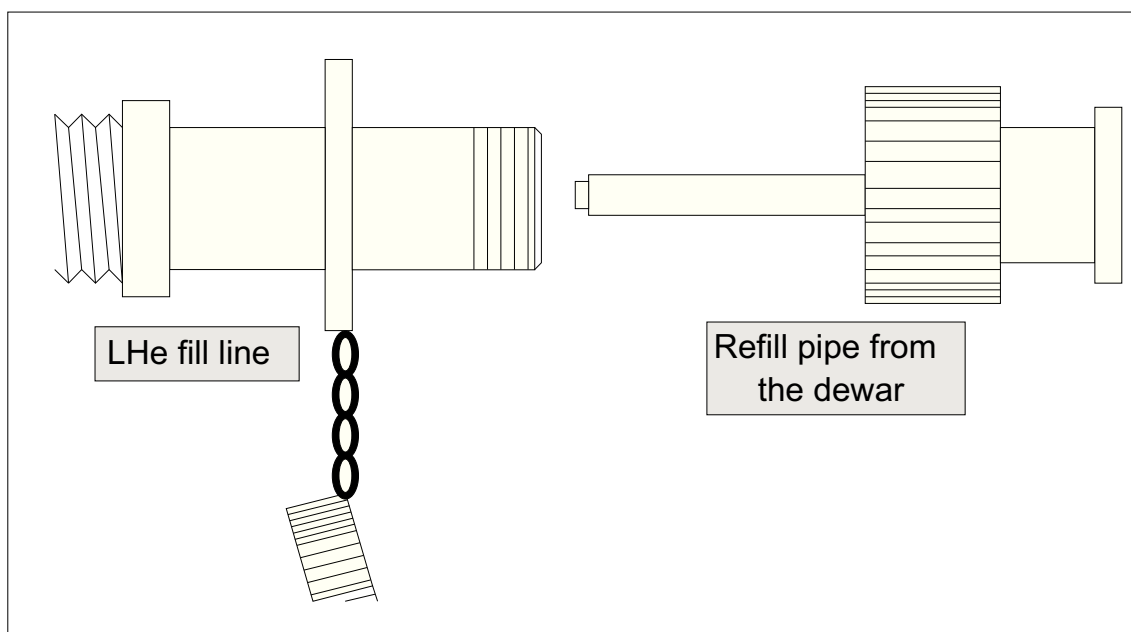
If either syphon is not fully cold and visibly plumbing before connecting together, with a HIGH FLOW of liquid from the dewar syphon.

The magnet will quench when the transfer is started.

» Connect syphon's with a HIGH FLOW of liquid from the dewar syphon.

- With system and dewar syphons fully cold, couple them quickly together and fasten the securing nut.
- If this joint is well made, ice will not form at this point. If after a few minutes gas is weeping from the coupling then warm with heat gun and re-tighten.

Fig. 75: Overview: Bayonet coupling

**! WARNING**

Transfer pressures lower than 2 psi [0.14 bar].

May cause the magnet to quench.

» Dewar pressure must be maintained at 4psi [0.28 bar] on initial connection.

**WARNING**

If either syphon feels cold on the outside, or becomes covered with condensation or frost, during initial cooldowns, the transfer must not be started.

During transfer if either syphon begins to frost on the outside, again stop the transfer.

- » The syphon must be replaced or re-pumped. There is a danger of quenching the magnet if a soft syphon is used.
- » Contact Uptime Service Center for advice.

NOTICE

Vibration of the pressure gauge needle of the magnet can occur.

- » This will stop when system pressure has settled.

NOTICE

Both excessive and too little pressure might cause poor efficiency.

Maintain a dewar pressure of about 4 psi [0.28 bar] for STATIC systems (with bypass valve open).

Maintain a dewar pressure of about 4 psi [0.28 bar] for MOBILE systems (with bypass valve open).

- » If available an electronic pressure meter fitted to the dewar will provide more accurate pressure monitoring.

- Start a Helium level measurement intermittently. Record the initial level. An increase should be seen within 15 minutes. If not, stop the transfer and investigate the reason for poor transfer efficiency.
- As soon as possible after connecting the syphons remove the 'U' tube from the vent down line and cap the end of the vent tube. Put 'U' tube in a safe place.
- During the transfer continue to monitor the Helium level and the dewar pressure.
- Maintain a constant 4 psi transfer pressure. The flow rate at this pressure will be about 10 liters/min. A 250 litre dewar will therefore empty in about 25 minutes.
- Never leave the transfer unattended.
- As soon as the level meter reads 100%, or the dewar empties stop the transfer immediately.
- Close the flow valve on the dewar syphon leg.

NOTICE

Helium dewar empty.

- » Good indications of the dewar becoming empty will be a sudden drop in the dewar pressure accompanied with a whistling sound from the syphon.

**WARNING**

It is important to know when the dewar is empty when transferring with a HES syphon.

If left open the magnet could quench.

- » The dewar flow valve must be CLOSED immediately.

- Continue by going to (→ Stopping the LHe Transfer and disconnecting the syphons / Page 98).

4.4.8 Helium Transfer with Smart Syphon

4.4.8.1 General Overview

Some OR24 and OR70 magnet systems are equipped with a Smart syphon (→ Fig. 76 Page 91)). The Smart syphon has an additional valve mechanism fitted internally that will open only when cooled to LHe temperature. The initial warm gas is diverted directly from the syphon into the system venting to prevent the magnet quenching.

The magnet system is designed to have the Smart syphon permanently fitted at all times. If the magnet needs to be filled from empty (Nitrogen temperature), or the syphon needs to be removed for any reason, contact Uptime Service Centre.

**CAUTION**

Damaged Smart syphon

Damage to ring nut, syphon leg or end cap could cause air ingress.

- » Contact Uptime Service Centre for advice.

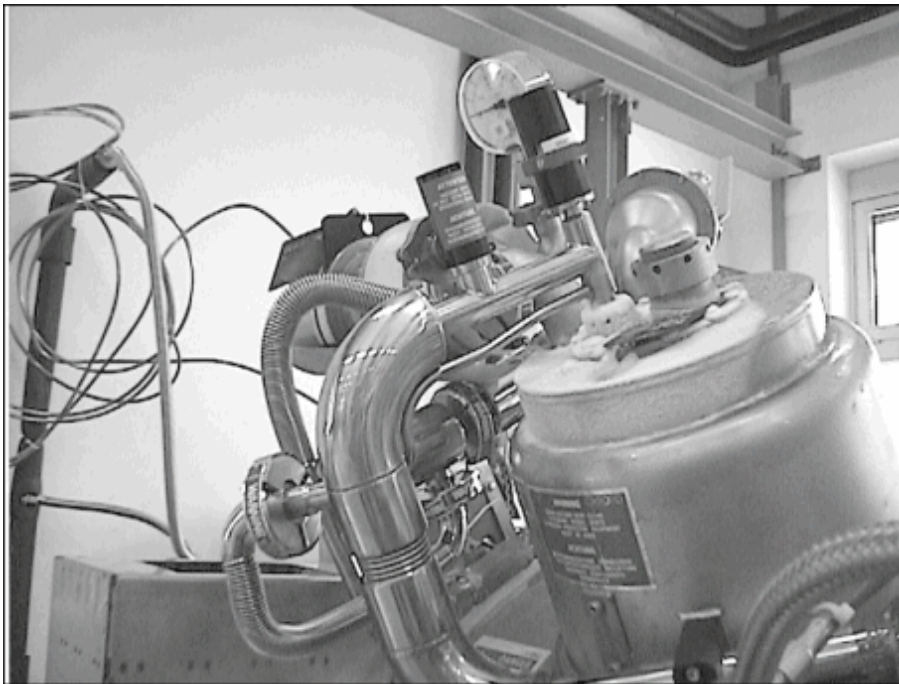


The Smart syphon does NOT require any initial precooling prior to connecting the dewar syphon.

The initial warm gas is vented away from magnet.

- It is recommended that the precool procedure is used at all times for the dewar syphon.

Fig. 76: Smart Syphon on the magnet OR70 or OR24



WARNING

Prevention of accidents to personnel and equipment.

Personnel injury could occur as well as causing damage to equipment and the magnet system.

- » Read all the health and safety instructions applicable to Helium filling and cryogenic handling before commencing this procedure.
 - » Refer to (→ Safety / Page 11)
-
- Check that the system pressure gauge is reading a positive pressure. If not, the transfer must not proceed (refer to (→ Check of magnet pressure / Page 80)).
 - Static and Mobile systems have different vent valve ratings. Before commencing LHe fill check the system type and operating pressure.

NOTICE

Mobile and Static systems are fitted with different magnet venting assemblies.

STATIC SYSTEMS

OR24/70 are fitted with a vent assembly with two 1/3rd psi valves in series with a bypass valve in parallel. When gas freely vents through the valves a pressure of 0.6 psi [0.041 bar] will be seen on the system pressure gauge.

OR41/42/35/32 systems are fitted with only one 1/3rd psi valve. When gas freely vents through the valve a pressure of 0.3 psi [0.02 bar] will be seen on the system pressure gauge.

MOBILE SYSTEMS

OR24/70 are fitted with a 1/3rd psi valve and 16 psia absolute valve which will give a higher differential pressure which can be as high as 4.2 psi [0.28 bar] on the system gauge at high altitudes. Standard atmospheric pressure will see the system gauge reading about 1.6 psi [0.11 bar].

OR41/42 are fitted with a single 16 psia valve. When gas vents freely through the valve a pressure of [1104 mBar absolute]

» Refer to (→ Fig. 84 Page 105) for pressure readings at different altitudes.

- Prepare the LHe dewar as instructed in section, (→ LHe Dewar Preparation / Page 81)
- Prepare & precool the dewar syphon as instructed in section, (→ Preparing the dewar syphon / Page 83) and (→ Precooling the dewar syphon / Page 83).

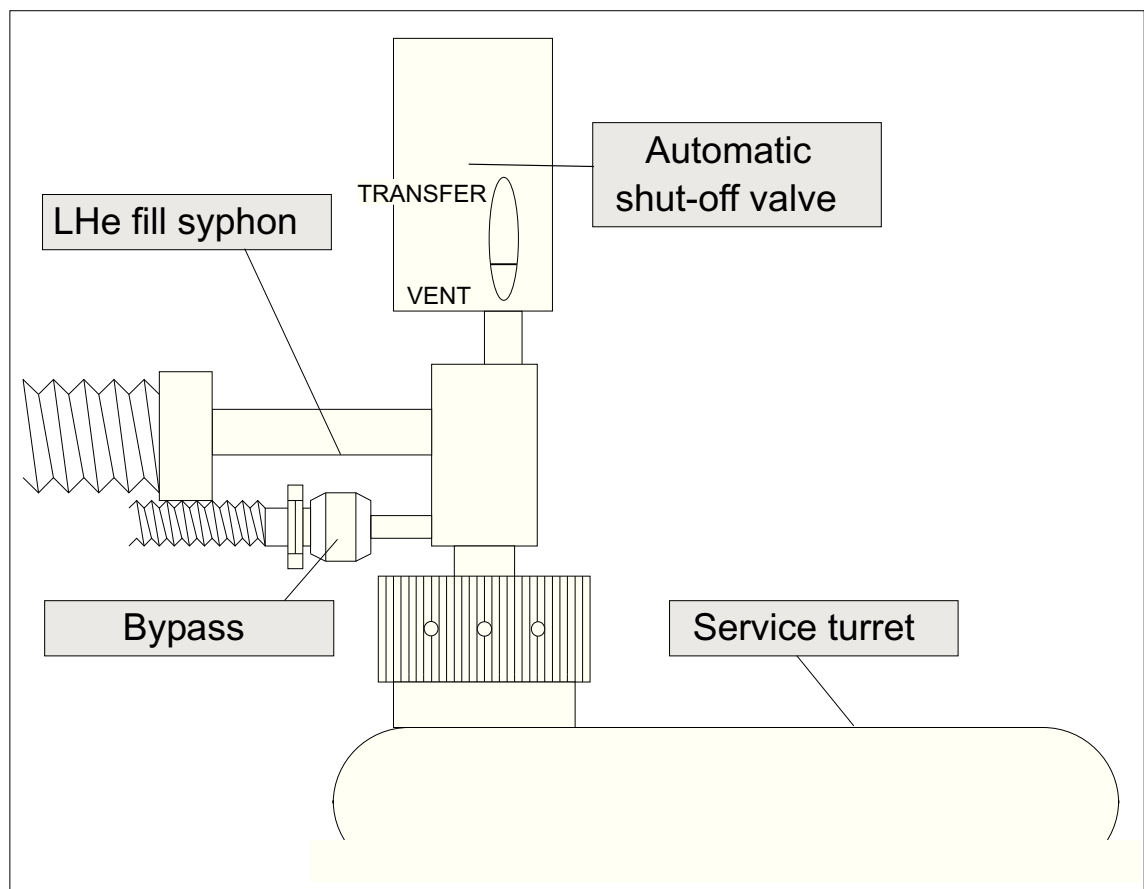
**WARNING**

Remove HDCL before starting LHe fill.

If the demountable current lead is left in the system when a Helium transfer is started, there is a significant risk of the magnet quenching.

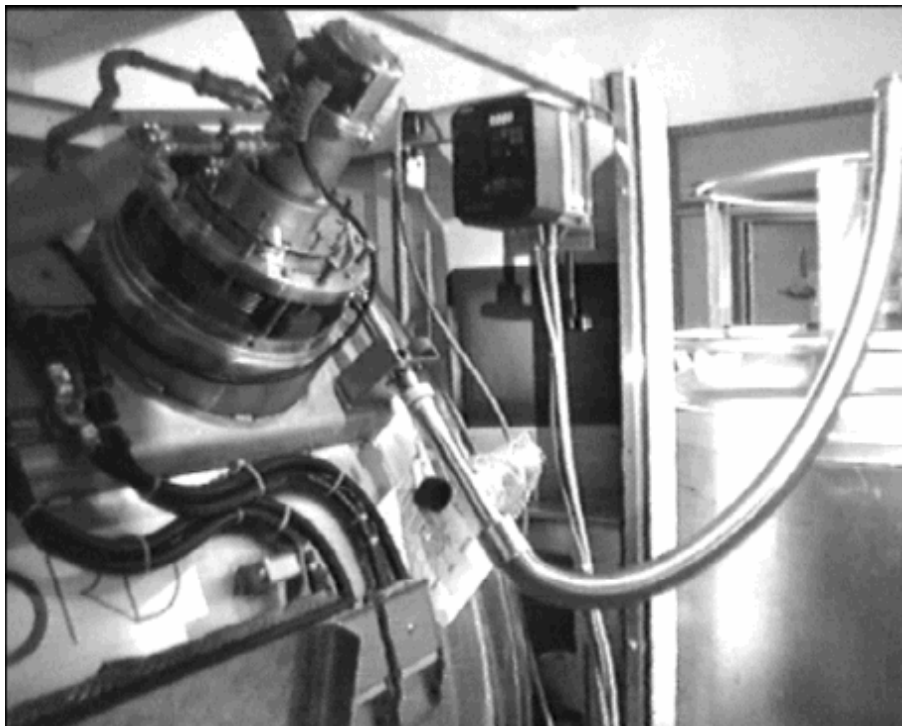
» Remove HDCL and replace with baffle assembly.

Fig. 77:



4.4.8.2 Smart syphon LHe fill procedure

Fig. 78: LHe dewar in the magnet room

**NOTICE**

Do not depressurize the magnet prior to transfer.

- » Fills with Smart syphons can proceed with pressure in the magnet.

4.4.8.3 Connecting to the magnet Smart syphon

! WARNING

Check that the Smart syphon valve is in the "vent" position. Physical damage may have occurred to the valve assembly.

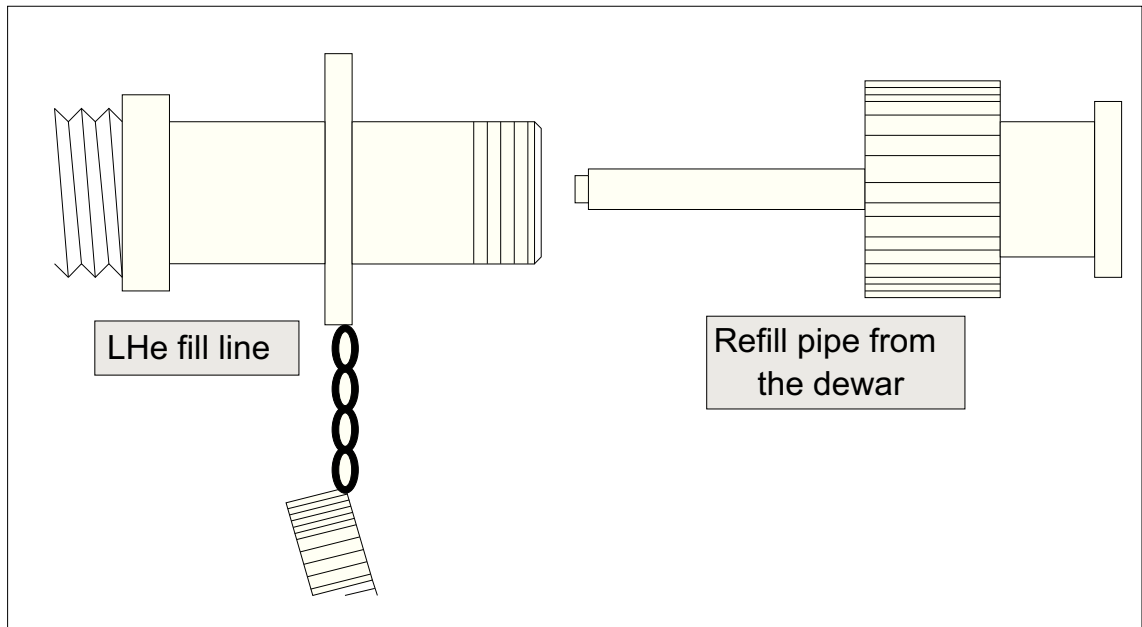
The magnet will quench if a transfer is attempted.

- » Do NOT proceed, continue with trouble shooting.

- Remove the end cap from the magnet syphon (if gas vents from the smart syphon replace the end cap and contact the Uptime Service Centre).
- Check that the "O"ring in the bayonet assembly is okay and securely located
- Check that the dewar syphon bayonet is clean and dry before connecting.
- Check there is no physical damage to the components ((→ Fig. 79 Page 95)).

- Open the dewar syphon flow valve fully to cooldown and create a flow of Helium before inserting it into the magnet syphon. (refer to (→ Precooling the dewar syphon / Page 83)).

Fig. 79: Overview: Bayonet coupling



CAUTION

If either syphon feels cold on the outside, or becomes covered with condensation or frost, during initial cooldown, the transfer must not be started.

During transfer if either syphon begins to frost on the outside, again stop the transfer.

The syphon must be replaced or re-pumped. There is a danger of quenching the magnet if a soft syphon is used.

- » Contact the Uptime Service Centre for advice.

NOTICE

The fill procedure is different for Mobiles and Static systems.

For MOBILE systems OPEN the system bypass valve.

- » For STATIC systems leave the bypass valve CLOSED.

- With the dewar syphon cold and 'pluming' (liquid will be seen white and cascading downwards) connect the syphons together and fasten the securing nut.
- Gas will flow through the Smart syphon, venting initially from near the top of the magnet syphon into the quench tube until the syphon is cooled to near 4.2 Kelvin. At this point the Smart Syphon valve will operate to let liquid into the Helium vessel.

- If this joint is well made, ice will not form at this point. If after a few minutes gas is weeping from the coupling then warm with heat gun and re-tighten.

NOTICE

Syphon cooling will take 5-10 minutes.

The syphon is fitted with a visual indication "TRANSFER" of the valve position. Initially the valve may vibrate until fully open

- » Maintain the dewar pressure at 4 psi [0.28 bar].

NOTICE

Gas escaping from around valve body.

- » Indicative of a broken seal on valve body. The transfer must be stopped and Uptime Service centre contacted.

NOTICE

Vibration of the pressure gauge needle of the magnet can occur.

- » This will stop when system pressure has settled.

4.4.8.4 What to do, if the Smart Valve does not open

If the Smart Syphon valve has not moved to the transfer position, it is probable that the smart valve has become stuck in the closed position.

- try warming smart valve body for 2-3 minutes while still maintaining dewar transfer pressure

Before moving manually the valve, check the following:

1. Syphon has been cooled for at least 10 minutes
2. The vent line from the top of the syphon has become very cold (visible fog falling, or liquid oxygen dripping - this occurs after about 10 minutes under normal circumstances).
3. The syphon on the dewar side is installed correctly (stinger fitted at the dewar end).
4. The shut-off valve at the syphon on the dewar side is completely open.
5. The syphon is covered with ice, check the vacuum of the syphon. If vacuum is bad, replace the syphon.
6. The pressure shown at the Helium dewar is o.k. (4 psi equals 0.28 bar)
7. Does the Helium dewar contain liquid Helium.

After you checked all items, the smart valve of the syphon can be opened manually.

- Apply even pressure to both sides of the ring, while lifting it gently to the top of the slot in the housing (about 10mm).
 - » If resistance is felt, do not continue!

- The syphon valve should remain open now, if Helium flow and the valve mechanism are o.k.
- At the end of the transfer, the valve should move to the closed (Vent) position.



WARNING

Manual operation of the lifter.

The magnet will QUENCH.

- » Do not manually move the valve unless the syphon has had adequate time to cool (5-10 minutes).

NOTICE

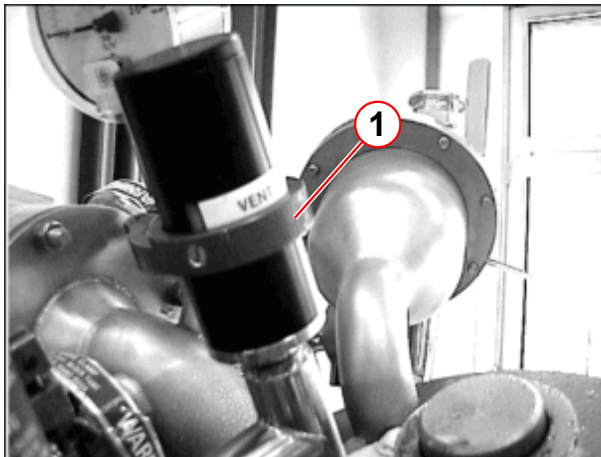
Both excess and too little pressure might cause poor efficiency

Maintain a pressure of about 4 psi [0.28 bar] for STATIC systems.

Maintain a pressure of about 4 psi [0.28 bar] for MOBILE systems. (with bypass open).

- » If available an electronic pressure meter fitted to the dewar will provide more accurate pressure monitoring.
- » If the syphon still does not open, remove the syphon, refer to (→ Flushing procedure for the Smart Syphon / Page 103).
- During the transfer continue to monitor the Helium level and the dewar pressure.

Fig. 80: Smart Syphon



(1) Lifting Ring

Maintain a constant 4 psi [0.28 bar] transfer pressure. The flow rate at this pressure will be about 10 liters/min. A 250 litre dewar will therefore empty in about 25 minutes.

- Never leave the transfer unattended.
- As soon as the level meter reads 100%, or the dewar empties, immediately stop the transfer.
- Close the flow valve on the dewar syphon leg.

- Wait until the Smart Syphon valve has returned to the "vent" position (1 to 2 minutes).

NOTICE

Manual operation of lifter.

- » If the Smart Syphon valve does not return to "vent" position, it is probable that the valve has become stuck in the "transfer" position. In this case, gently warm the ring on the top of the magnet syphon which can then be used to close the smart valve.

4.4.8.5 Stopping the LHe Transfer and disconnecting the syphons

- Close the flow valve on the dewar syphon leg.
- Close the gas supply to the dewar on the cylinder regulator.
- Slowly open the depressurisation valve on the dewar.
- If the dewar is empty, refer to (→ Changing supply dewars / Page 99).
- Warm the syphon joint to room temperature using a heat gun.
- Stand clear of the syphon joint, and disconnect the syphons. **Take care.** As soon as the syphons are disconnected, cold gas will exhaust. Wipe away any moisture from the end of the magnet syphon with a clean, lint-free cloth or paper, and cap the end of the syphon.

NOTICE

The syphon end cap must be refitted and leak checked immediately after disconnecting the two syphons.

The end cap must be leak checked.

- » If it is left off, loose, or the "O" ring/seal is missing, air will enter the syphon with the possibility of causing air ingress.
- » Check that the end cap is fully tightened by applying gentle heat to the area. **DO NOT** apply heat continuously to the plastic cap.

- Check the dewar is fully depressurized and then remove the dewar syphon.
- Immediately close the access valve on the top of the dewar; then close the top vent valve and dewar depressurisation valve.
- Open the valve leading to the low pressure relief valve.

WARNING

Dewar valve positions.

If the access or vent valves on the dewar are left open, air will freeze in the neck of the dewar with the strong possibility of causing an explosion.

- » See LHe supplier's manual for safe operation.

WARNING

Danger of air ingress.

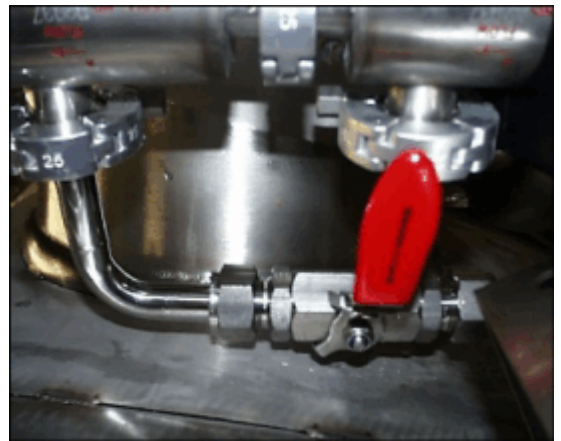
CLOSE THE SYSTEM BYPASS VALVE.

- » This must be warmed before closing to ensure excessive force is not used (see (→ Fig. 81 Page 99) and (→ Fig. 82 Page 99), open and closed bypass valve positions).

Fig. 81: Bypass valve open



Fig. 82: Bypass valve closed



- Warm the dewar syphon and store it in a safe place.
- Changing to a new full LHe - dewar: refer to (→ Changing supply dewars / Page 99).

4.4.8.6 Changing supply dewars

- If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one. Care will be necessary during this operation to ensure that efficiency is maintained and the dewar syphon is not iced.
- When the dewar empties (whistling of the syphon or dewar pressure drop noted), stop the transfer immediately, by closing the dewar syphon flow valve.
- Turn off the gas supply and vent the storage dewar down to atmospheric (zero) pressure. Close the dewar valve after venting down and open the low pressure relief valve.
- Warm the syphon joint to room temperature.
- Stand clear of the syphon joint and disconnect the two syphons. **Take care.** As soon as the syphons are disconnected, cold gas will exhaust from the cryostat syphon. Wipe

away any moisture from the end of the cryostat syphon with a clean, lint-free cloth or paper, and cap the syphon end.

- Ensure that the full dewar is to hand and is depressurized and ready to receive the syphon (vent valve and low pressure relief valve closed).
- Some pressure might have built up in the old dewar. Open the vent valve and ensure that the dewar is depressurized. Close the vent valve.
- Remove the syphon from the empty dewar, and immediately put it into the full dewar. If necessary, transfer the dewar adaptor with the syphon. If a different adaptor is used, the gas hose must previously have been changed to the full dewar. As the syphon is entered into the dewar wipe off the frost from the surface of the down leg. Allow the syphon to cool, as previously, until a cold "plume" appears.

NOTICE

Avoid air ingress to syphon and dewar.

- » **Quickly exchange the syphon. The dewar syphon may be pushed down into the dewar more quickly than when it was fully hot, as there will not be so much liquid boiled off.**

- As soon as possible after removing the syphon from the empty dewar, close the dewar access valve, and open the valve leading to the low pressure relief valve.
- When the dewar syphon is cooled, close the flow valve and move the dewar near to the magnet so that the dewar syphon will reach the cryostat syphon without excessive bending.
- Continue fill procedure by following either (→ Connecting to the HES syphon / Page 88) or (→ Connecting to the magnet Smart syphon / Page 94).

4.4.8.7 After the transfer

For correct and safe operation of the magnet after a Helium transfer it is a requirement to ensure the system is fully leak checked so that the system will maintain a safe working pressure.

NOTICE

Ramping the magnet.

- » **After a Helium fill it is advisable to wait one hour before ramping the magnet.**

- Use a heat gun to warm up the turret, quench valve and vent valves. Use paper towel-ling to remove excess water.
- To help pressurize the magnet system the EIS switch heater circuit may be used.
- Use a suitable leak check solution or electronic detector to check turret, venting, valves and the syphon end cap for leaks.

- Before leaving the site the system MUST be checked to ensure that it has pressurized back to its standard operating pressure.

4.4.9 Magnet System leak check

The magnet system must be leak checked after work has been performed during a Service visit or a Helium fill visit if the system turret and venting has been subjected to extremes of temperature.

The magnet system must remain pressurized at all times. Before leaving an installed system site a full leak check must be performed and the system seen to be returning back to its normal operating pressure.



WARNING

There must always be a positive pressure in the system.

Internal "icing" of the magnet may result causing the magnet to quench the Helium vessel to rupture.

- » Follow leak check procedures during all service operations.

If the system pressure remains at zero but there is no apparent leak, check the quench valve integrity. Refer to the (→ Procedures / MR-000.840.17).

- For Static systems the normal operating pressure is 0.6 psi [40 mbar] at sea level.
- For Mobile systems the normal operating pressure is 1.6psi [1.1 bar] at sea level.



WARNING

The system operating pressure for mobile and static systems will vary with changes in local atmospheric pressures.

- » Mobile systems taken to high altitudes (above 6000 feet) will see operating pressures up to 4.5 psi [300mbar].

4.4.9.1 Procedure

An electronic Helium leak detector should be used to search for leaks around all of the joints on the service turret. (Follow the manufacturer's instructions for correct use of the electronic detector.)

- The leak rate using an electronic device should not exceed 1×10^{-4} mbar-l/sec.

**WARNING**

Take care when using an electronic leak detector with the magnet at field.

Accidents to personnel and equipment damage will occur.

» Check for magnetic parts before use.

- If electronic equipment is not available a suitable leak detection fluid must be used.
- Leak check fluid must be applied liberally by either brushing or squirting onto the joints and seals. Leaks will be detected by observing bubbles where the fluid has been applied.
- The turret, quench valve and venting must be warmed to room temperature and dried before commencing the leak check. Remove condensation and water with paper towelling.

The specific area's to leak check are:

- Clamps and seals on vent lines.
- HDCL and syphon seals.
- System pressure gauge.
- Turret top plate.
- Syphon end cap.
- The turret electrical connections.
- Flexible tubing.

Where leaks are detected you should:

- Tighten or replace clamps if damaged.
- Check that the correct type of seals are fitted.
- Check seals faces for damage. Replace if damaged.
- Tighten syphon end cap and the central retaining screw for the chain.
- Tighten Syphon / HDCL seals. If leak persists replace the copper seals.
- Check Quench valve burst disc and "O" ring. Replace if necessary.
- Check bolted flanges are correctly torqued.
- Seals covered in vacuum grease should be cleaned and inspected. Refit the seal or "O" ring using NO vacuum grease.

If problems persist with leaks to the turret or venting then your local Uptime Service Centre should be contacted.



Do not use any form of vacuum grease on O rings or seals.

It is recommended that they are wiped clean and checked for damage prior to fitting. Vacuum grease with time dries and can cause a seal to leak.



Replace used high load seals.

The high load seals (copper (Cu) or Heli-flex types) must be replaced with a new one each time the joint is broken.

4.4.9.2 Flushing procedure for the Smart Syphon

The syphon may be partially blocked with ice. To clear any ice the Smart syphon must be removed from the system.

- Depressurize the system to zero by opening the system bypass valve.
- Using the syphon ring nut spanner, release the syphon nut and remove the syphon.
- Replace the syphon baffle plug into the magnet, to prevent air ingress into the system.
- Close the system bypass valve.
- Lay the Smart syphon and dewar syphon onto the floor and connect together.
- Connect a gas supply of pure Helium (99.999%) gas onto the end of the dewar syphon.
- If still frosty, gently warm up the Smart syphon system leg and valve assembly.
- Purge the two syphons with the Helium gas for 10 minutes at a pressure of 5 psi.



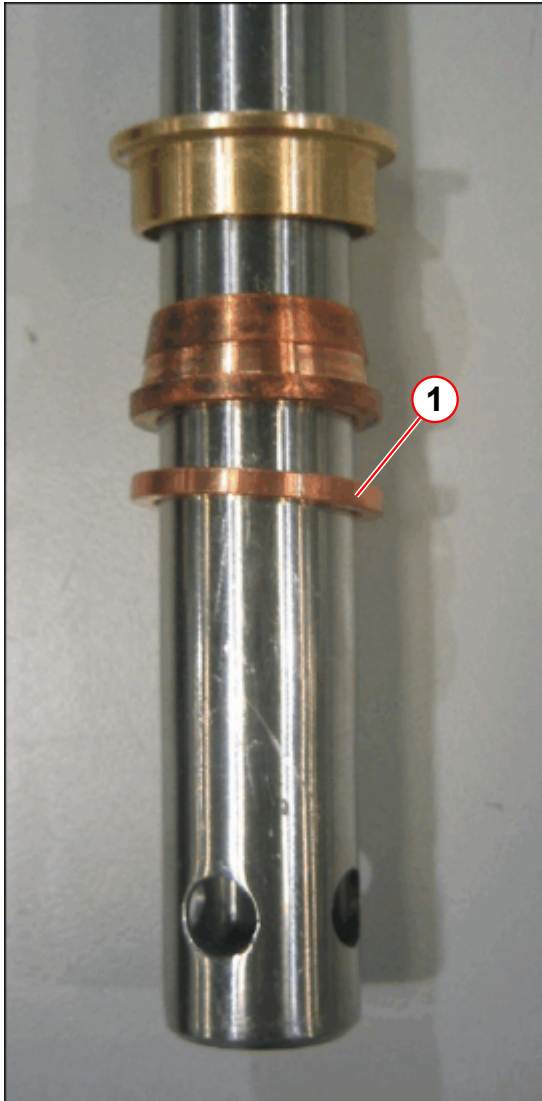
The Helium gas will vent through the Smart syphon vent valve.

- After 10 minutes stop the gas purge and disconnect the gas line from the dewar syphon.
- Disconnect the two syphons and refit the syphon end cap.
- Remove the syphon baffle plug from the system and re-fit the Smart syphon.



A new copper seal will be required for the syphon before refitting ((→ Fig. 83 Page 104)). Only the flat copper ring must be replaced, the other parts of the sealing are compressed and should not be removed !

Fig. 83:



(1) Copper seal to be replaced after removal of syphon (before re-inserting)

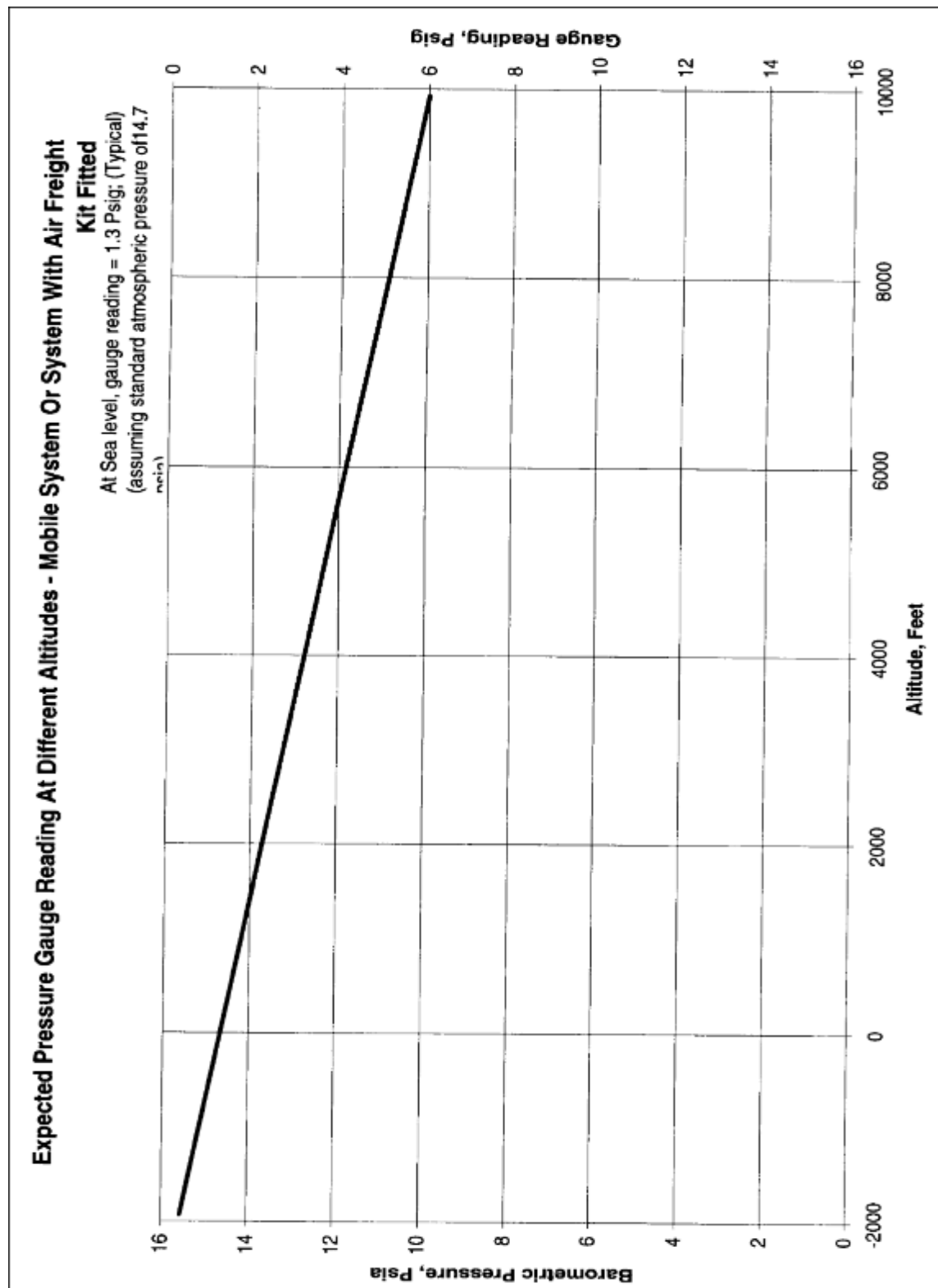
- Tighten syphon ring nut hand tight and complete using the ring nut spanner.



The system must be allowed to repressurize and the syphon seal nut leak checked using a suitable leak check solution.

- If the Smart syphon still fails to operate, a replacement is recommended.

Fig. 84:



5.1 Information

- Please read the following information carefully to avoid personal injury and/or damage to the magnet.
 - Only Siemens technicians trained in ramping magnets are allowed to perform this work.



Read the entire document including the flowchart before you begin any of the procedures.

5.2 Safety



Adhere to the general safety information and the special safety information included in this chapter when performing the steps indicated.

5.2.1 Always

- **Always** wear the protective clothing listed below when ramping the magnet up or down.
- **Always** restrict access to the magnet room when ramping the magnet. Use flashing lights and signs.
- **Always** inspect all equipment and tools for defects prior to use.
- **Always** ensure that tools and auxiliary materials are in satisfactory condition and conform to manufacturer's specifications (comply with legal regulations).

5.2.2 Never

- **Never** use defective or damaged equipment.
- **Never** allow people to stand too close to the magnet during ramping.
- **Never** block emergency exits.
- **Never** smoke or use open flames in the examination room.

5.2.3 Protective clothing

The following protective clothing must be available on-site for all technical personnel (for direct purchase refer to PMTD for the item number):

- Face mask
- Protective gloves (cryogenics) and leather work gloves
- Lab coat or overalls (cotton or linen)

5.3 Ramping the magnet



CAUTION

Ramping with helium lower 70 %

Risk of quench

- » When initially ramping the magnet the LHe level **MUST** be at least 70%.

5.4 Materials and Tools required

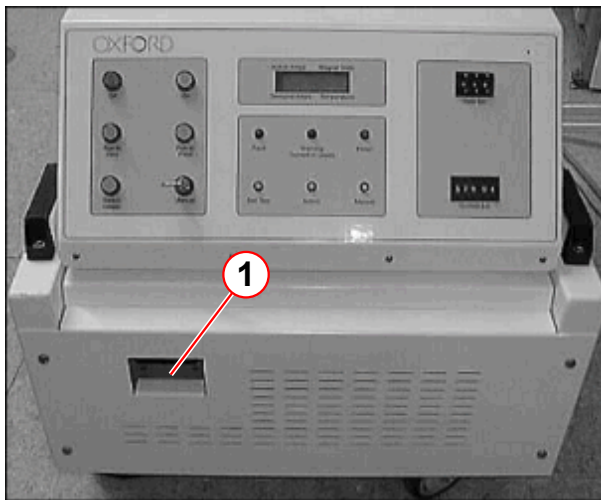
- Six to ten liters of distilled water
- Current probe and baffle copper sealing assembly
- MPS (Magnet Power Supply), (p.n.: 47 61 412)
- Heat gun
- Compass or magnetic pole indicator
- Non-magnetizable tools
- Standard tool set
- Tape measure or yard stick
- Torch light
- Acrylic window (p.n.: 31 08 979)
- Additional ring nut

5.5 Magnet Power Supply

Your new system does not have its own MPS (Magnet Power Supply). Since the MPS is seldom needed for ramping the magnet up and down, it is now a separate transportable service tool consisting of:

- a step-up /down transformer with power cable
- the ramping cables
- water hoses for cooling the MPS and special service tools
- HDCL (Hardwired Demountable Current Lead)

Fig. 85: The new MPS (Main switch in the front panel)



(1) Main switch

5.6 HDCL Inspection and Repair

5.6.1 Introduction

Recent magnet installation issues have led to an Engineering review of the HDCL and the relevant procedures that apply to the product. The HDCL is a key component in the successful ramping of Oxford Magnet Technology's super conducting magnet systems.

The conclusion and recommendations from the review have found it essential for the HDCL to have improvements made to its manufacturing process and also for all existing HDCL's to be visually checked and for the maintenance procedure to be performed prior to ramping a magnet system.

5.6.1.1 Scope

The Inspection and Repair procedure applies to all HDCL's supplied with Mobile Power Supply Units, (MPSU), and also any HDCL supplied directly to the customer with each individual magnet system.

Existing HDCL's that do not meet the criteria of the Inspection and Repair procedure should be reported to the OMT Service department, who will advise on the correct action to take.

5.6.1.2 SUMMARY

A number of changes have been made to the manufacturing process of the HDCL.

These changes will improve:

- Improved gas flow through the central pin connector, reducing heat during ramping.
- Ensure better contact of central pin to the magnet connector.
- Ability to change the outer contact bands.
- Lower resistance across lead when ramping.

The procedure also provides details of the key components of the HDCL that must be checked prior to insertion into a magnet system. The critical checks and measurements are:-

- Ensure there are THREE (3), inner multi-contact bands.
- Length of inner contact conductor pin.
- Condition of Nomex conductor sleeving.
- Depth of inner conductor pin central vent hole.
- General cleanliness of the top St St. collar.
- General slackness of joints and locking collars.
- Condition of inner and outer contact bands.
- Condition of fly leads.

In addition the inner and outer contact bands are now available as a kit, part number is 7461580. The kit includes 3 inner bands and 1 outer band.

A quantity of these kits will be added to all future BOM's of the MPSU.

5.6.1.3 Procedure

Fig. 86:

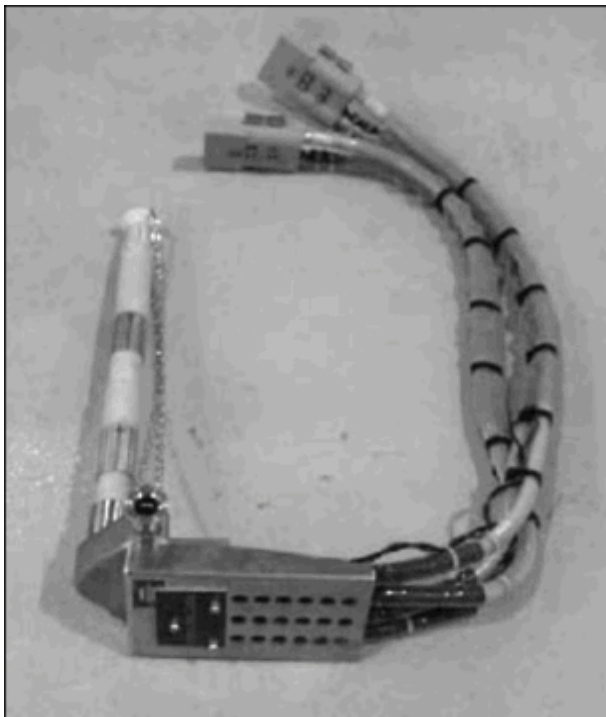
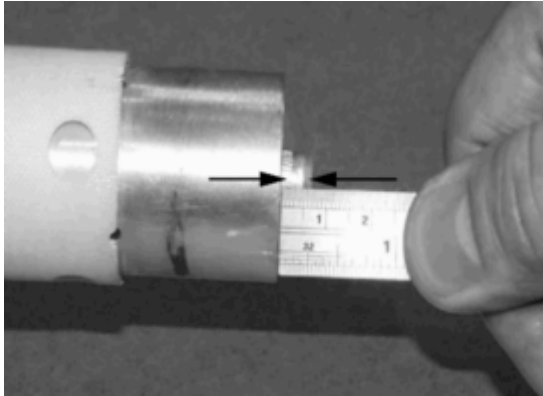


Fig. 87:



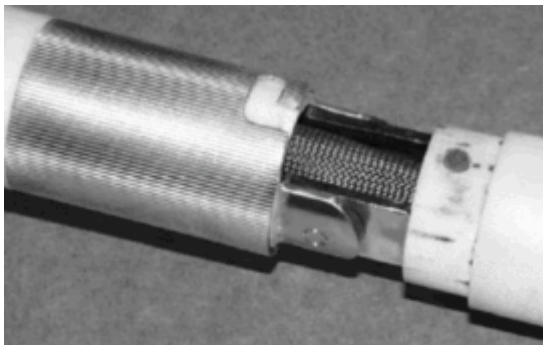
- Check the multi-contact inner conductor pin, it should be of the three-conductor band type.
- If it is the older two-conductor band version, then the HDCL should not be used

Fig. 88: Inner pin measurement



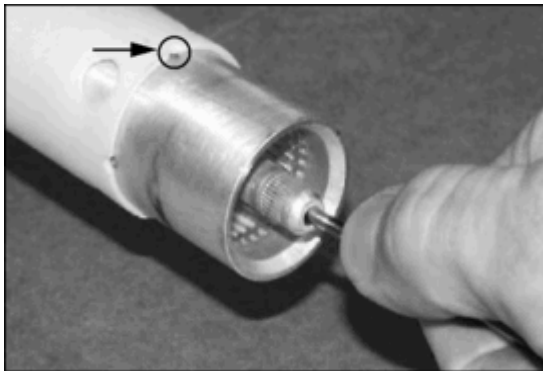
- Measure the distance that the inner conductor protrudes beyond the outer conductor.
- The tolerance range is 4.5 mm to 8 mm.
- The HDCL should not be used if measurements are outside the range

Fig. 89: Checking the inner isolation



- Slide the locking collar back and check that BOTH the inner conductors are covered with 'Nomex' sleeving. If they are not, then the HDCL should not be used.

Fig. 90: Checking the inner vent hole



- Check the depth of the central vent hole in the inner conductor using a thin rod, e.g. 2 mm welding wire. It must be 82-84 mm.

If a GRP pin is obstructing the vent hole, then it should be drilled through (via the vent hole) using a 4.4 mm drill.

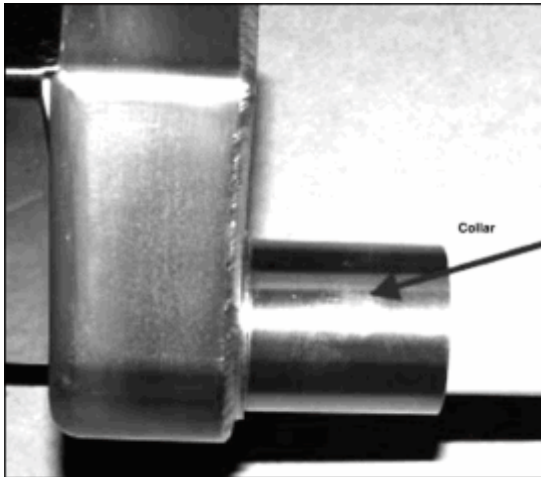


To avoid obstructing any of the coaxial connectors venting, vacuum out the swarf. Do not blow swarf up into the breather hole and housing.

5.6.1.4 General Cleaning and Examination

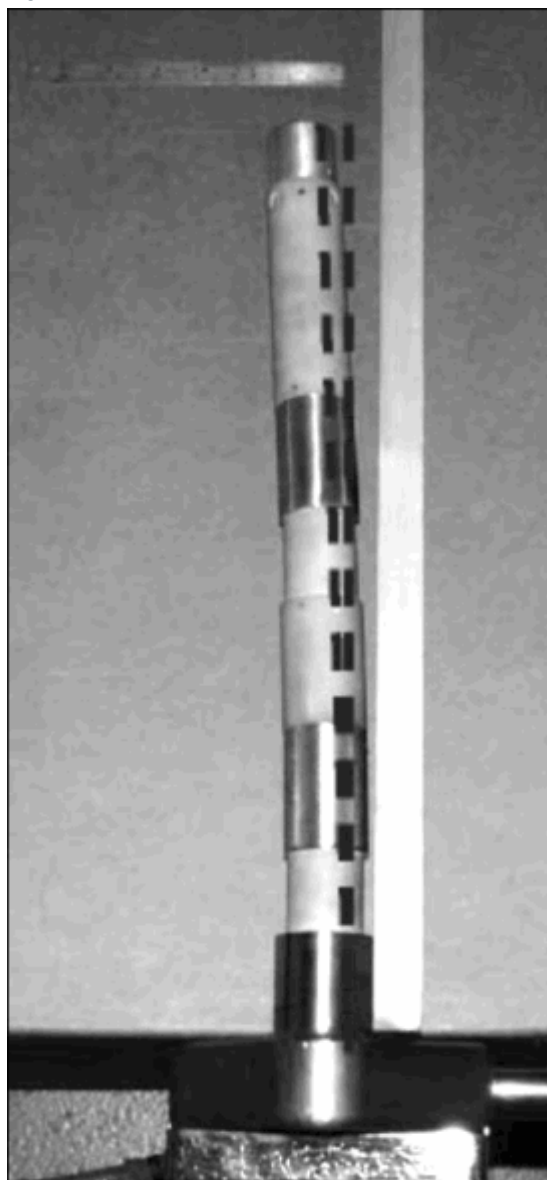
Clean off all the external and internal parts (undo sliding joints to gain access to internal parts), visible dirt and contamination with clean lint free cloth.

Fig. 91:



- Check for any scratches or marks on the collar. These should be 'dressed off' with fine/medium wet and dry emery paper.
- Finish off with 'Scotchbrite', (Polishing cloth)

Fig. 92:



- Check for any excessive slackness in the joints and between the GRP and the locking collars
- Slackness is defined by the distance measured from the centre of the inner conductors position when the lead assembly is straight, to the furthest position that it can travel/bend away from this position.
- If there is more than 20mm of travel then the HDCL should not be used.

5.6.1.5 Contact bands

NOTICE

Bad contact bands

- » For the first use on a site, the contact bands of the inner and outer HDCL contact have to be replaced.

Carefully remove contact bands from inner conductor and outer conductor without damaging the connector base material or silver plating.

- Place the HDCL on a soft surface.

- Use two small screwdrivers to pull the spring apart at the opening. This allows the springs to be easily removed from the contact.

Fig. 93: The spring can be removed using two small screwdrivers

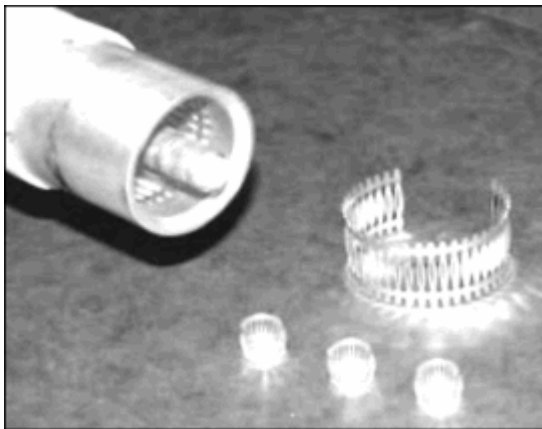


5.6.1.6 Contact Band Refitting

Great care must be exercised when fitting the contact bands; they must not be bent or distorted in any way.

Inner and Outer contact bands need to be replaced regularly, and a record kept.

Fig. 94:

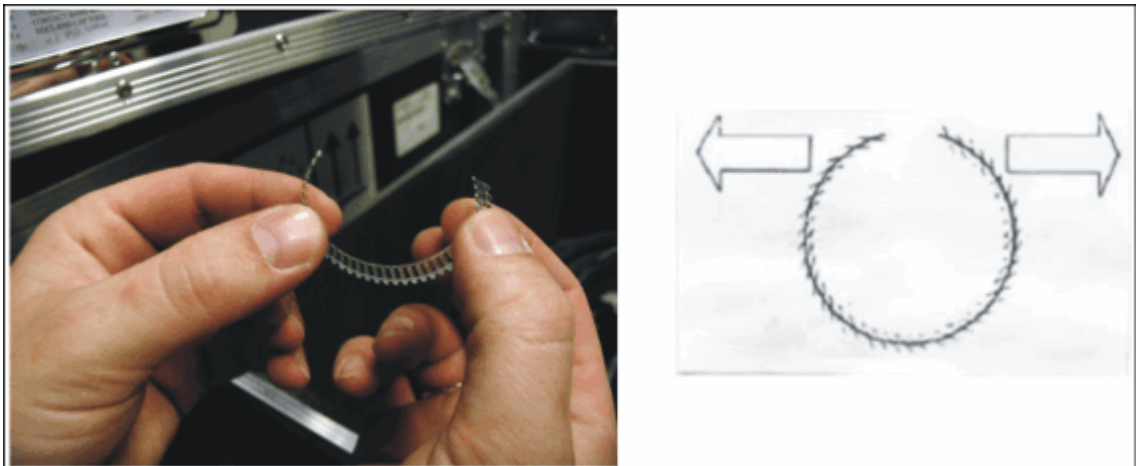


- Check for any scratches or marks on the collar. These should be 'dressed off' with fine/medium wet and dry emery paper.
- Finish off with 'Scotchbrite', (Polishing cloth)

The outer band should be fitted first, to avoid possible damage to the inner contacts.

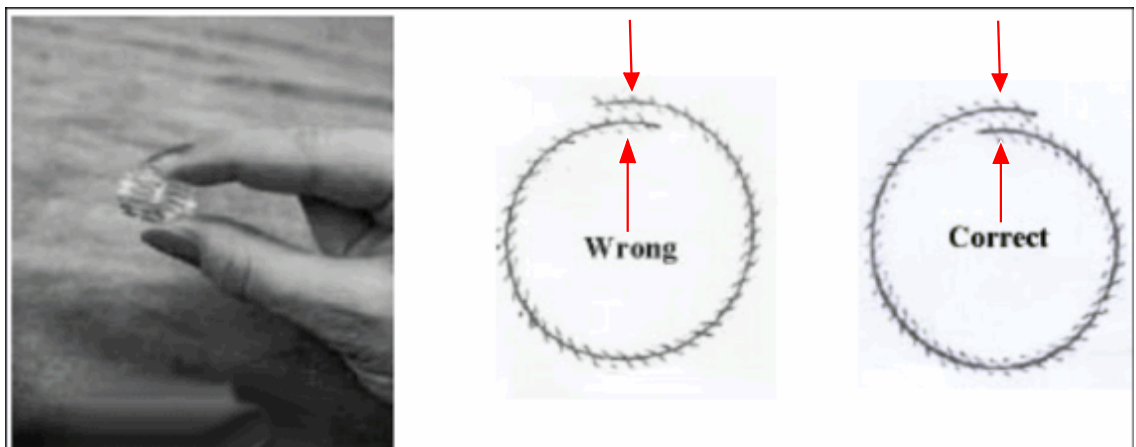
If you notice that the multi-lam band for fitting into the coaxial conductor socket groove has an insufficient spread effect, then the multi-lam band should, before fitting, be stretched out a little bit at the separating points (→ Fig. 95 Page 117) in order to reach the spread strength in the socket. By doing so it is also possible to correct easily slightly twisted multilam bands

Fig. 95:



Afterwards the multi-lam band is being rolled by hand, so that the multi-lam vertical planes are not being caught into each other (⇒ Fig. 96 Page 117)). The multi-lam band should only be rolled as small as necessary (diameter of the socket entry), so that it can be put sensibly into the groove.

Fig. 96:



Insert the multi-lam band into the socket recess part. The inside of the socket must be clean.

Insert the multi-lam band into the provided multi-lam socket groove, (back-turning in the inside of the socket) with the help of fitting tool. The multi-lam band should be clean and with sufficient tension to hold it against the socket wall by itself.

Fit the inner bands by sliding them over the centre pin. A thin screwdriver will help to locate them in their grooves.

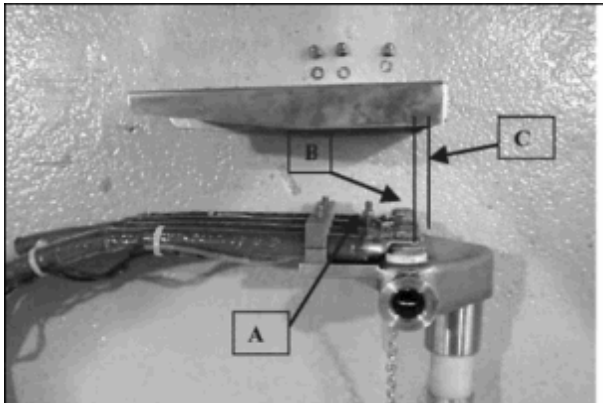
Be careful not to scratch or bend the bands.

5.6.1.7 HDCL Fly Leads

Inspect the condition of the cable insulation especially where they enter the end of the four brass tubes. Do not use if damaged.

5.6.1.8 Terminal lead checks

Fig. 97:



Prior to fitting the HDCL several checks are required of the terminals by removing the top cover plate.

- A) Check that the M6 full nut is lock-tightened onto the centre stud to prevent the cover plate from being over tightened onto the terminals. A full nut can be glued into place using a thread lock solution. The top of the nut must be level with the top of the Tufnol lead insulator bracket.
- B) Check that the lock nuts fitted to the terminals are 22 or 24 mm across flats.
- C) There must be a minimum air gap of 2mm between the lock nuts and the cover plate. They must not protrude beyond the edge of the PTFE spacer.

5.6.1.9 Packing

Fig. 98:



- Warm up and dry after use. Fit the plastic plug protection cap into the connector end.

5.7 Preliminary steps for ramping

5.7.1 List of activities which should have been completed

Completed activities	YES / NO
Magnet installation has been completed	
Cold head is in operation	
The HDCL inspection and repair have been done ¹	
The waterhoses are installed and the RCA is operating	
The LHe level must be higher than 50% ²	
He exhaust system has been installed and is operational	
Ventilation system has been installed and is operational	
The ERDU function has been started up successfully	
RF-room integrity has been tested (not possible after ramping the magnet)	
All magnetizable material must be removed from the RF-room	

1. Refer to .(→ HDCL Inspection and Repair / Page 111)

2. For more information, refer to the (→ Footnotes / Page 165) "LHe Level"

All of the items above should have been confirmed with **"Yes"**.

5.7.2 Restricting hazardous areas



CAUTION

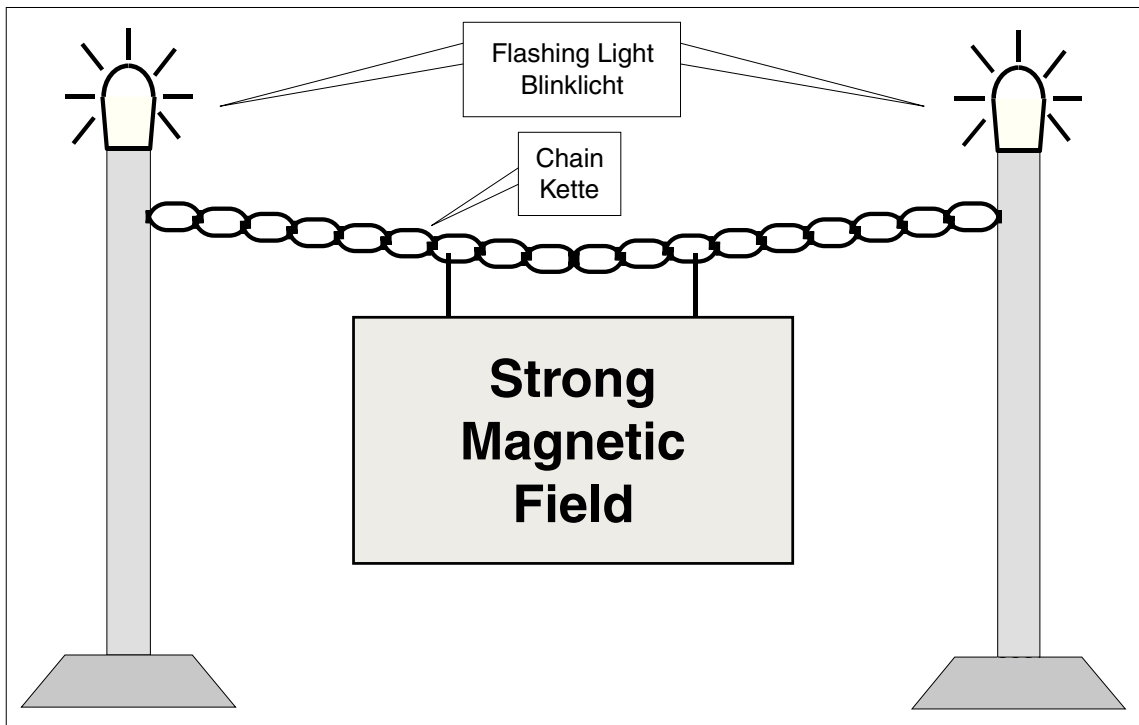
Strong magnetic field

If not observed, physical injury can occur

- » All magnetic field warning signs (especially "Strong Magnetic Field" and "Pace-maker-Warning") have been posted at appropriate locations. Restrict access to the magnet room. Use chains and flashing lights as well as signs.

- Switch off the GPA and the RFPA cabinets before ramping the magnet.

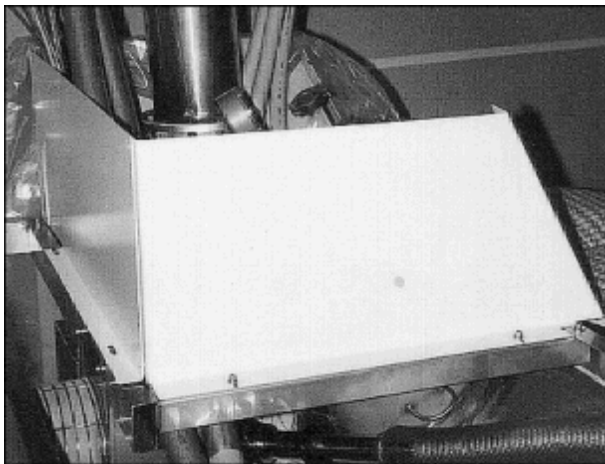
Fig. 99: Use flashing lights and chains to restrict access to the magnet field



5.7.3 Removing the covers

- Remove the magnet side and back covers.
- Remove the magnet turret covers.

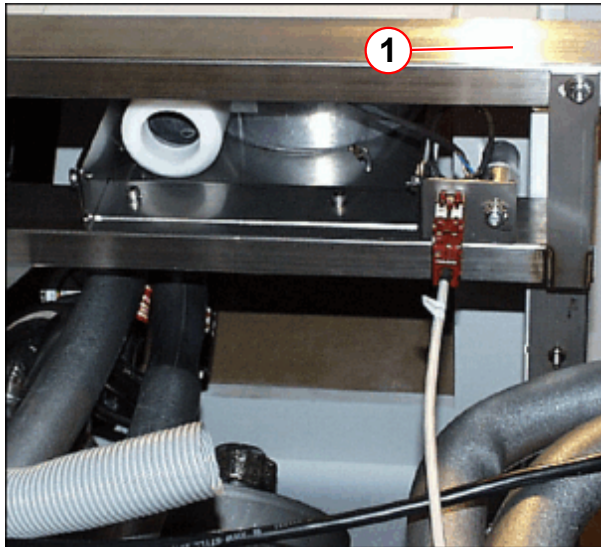
Fig. 100: Remove these covers



5.7.4 Tilting the patient air ventilator

- Remove the front rail of the cover frame (three screws).

Fig. 101: Patient air hose removed off the ventilator



(1) Front rail

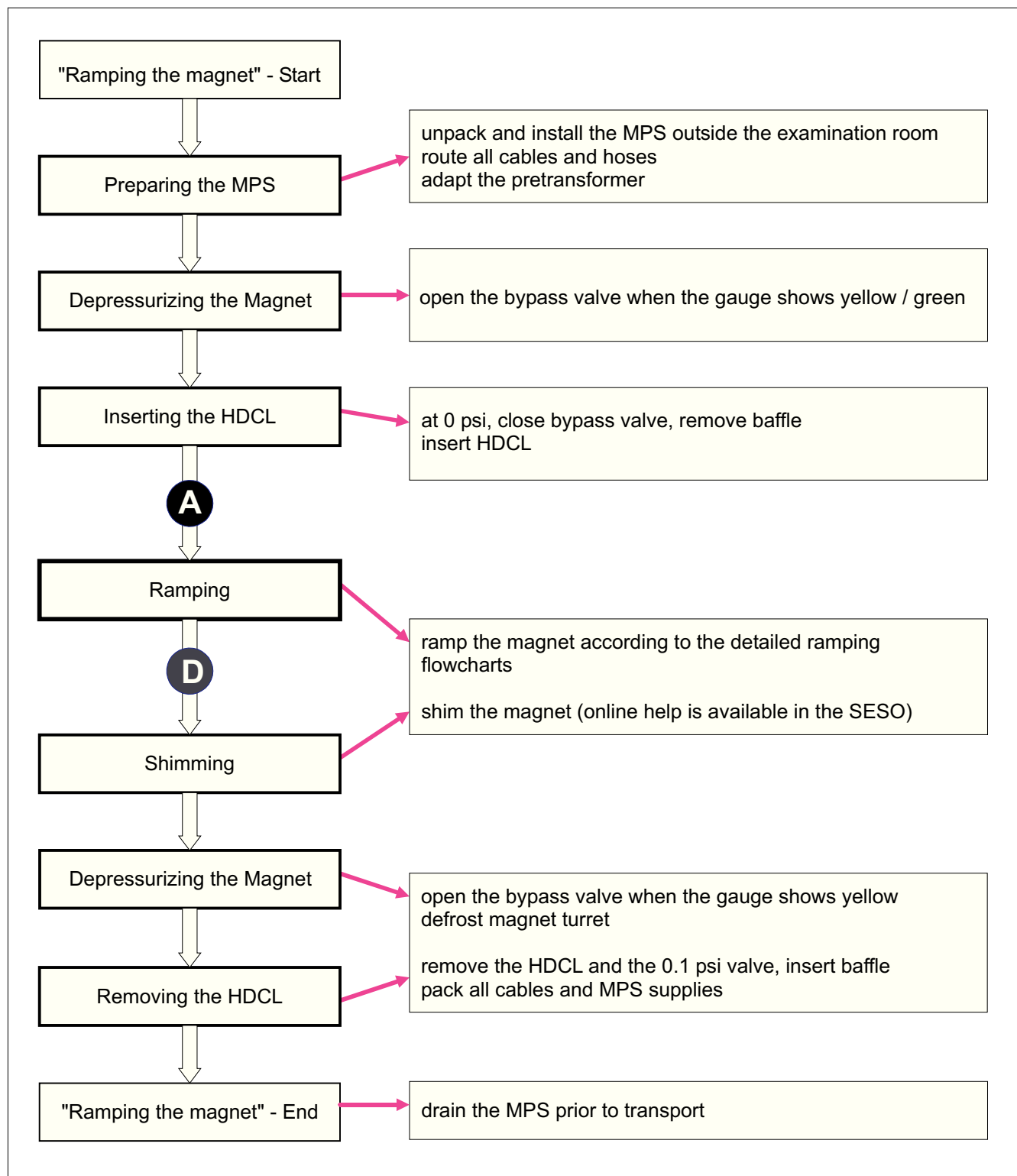
- Tilt down the ventilator by removing two screws (→ Fig. 102 Page 121)) to access the service turret.

Fig. 102: Patient air ventilator



5.8 Complete Ramping - flowchart

Fig. 103: General flowchart

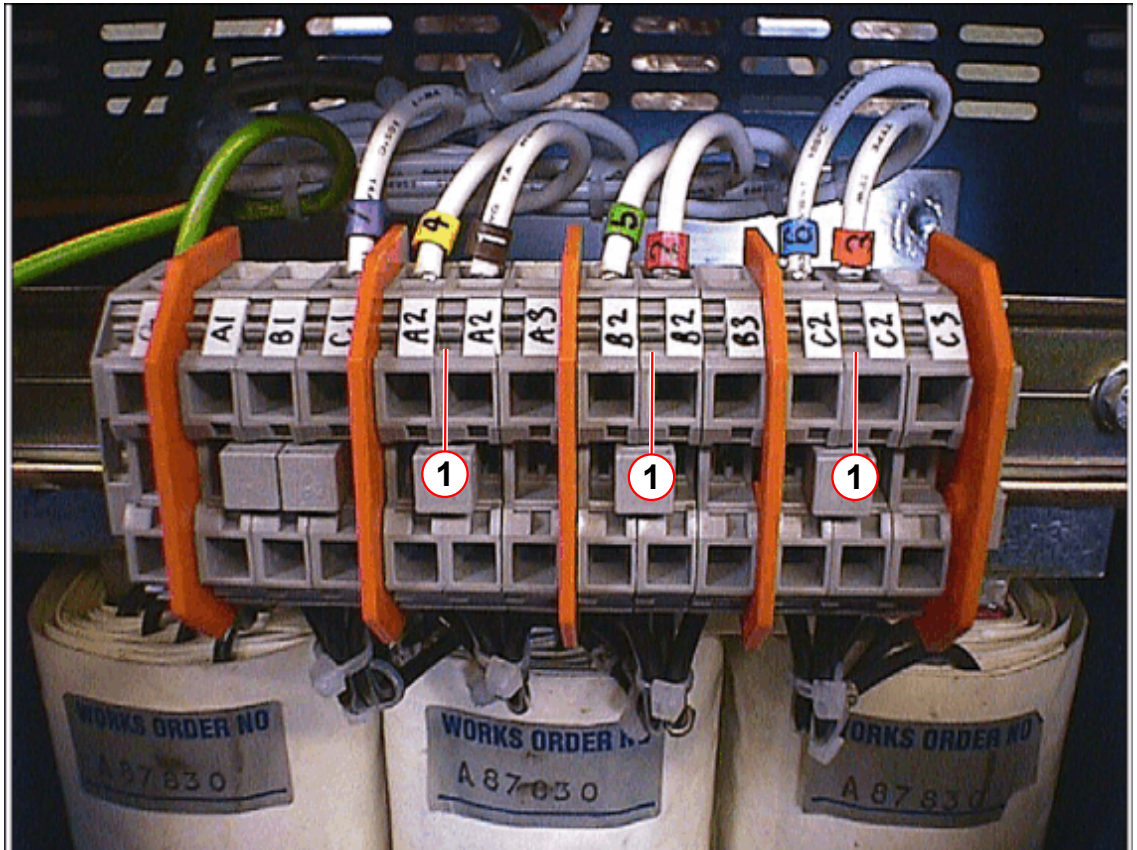


5.9 Preparing the MPS

Unpack all parts of the MPS from the transport boxes.

- Place the MPS control box on top of the power box (if not yet done).
- Position the MPS outside the RF room door and engage the wheel stops. Position the pretransformer next to the CCA.
 - Refer to subtitle (→ MPS cabling / Page 164).

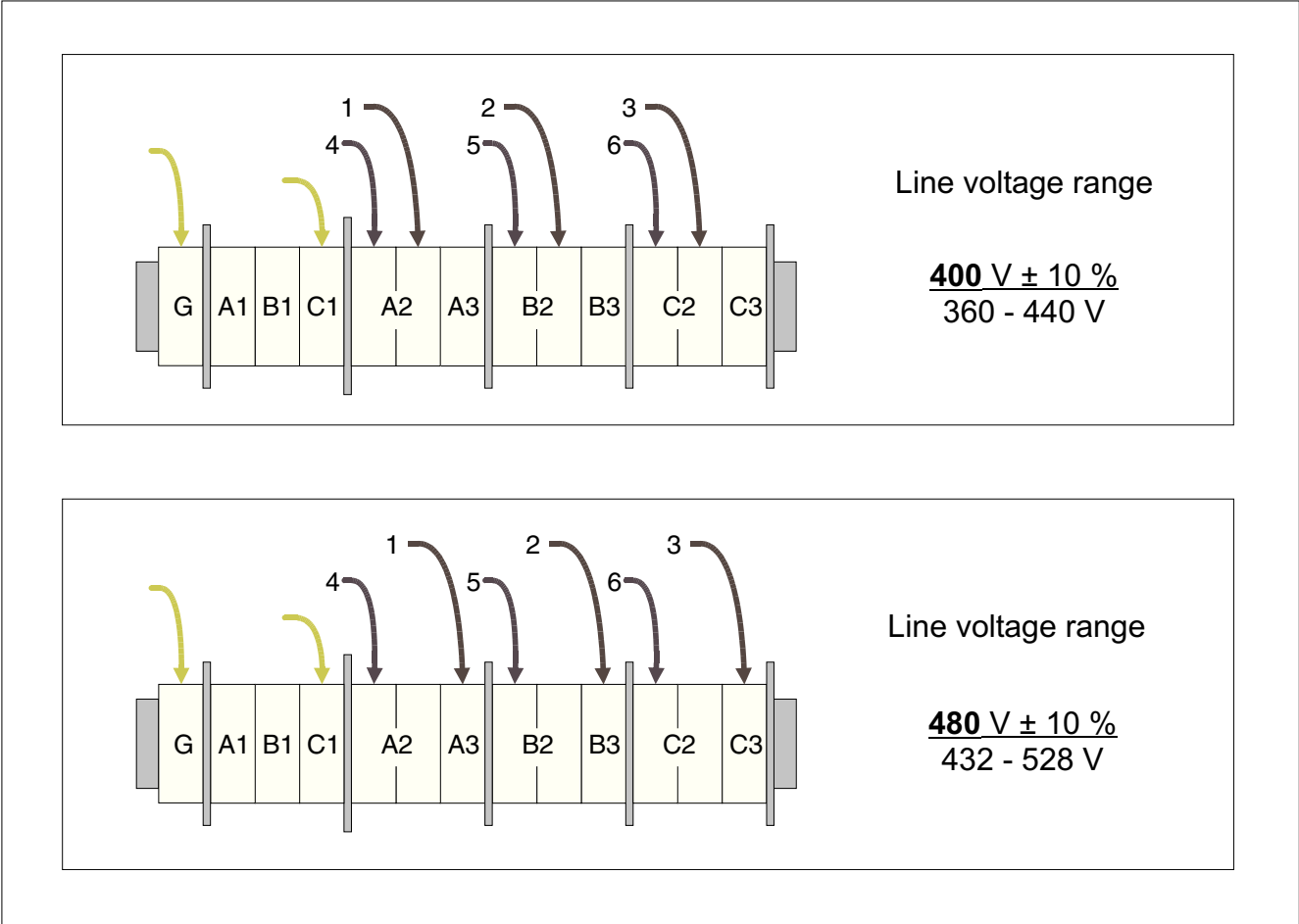
Fig. 104: Pretransformer of the MPS



(1) Wires 1, 2, 3 adapted for 400 V

- Switch off fuse F5 at the line power box on top of the CCA.

Fig. 105: Pretransformer terminal block



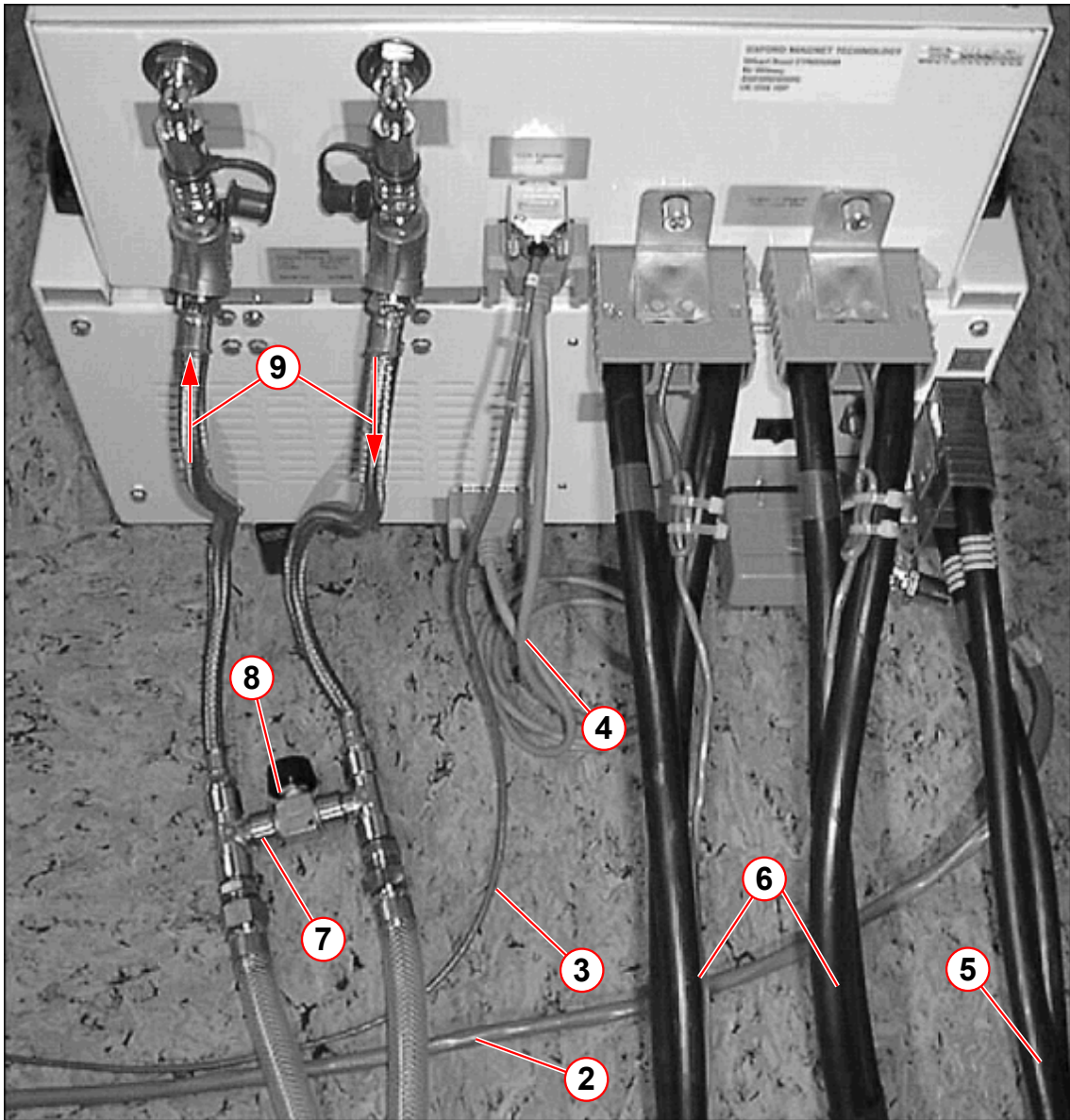
- Adapt the wires at the pretransformer to the on-site line voltage according to the figure above.
- Establish the cable connection between X1 of the MPS transformer and X102 of the CCA.
- Switch on fuse F5 at the line power box on top of the CCA.
- Use a multimeter to measure the output voltage at connector X2 of the MPS transformer.
 - » It must read $400\text{ V} \pm 10\%$.
- Switch off fuse F5 again.
- Establish the following connections (refer to (→ Fig. 106 Page 126), (→ Fig. 134 Page 164), (→ Fig. 135 Page 165) - cable numbers):

No	Cable / hose	from		to
1	Power cable (already done)	X102.CCA	-	X2.Transformers
2	Power cable	X1.Transformers	-	MPS
3	9 pin (D-sub) control cable	MPS.K2200	-	X6.K2201.CCA

No	Cable / hose	from		to
4	25 pin computer cable	Control box of the MPS	-	Power box of the MPS
5	Ground cable	MPS	-	Magnet
6	Two ramp cables	MPS	-	Magnet
7	Waterhose	MPS	-	Magnet

- Switch off the RCA cooler to reduce the water pressure in the hoses (F1 at the refrigerator).
- Verify that the system water hoses are labeled at the magnet.
- Connect the end of the water hose with the bypass at the MPS and the other end with the quick connect couplings at the magnet. The MPS will now be water-cooled instead of the gradient coil. Check the water flow direction according to (→ Fig. 106 Page 126).

Fig. 106: Back of the MPS with all cables connected



- (2) Power cable
- (3) 9 pin (Dsub) control cable
- (4) 25 pin computer cable
- (5) Ground cable
- (6) Two ramp cables
- (7) Waterhose
- (8) Flow directions
- (9) Bypass

**CAUTION**

Incorrect lock of the current lead

If not observed magnet and MPS can be damaged

» Connect and secure the cables

Fig. 107: Lower connector for MPS current leads

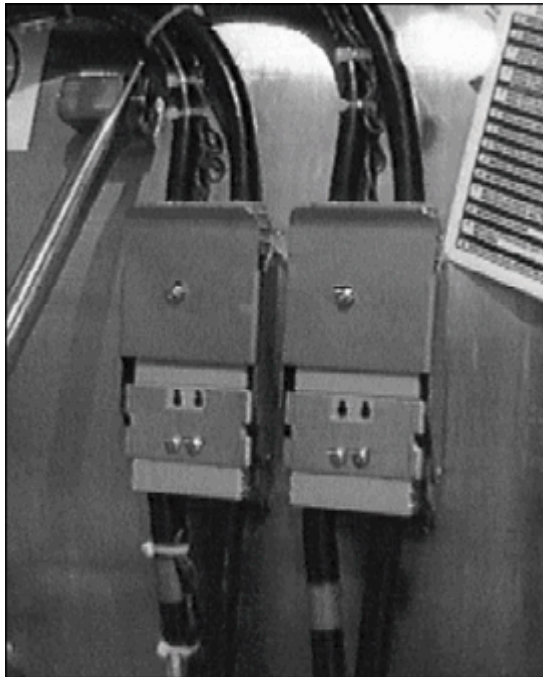


Fig. 108: Magnet - ground connection

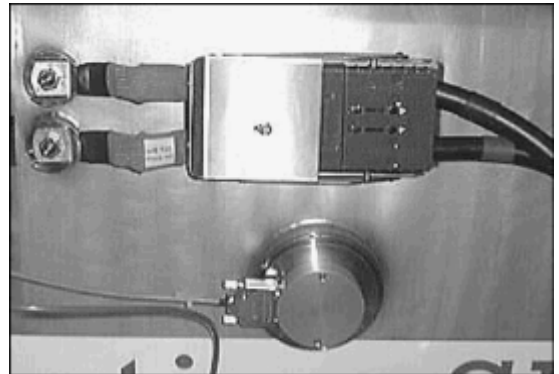
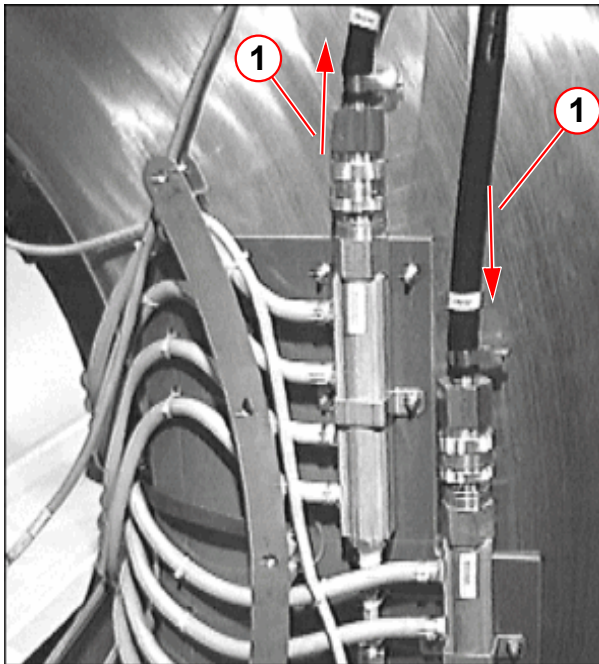
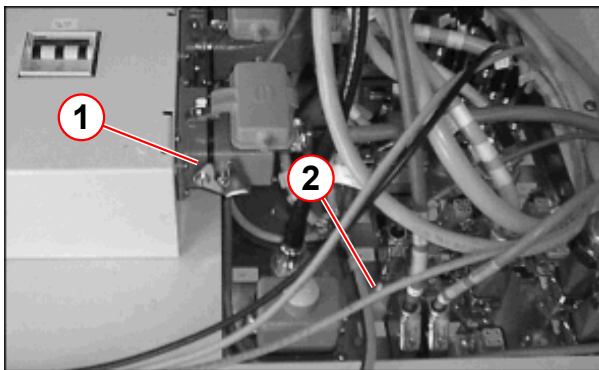


Fig. 109: Water-cooling reconnected



(1) Flow directions

Fig. 110: CCA connections



(1) X102

(2) X6



Make sure that no water enters the electronic assemblies.

Use a towel or a cloth to catch the water that leaks out when opening and closing the quick connects

- Switch on the cooler again (F1 at the refrigerator).



Make sure that the water pump is running and that there is no water leaking from the hoses.

- Check the water level at cooler 1. If it is not normal, fill the cooler with distilled water, until the yellow lamp goes out.

- Access the service menu in the SeSo (Service Software) as follows at the MRC:
- Enter your password and select menu item **"Magnet & Cooling"**.
- Select **"Status"** in the "Action" menu.
- The current flow rates are listed in the field **"Actual Cooler Values"** under **"Water-flow"**.
- If required adjust the bypass valve (→ Fig. 106 Page 126)) so that the flow rate is:
 - 30 l/min -30% +0% (21 l to 30 l) in the "gradient coil".



When filling the secondary water circuit, adhere to the instructions (→ Filling the secondary water circuit / Page 50).

The flow rates at the RCA do NOT have to be changed for this purpose.

5.10 Depressurization of the Helium Vessel (Magnet)



WARNING

Cryogenic liquids

If not observed cold burns can occur

- » **Do not work on the cryostat or disconnect any of the fittings without first depressurizing the helium vessel.**

The helium vessel will normally be pressurized to approx. 40 mbar (2/3 psi) overpressure (a different pressure is possible at mobile magnets).

Before depressurizing the helium vessel, ensure that protective clothing, face mask and gloves are worn. Also check that the area is well ventilated.

The helium vessel is depressurized via a tube and ball valve which bypass the overpressure valve assembly.

NOTICE

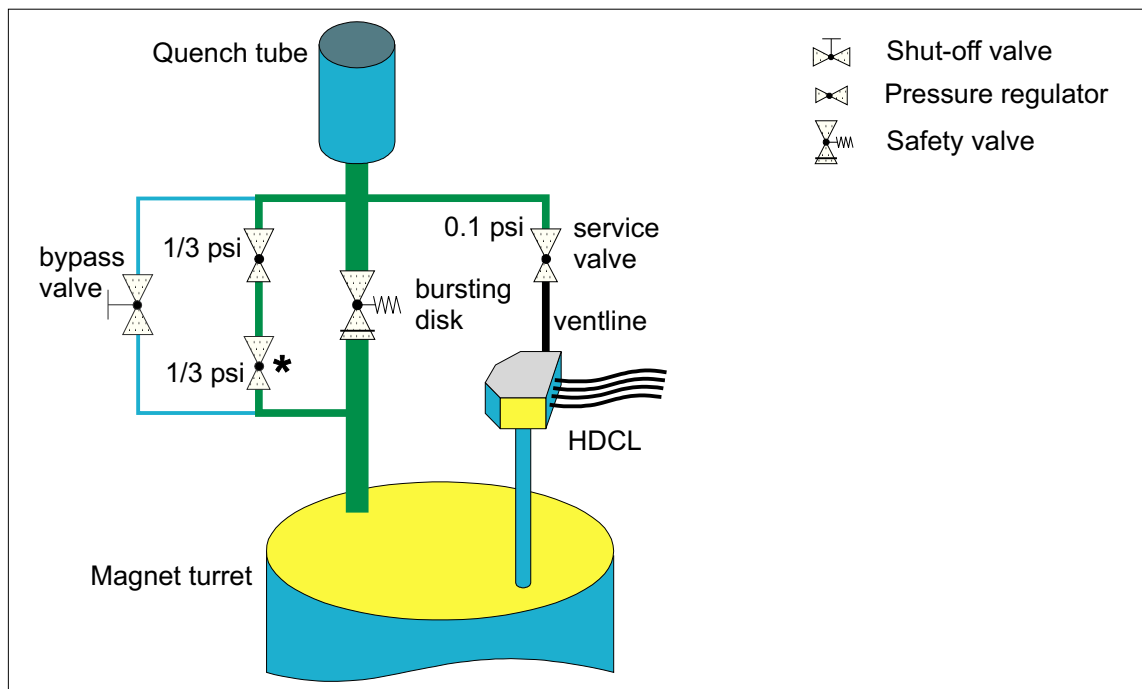
Low pressure in the helium vessel

Before opening the bypass, the pressure gauge reading must be in the yellow or green sector or at least 0.1 psi in the helium vessel.

- » **To avoid ice buildup in the magnet, the bypass must never be opened when the pressure gauge reads negative (red sector). If this is the case the EIS can be energized until the pressure gauge indicates at least 0 psi. Refer to the section (→ Energizing the EIS circuit / Page 160).**

- Open the depressurization valve (bypass valve).
 - » Helium gas will flow through the vent line and into the quench tube (venting system).

Fig. 111: Block diagram of the magnet and service valves

**WARNING**

Risk of asphyxiation

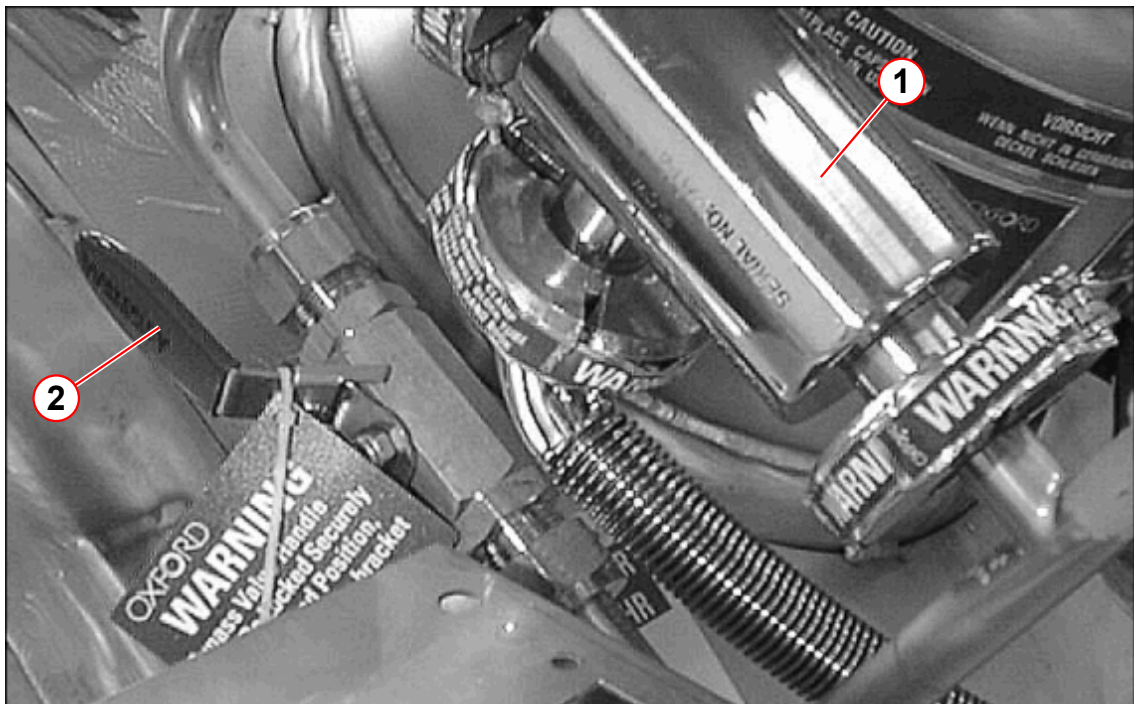
If not observed person working with cryogenics becomes dizzy or loses consciousness

» Helium gas must not be vented into the magnet room.



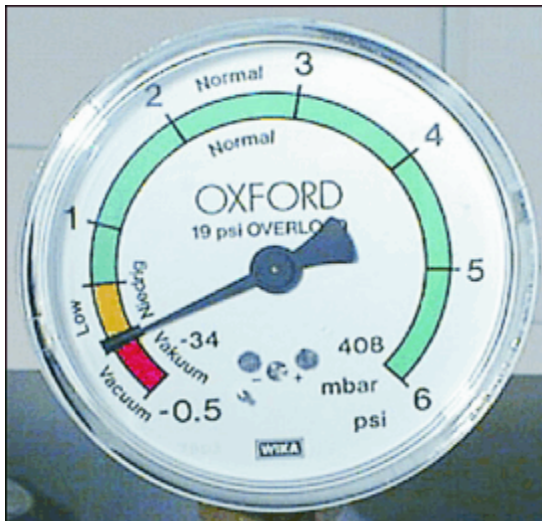
The magnet turret and the quench pipe will get very cold as a result of the escaping cold gas.

Fig. 112: The depressurization valve opened



- (1) Standard 1/3 psi valve
- (2) Depressurization valve opened

Fig. 113: System at atmospheric pressure (fully depressurized)



- Allow the system to depressurize until the gauge reads zero psi (black region). The magnet is now at atmospheric pressure and may be opened.

**CAUTION**

Depressurized helium vessel

If not observed ice formation in the helium vessel and neck tube will occur

- » Do not leave the system depressurized longer than necessary.

- You can now open one of the turret openings into the LHe vessel.
- Immediately close the depressurization valve after opening any turret openings.



After each operation has been completed on the turret, in particular fitment or removal of the HDCL the seals must be checked for leaks using a suitable leaker checker or solution.

5.11 Inserting the hardwired demountable current lead (HDCL)

The HDCL (Hardwired Demountable Current Lead) is a new current probe with a compact design and hard-wired current leads. These wires are about 1 m long and will be connected at the top of the magnet



Adhere to the cryogenic safety procedures during the following operations.



CAUTION

High pressure in the helium vessel Cryogenic liquids

If not observed cold burns can occur

- » Do not work on the cryostat or disconnect any of the fittings without first depressurizing the helium vessel.



WARNING

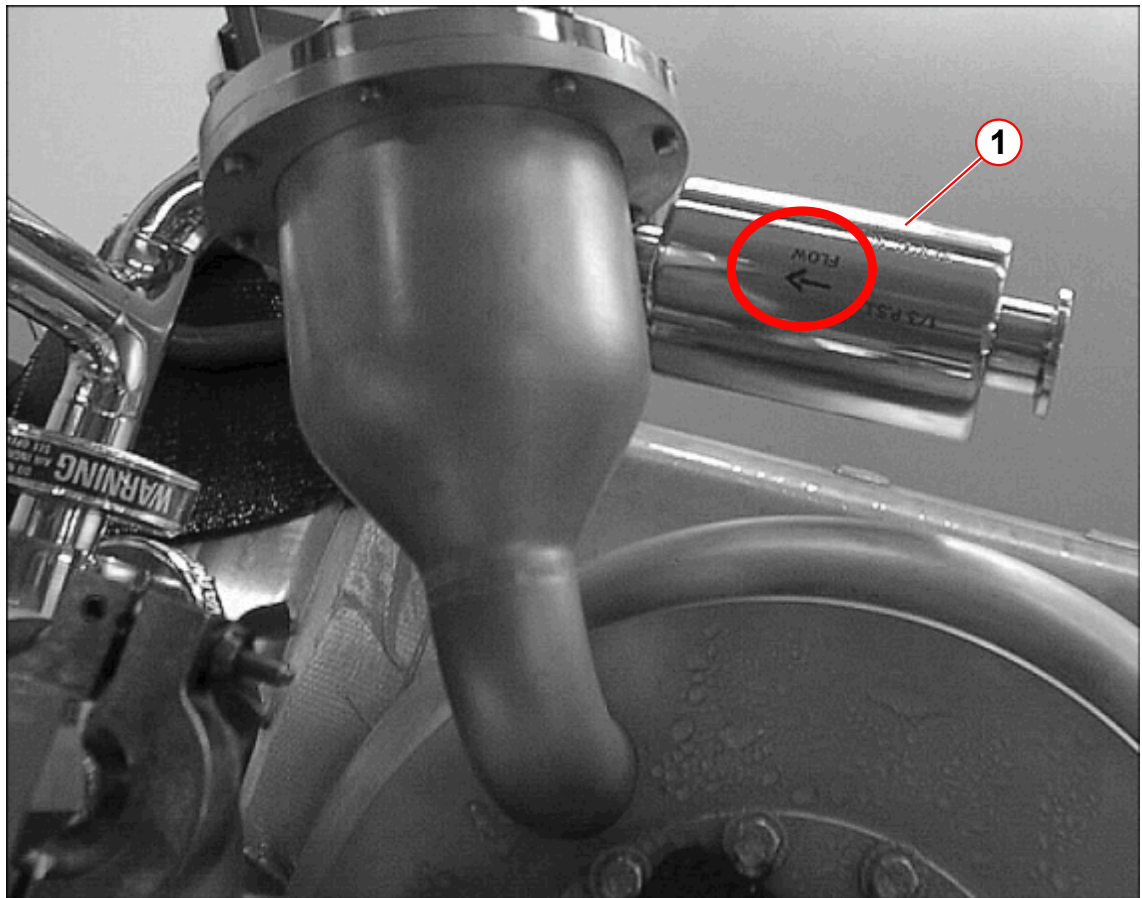
Current in leads

If not observed, death or serious physical injury can occur.

- » The MPS and ramp cables must remain connected to the HDCL as long as it is inserted in the magnet. This applies to all procedures - including insertion or removal of the HDCL. when the magnet is at field.

- Mount the 1/3 psi valve at the vent port. Use a seal and observe the flow direction. The arrow must point towards the quench tube.

Fig. 114:



For the first use on a site the contact springs of the inner HDCL contact have to be replaced. (→ HDCL Inspection and Repair / Page 111)

- Connect the HDCL cables to the top of the magnet. Ensure that the connection is properly made, and lock the lever with the nut provided.

- Unpack the HDCL and connect the vent line with the angled side towards the HDCL using a seal.

Fig. 115: HDCL with ventline mounted

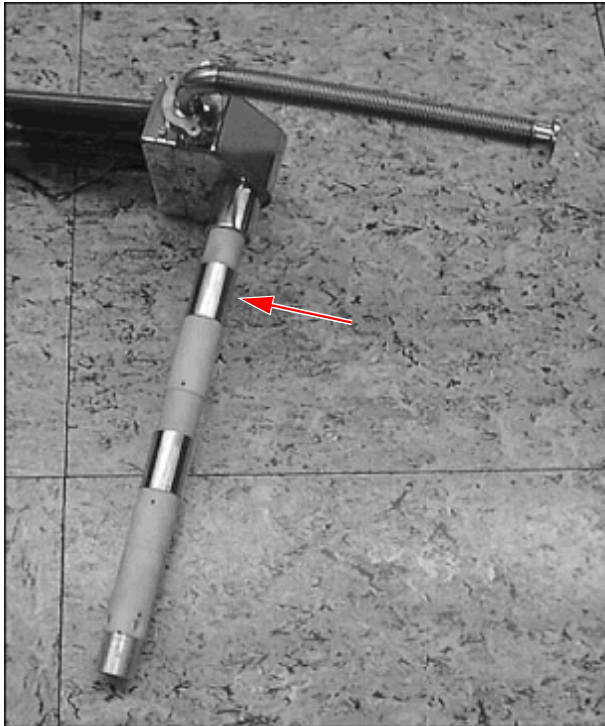
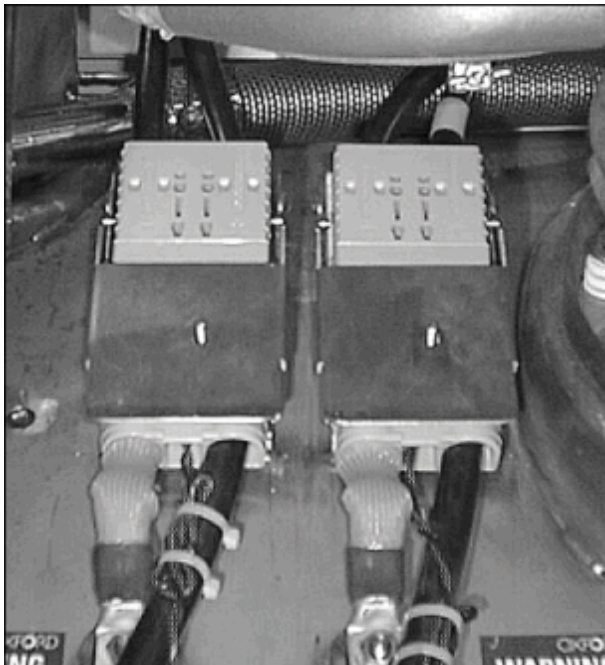


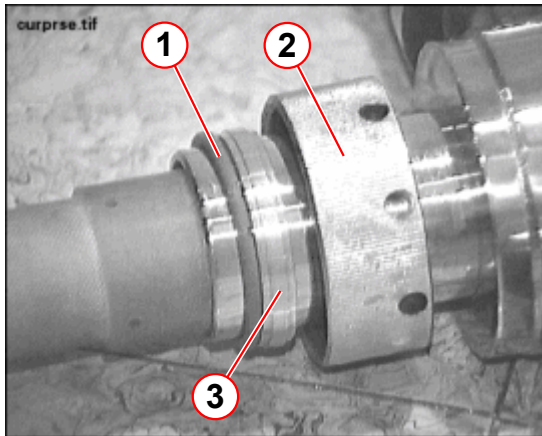
Fig. 116: HDCL current leads connected at the magnet



- Verify that the MPS current leads are connected to the MPS and to the lower connectors at the magnet.
- The HDCL vent line should remain connected to the HDCL at all times so as not to break the seal between the HDCL and the vent line.

- Slide the additional ring nut, brass compression ring, and 'O' ring onto the HDCL

Fig. 117: HDCL prepared with ring nut, brass compression ring and O-ring



- (1) O-Ring
(2) Ring nut
(3) Brass compression ring

- » If an additional ring nut is not available you must use the ring nut that holds the baffle set in place.



CAUTION

HDCL vent is not open and inserting the HDCL too quickly

The magnet may quench

- » Open the HDCL vent port and insert the HDCL slowly



CAUTION

Top of the HDCL will get cold.

If not observed cold burns can occur

- » Wear gloves for protection



If a strong mechanical resistance is felt, call the hotline. A large amount of ice may be present. If anything but minimal force is needed to push the HDCL down, withdraw it and try to de-ice the port by warming and re-inserting the probe. If the HDCL is removed the port must immediately be closed using the baffle or the plexiglas.



Do not attempt to force anything down the port to dislodge the blockage. This will only result in ice dropping into the socket - a condition which is more difficult to clear because it is in the helium reservoir.

- Warm up the turret and valves (including the 1/3 psi valve indicated with a *, refer to (→ Fig. 111 Page 131)), until no moisture is visible on the metal.
- Immediately close the depressurization valve after opening any turret openings.

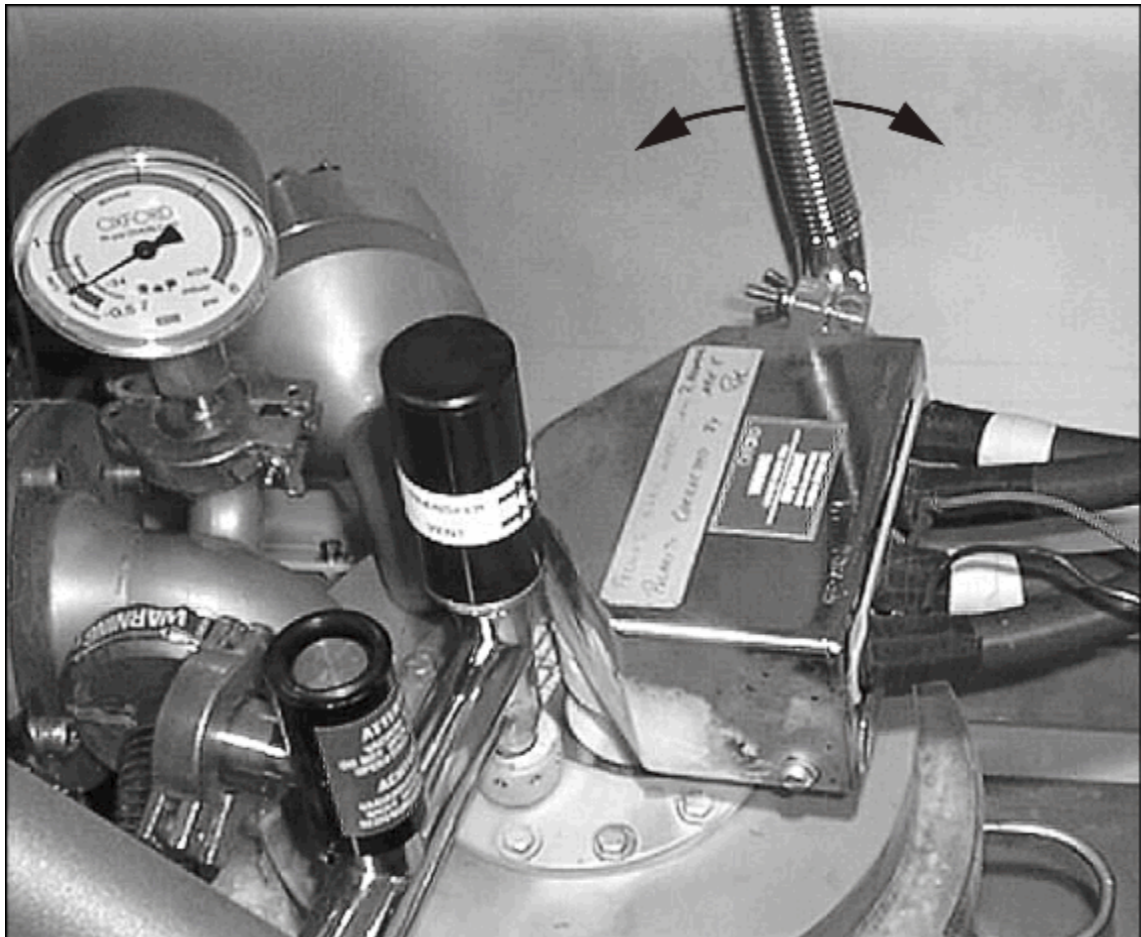
- Ensure that the end of the HDCL vent line is not capped. Remove the baffle set from the current lead port. Remove the copper gasket and destroy it (the gasket must not be used again).
- Before the gas flow subsides, insert the HDCL into the port. Move it down until the middle of the upper hinge is half way in (refer to arrow of (→ Fig. 115 Page 136). Hold it down in this position until the gas flow out of the vent pipe subsides. This is the indication that the HDCL has cooled down enough.



In low ceiling installations it may be necessary to bend the HDCL at the hinges provided. Always ensure that the hinge steeves are positioned over the hinge before the HDCL is inserted into the port.

- Ensure that the 'O' ring and the brass compression ring are seated correctly.
- Insert the HDCL fully into the magnet and secure it by screwing down the ring nut by hand. There is no need to use the 'C' spanner. If the ring nut cannot be screwed down use the heatgun to warm up the metal and try again.

Fig. 118: Turn the vent line so that it can be connected to the 1/3 psi valve



- Connect the vent pipe to the 1/3 psi valve already attached using a teflon seal. Note that the cryostat will continue to be vented through the HDCL and will not fully pressurize until the HDCL is removed and the baffle set is sealed back in place.
 - » The pressure in the magnet should now rise to approx. 1/3 psi (yellow range) within a few minutes.
- If the pressure does not rise or if the pressure rises too high, evaluate the problem first before ramping (e.g.: check the O-ring).

NOTICE

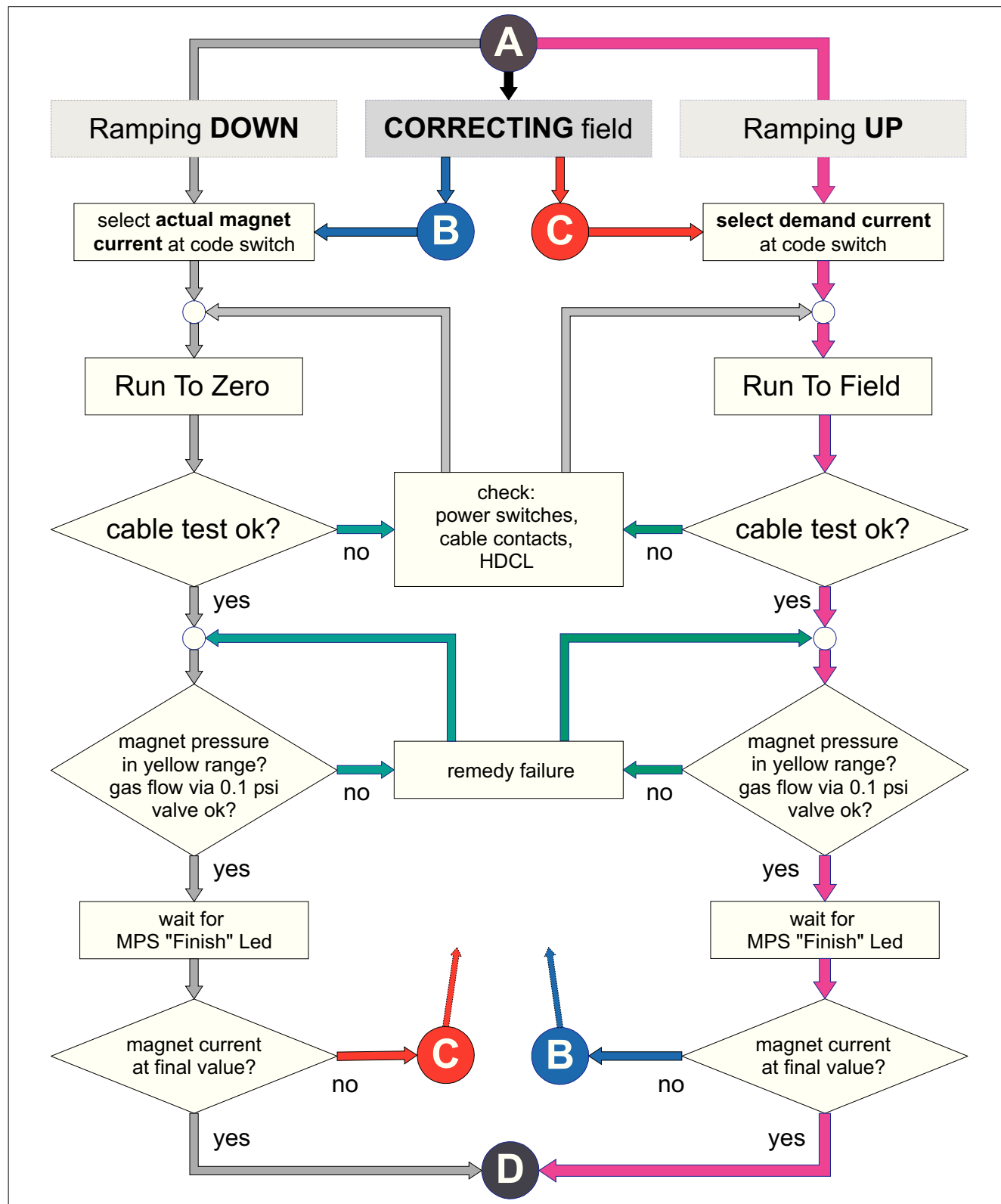
HDCL is inserted for longer than 4 hours without ramping and never over night.

Do not leave the HDCL inserted for longer than 4 hours without ramping and never over night.

- » **During shimming, when numerous ramps are needed, the HDCL may be left in place and venting through the 1/3 psi valve. The vent must NEVER be blanked.**

5.12 Ramping - flowchart

Fig. 119: Ramping flowchart



5.13 Ramping up the magnet



CAUTION

Helium level

Magnet can quench

- » Ensure that the helium level is higher than 50% (initially ramping 80%) before ramping up the magnet.



The quench valve must be completely defrosted prior to ramping. The HDCL may be left venting through the 1/3 psi valve if further ramping or shimming is needed. Otherwise remove the HDCL (refer to (→ Removing the hardwired demountable current lead (HDCL) / Page 155)).



During magnet ramping, the operator must be within clear view of the MPS control panel to monitor and report any conditions or failures.

5.13.1 Setting the current and voltage values

- Set the demand values "Demand Amps" and "Magnet Volts" at the code switches (refer to the "Oxford - Installation data sheet" document, subtitle "Operating parameters").



Current Set = magnet operating current.

5.13.2 Switching on the magnet electronics

- Switch on circuit breaker F5 on top of the CCA.
- Press the switch on the back of the MPS.
- Switch on the circuit breaker in the front (refer to (→ Fig. 85 Page 110)) of the power box to turn the MPS on.



CAUTION

Starting ramp procedure

Risk of quench

- » Before starting the ramp procedure, make sure you have positive pressure in the magnet.

Fig. 120: MPS switches, display and Leds

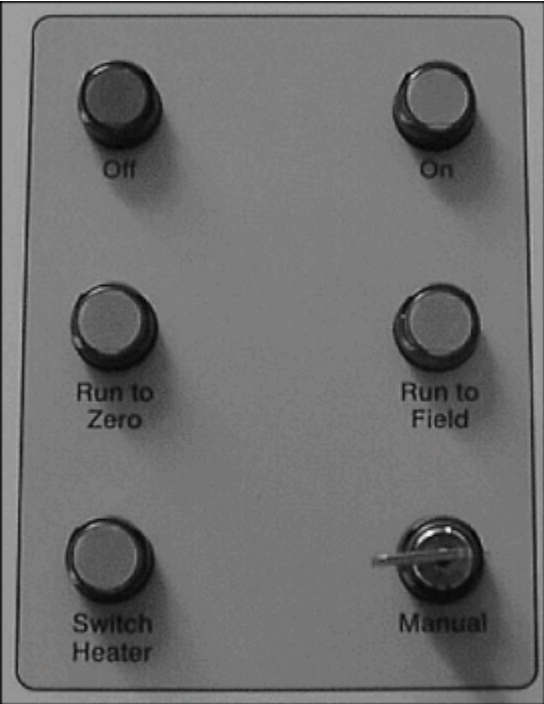
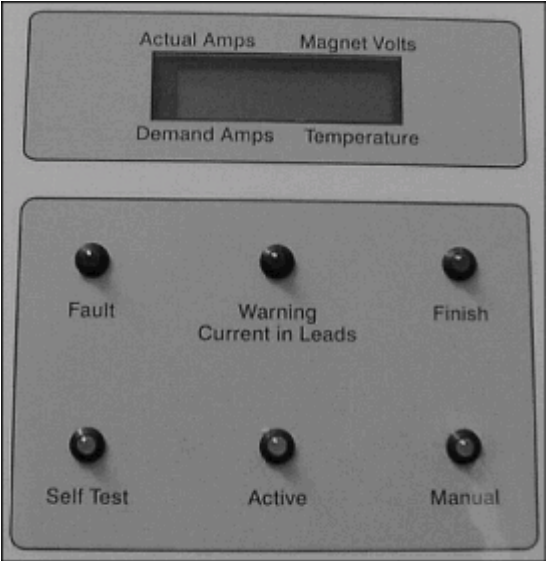


Fig. 121: MPS switches, display and Leds



- Switch the key switch to automatic mode (horizontal position).
- Press the **"ON"** button on the control panel.
 - The lamp in the ON button is illuminated.
 - The lamps in the "Run to Field" and "Run to Zero" buttons flash alternately.

The display of the MPS shows several status infos for **MAGNETOM**

Fig. 122: MPS initialization

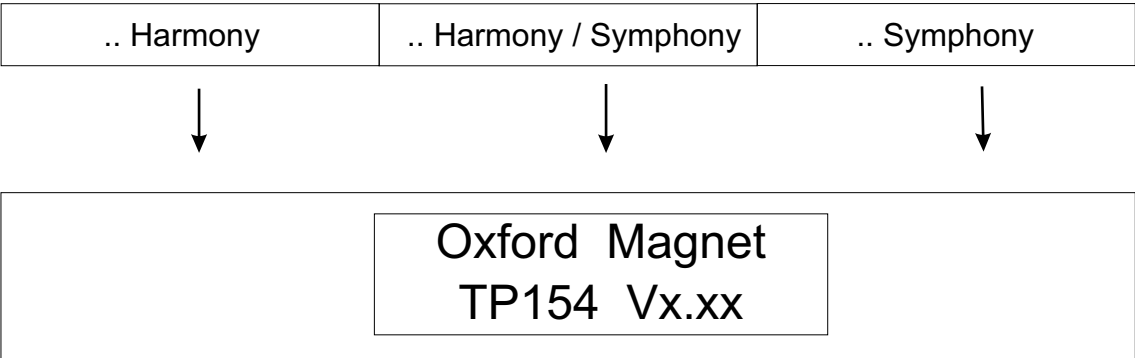


Fig. 123: MPS idle

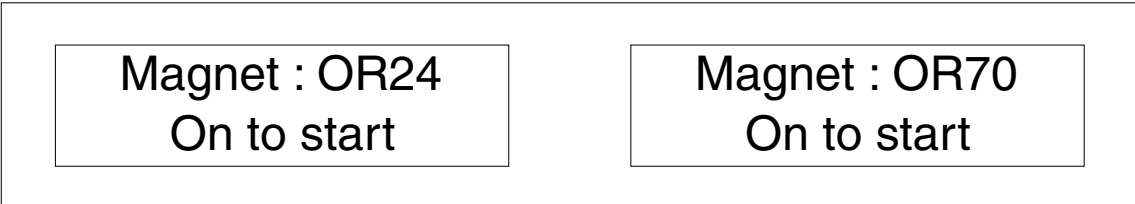
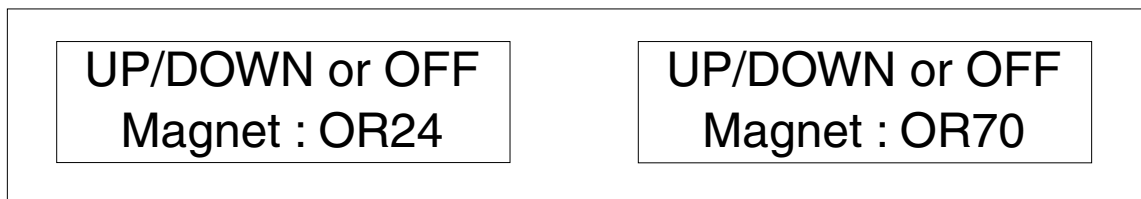
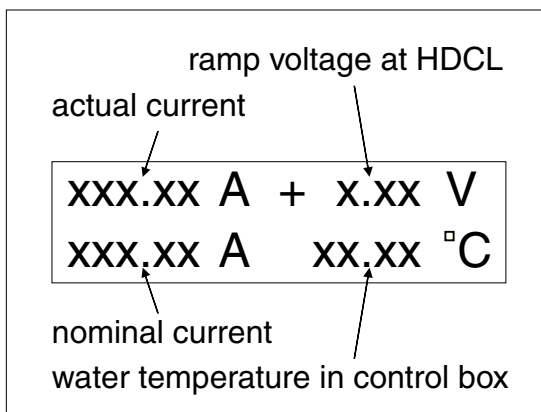


Fig. 124: MPS display, prepared to operate



5.13.3 Ramping up the magnet

Fig. 125: MPS - display explanations



- Check all cable connections again (especially the lower magnet connectors in (→ Fig. 107 Page 127)). They **must make full contact**.
 - » Loose connections can cause voltage losses especially in the sense lines.
- Press the **Run to Field** button.
 - The lamp in the "Run to Field" button is illuminated.
 - The "Self Test" LED > "ON" indicating that the self-test of the magnet electronics is in progress.

After the self-test has been completed:

- LED **Self Test** > "OFF"
- LED **Active** > "ON"

The cable test follows:

- » The cable test ramps up the cable to the "Current Set", checks the HDCL voltage and ramps down again. Refer to (→ HDCL contact checks / Page 145)",, if voltage problems occur and the cable test does not end successfully (error code 206 in MPS display).
- » As soon as there is current in the ramp leads the "Warning Current in Leads" Led will be illuminated.

 **CAUTION**

Pressure in the magnet is too low

Risk of magnet quench

- » Check that the pressure in the magnet during the cable test is in the upper yellow range.
- » The two standard 1/3 psi valves (refer to (→ Fig. 111 Page 131) (*) and (→ Fig. 112 Page 132)) must NOT get cold. If they do, stop the ramp by pressing the OFF button and warm up the valve assemblies. If they frost up during a ramp the gas is not flowing through the HDCL .Check the by-pass valve, it must be closed. Check if the HDCL 1/3 psi valve is mounted correct flow direction and if it is working.

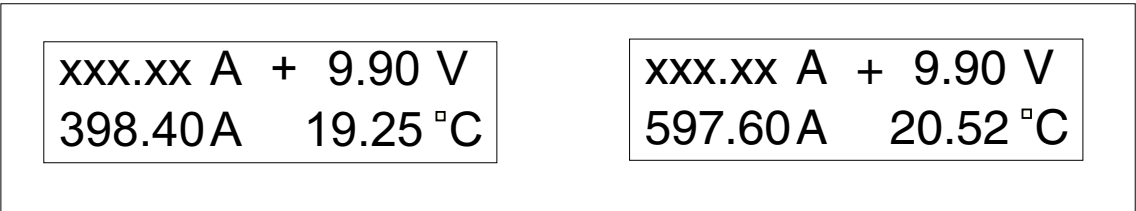
Fig. 126: MPS prepares to "run to field"



After the cable test, the magnet will ramp up to the current value selected:

- Overfielding > Current Set + 3 A
- Current Set

Fig. 127: Ramping up to the "Current Set"



After the magnet is ramped up, the magnet ramping cables discharge automatically.

- LED **Finish** > "ON".
- Press the **"OFF"** button and switch off both circuit breakers at the MPS.
- Switch off circuit breaker F5 on top of the CCA.

NOTICE

Power failure

- » In the event of a power failure during magnet ramping, follow the procedure (→ Power failure during magnet ramping / Page 162)

■ Note the demand magnet current set at the code switch:

[A]

5.13.4 HDCL contact checks

On new twin turret products fitted with the HDCL, the lead check voltage is set to a value [125 mV for 1 T, and 170 mV for 1.5 T magnets]. This is because the voltage tapping point is taken from the top of the HDCL so that the only joint measured is the connection to the terminal plate on the magnet itself. In addition, it has been found that higher voltages will cause the magnet to quench. It is possible that this setting might occasionally cause problems, with the MPS tripping out on over-voltage during the lead check.



WARNING

Current in leads

If not observed, death or serious physical injury can occur.

- » **Ensure that there is no current flowing in the circuit before removing any connections.**

The following is a list of actions which can be taken to reduce the lead check voltage.

Actions to reduce the lead check voltage

1. Ensure that the gas flow up the lead is fully established so that the HDCL is venting properly. The boil-off gas cools the lead and reduces its resistance. The vent tube from the lead must be very cold (icey). There must be no other venting route.



The lead check assists in generating boil-off, so a second attempt immediately after the first will start with a higher boil-off and should give a lower voltage. Further attempts will not improve the readings.

2. Ensure that the lead is fully engaged with the magnet connector. There should be a gap of not more than 10 mm between the bottom of the HDCL housing and the top of the ring nut on the port [typically about 5 mm].
 - » The connection of the contact springs to the magnet connector can sometimes be improved by rotating the HDCL. Two to three attempts may be required since this does not always improve the situation. No upwards force may be applied.
3. Ensure that the volt sense lead contacts in the Anderson connector are mating well. If there is no contact for the volt sense leads, the MPS will measure the voltage across the complete cable set. If there is poor contact, the MPS will measure a voltage between the full cable voltage and the voltage across the top of the HDCL. Moving the Anderson connectors might improve the contact. The resistance of the volt sense circuit should be in the range of 0.1 to 0.2 ohms. Do not attempt this action during a lead check.
4. Disconnect the Anderson connectors and ensure that the volt sense contacts are approximately in the center of their housing and have some free movement. If this is not the case, contact the MR-hotline.
5. Remove the HDCL and completely dry the bottom contacts. Check that the contact springs are all present and in good condition, and that they are clean and silvery [not dark with oxide]. Any damaged or dirty contact springs must be replaced. " if the con-

tacts need to be replaced. (→ HDCL Inspection and Repair / Page 111) if the contacts need to be replaced.

6. Check that the magnet connector is free of ice.

5.14 Ramping the magnetic field to the required field strength

To avoid magnet quenches during short ramp cycles, the field must be reduced to zero first and then set to the calculated final magnet current.

- Check the actual frequency (**f** actual).
- Calculate the **final current** using the **highest frequencies possible** listed in the formulas below: The formula takes the tolerances of the different magnet power supplies in consideration

Fig. 128: MAGNETOM Harmony / MAGNETOM Symphony

$I_{\text{final}} = \frac{f_{\text{nominal}} \times I_{\text{demand}}}{f_{\text{actual}}}$	
$I_{\text{final}} = \frac{40.53 \text{ [MHz]} \times I_{\text{demand}}}{f_{\text{actual}}}$	$I_{\text{final}} = \frac{63.68 \text{ [MHz]} \times I_{\text{demand}}}{f_{\text{actual}}}$
MAGNETOM Harmony	MAGNETOM Symphony

	1 Tesla	1.5 Tesla
lower limit of frequency	40.35 MHz	63.50 MHz



Before the lower limit of the frequency range is reached, the magnet must be scheduled for re-ramping.

Field decay for Harmony: approx. 3 kHz / month

Field decay for Symphony: approx. 4.6 kHz / month

- Note the **final magnet current:** [A]
- Switch on circuit breaker F5 on top of the CCA and switch both circuit breakers at the MPS.
- Press the **"ON"** button at the control panel.
 - The lamp in the ON button is illuminated, the lamps in the "Run to Field" and "Run to Zero" buttons flash alternately.
- Press the **"Run to Zero"** button.
 - The lamp in the "Run to Zero" button is illuminated, LED Self Test > "ON" indicates that the self-test of the magnet electronics is in progress.
- After the self-test has been completed:
 - LED **Self Test** > "OFF", LED **Active** > "ON"
 - The magnet will ramp down to zero.

- Set the **final magnet current** at the code switch.
- Again press the **"Run to Field"** button to repeat the procedure and ramp up to the final magnet current.
- Press the **"OFF"** button.
- Switch off both circuit breakers at the MPS and switch off circuit breaker F5 at the CCA.

5.15 Co-Siting

Co-Siting means installing two MR systems with identical field strengths in close proximity.

NOTICE

Co-sited systems have to be ramped to different field strengths.

» The nominal frequencies should differ by 100kHz.

Example:

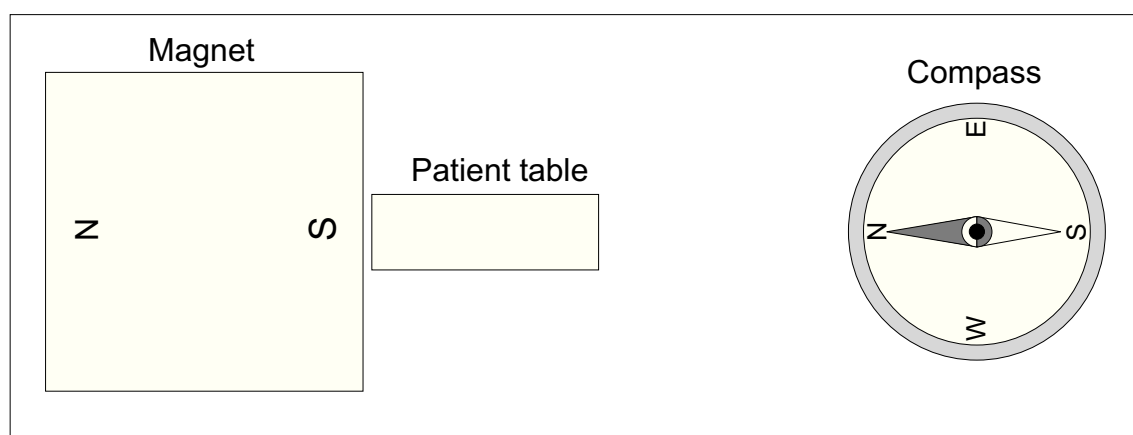
	1 Tesla Systems	1.5 Tesla Systems
System 1	40,53 MHz	63,68 MHz
System 2	40,43 MHz	63,58 MHz
Lower limit for re-ramping	40,35 MHz	63,50 MHz

5.16 Checking the magnetic field polarity

5.16.1 Checking the field direction with a magnetic compass

- Align the compass to geographic north outside the building, well away from all magnetizable structures:
 - Note which end of the needle points to **geographic north** (**geographic** north pole coincides with **magnetic** south pole). In the following, this G17 end of the needle will be referred to as the "N end of the needle".
- Enter the RF room with the compass, continuously observing the compass needle. Keep the compass away from magnetized elements, since they could demagnetize the compass needle.
- Proceed as described above. The "N end of the needle" must point to the magnet. This ensures that the south pole of the magnet points to the patient table.
- Again align the compass to geographic north outside the building, well away from all magnetizable structures:
 - Ensure that the **same** end, the "N end of the needle" points to **geographic north** as before. If this is the case, you can be sure that the compass wasn't demagnetized and the measurement was successful.

Fig. 129: Magnetic field check



5.17 Shimming the magnet

The magnet is now prepared for shimming. Follow the procedure below only if the magnet is to be shimmed.



Reinstalled systems and local moved systems must be shimmed from bare (all existing iron must be removed from the shim trays).



If the system is left over night between shimming iterations, the current probe should be removed and the port blanked (insert the baffle).



Before plotting the field, wait about 30 minutes to one hour to let the field stabilize.

The shim procedure is part of the service software ("SeSo"). All relevant work steps are to be performed "on-line" (on the screen). On-line Help will explain the individual work steps.

- Start Numaris at the MRC.
- Start the "NUMARIS Service Software".
- Enter your password and select the menu item **"Shim"**.
- Perform the shim process by following the instructions on the screen.



After shimming is done, continue with (→ Removing the hardwired demountable current lead (HDCL) / Page 155)

5.17.1 Ramping down the magnet



CAUTION

Cooling of the HDCL

Risk of a quench

- » The 1/3 psi valve indicated with a *, (→ Fig. 111 Page 131), must be completely defrosted prior to ramping. The DCL may be left venting through the 1/3 psi valve, if further ramping is needed. Otherwise remove the HDCL (refer to removing the demountable current lead). The HDCL vent **MUST NOT** be blanked with the magnet at field.

- The HDCL must be connected (refer to (→ Inserting the hardwired demountable current lead (HDCL) / Page 134)).

- Check the LHe level. It must be higher than 30%. ¹
- Turn the system on.
- Switch on circuit breaker F5 on top of the CCA and power on the MPS.
- Press the **"ON"** button.
 - The lamp in the ON button is illuminated.
 - The lamps in the "Run to Field" and "Run to Zero" buttons flash alternately.
- Set the magnet current (refer to the "Installation Protocol" in the Service Software), if not yet done.
- Press the **"Run to Zero"** button.
 - The lamp in the "Run to Zero" button is illuminated.
 - The Self Test LED > "ON" indicates that the self-test of the magnet electronics is in progress.



When ramping down the magnet, the magnet electronics runs a self-test and a cable test as well.

- Check the 1/3 psi valves



CAUTION

No sufficient cooling through the HDCL

Risk of a quench

- » If the 1/3 psi vent valves below the Quench valve frost up during self test, the ramp should be stopped. This indicates that there is not sufficient cooling through the HDCL. The 1/3 psi valves may need to be warmed up before continuing.

- After the self-tests have been completed:
 - LED **Self Test** > "OFF"
 - LED **Active** - > "ON"
- After the magnet has been ramped down: (wait approximately 3 min.)
 - LED **Active** - > "OFF"
 - LED **Finish** - > "ON"
- Press the **OFF** button and switch off the MPS.
- Switch off circuit breaker F5 on top of the CCA.

1. For more information, refer to the footnote "LHe Level" in the Appendix.



In the event of a power failure during magnet ramping, follow the procedure (→ Power failure during magnet ramping / Page 162)

5.17.2 Power failure during magnet ramping

A power failure when ramping the magnet up or down will cause the magnet power supply to trip out of RUN mode, closing the output shorting contactor. The magnet current will then run very slowly down through the magnet leads and shorting contactor.



WARNING

Current in leads

If not observed, death or serious physical injury can occur.

- » The MPS must remain connected to the HDCL during and following a power failure. The magnet leads and the shorting contactor are not indicated. Take no action until the power has returned and the following procedure is carried out.



After a power failure, there is NO indication on the power supply until mains power is restored.

When power returns, or after a momentary power failure, the red LED "Warning - Current in leads" will be illuminated as a warning that current is still flowing in the magnet, magnet leads and shorting contactor circuit.

- Press S1 and confirm that mains power has returned (the green lamp in S1 should be illuminated).
- Check that the MPS has powered up in the OFF state by verifying that the LED in the **ON** button is extinguished.
- Check whether the red LED "Warning - Current in leads" is illuminated or not.
 - If it is not illuminated, it is safe to switch off the MPS and circuit breaker F5 and proceed with removal of the HDCL.
- If it is illuminated, press the **ON** button on the MPS.
- Then select **Run to Zero** or **Run to Field** to continue ramping under MPS control.
- Wait until the green LED on the MPS illuminates in **Finish**, then press the **OFF** button.
- Check that the red LED "Warning - Current in leads" is not illuminated before switching off the MPS and circuit breaker F5.
- It is now safe to remove the HDCL and disconnect the magnet leads (follow the procedure (→ Removing the hardwired demountable current lead (HDCL) / Page 155))

NOTICE**Removing the HDCL**

- » The MPS must remain connected until after the HDCL has been removed. The HDCL vent port **MUST NOT** be blanked.

5.18 Removing the hardwired demountable current lead (HDCL)

5.18.1 Safety checks

Before removing the HDCL, ensure that no current is flowing in the magnet leads and shorting contactor circuit by carrying out the following procedure:

- Ensure that circuit breaker F5 on top of the CCA and the switch on the back of the MPS are turned on (the lamp in the ON key is illuminated green).
- Check if the red LED "Warning - Current in leads" at the MPS control panel is not illuminated. If it is illuminated, hazardous currents may be present and the procedures in (→ Power failure during magnet ramping / Page 162) must be carried out.



CAUTION

Removing the HDCL

This can result in damage or personal injury.

- » Do not remove the HDCL if the red LED "Warning - Current in leads" is illuminated. . The MPS must be used to fully discharge the magnet leads by following the procedure described in (→ Power failure during magnet ramping / Page 153)

NOTICE

Negative pressure in the helium vessel

- » Never open the HDCL port with negative pressure in the helium vessel.



CAUTION

Disconnecting the HDCL during a power failure.

This can result in damage or personal injury.

- » Never remove or disconnect the HDCL during a power failure. There will be no warning of hazardous currents.



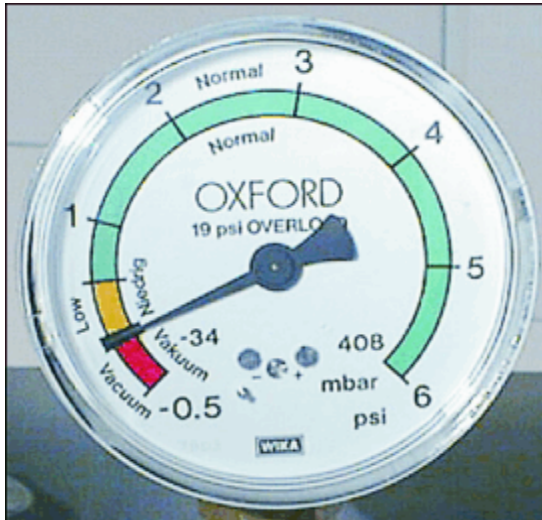
WARNING

Current in leads

If not observed, death or serious physical injury can occur.

- » The MPS and ramp cables must remain connected to the HDCL as long as it is inserted in the magnet. This applies to all procedures - including insertion or removal of the HDCL. when the magnet is at field.

Fig. 130: System at atmospheric pressure (fully depressurized)



- Depressurize the magnet (if it is pressurized) to zero psi as described in (→ Depressurization of the Helium Vessel (Magnet) / Page 130).

- Remove any water/ice build-up from around the current lead entry port using a hot air gun and paper towels.
- Shut off the switches at the MPS and switch off circuit breaker F5 on top of the CCA.



CAUTION

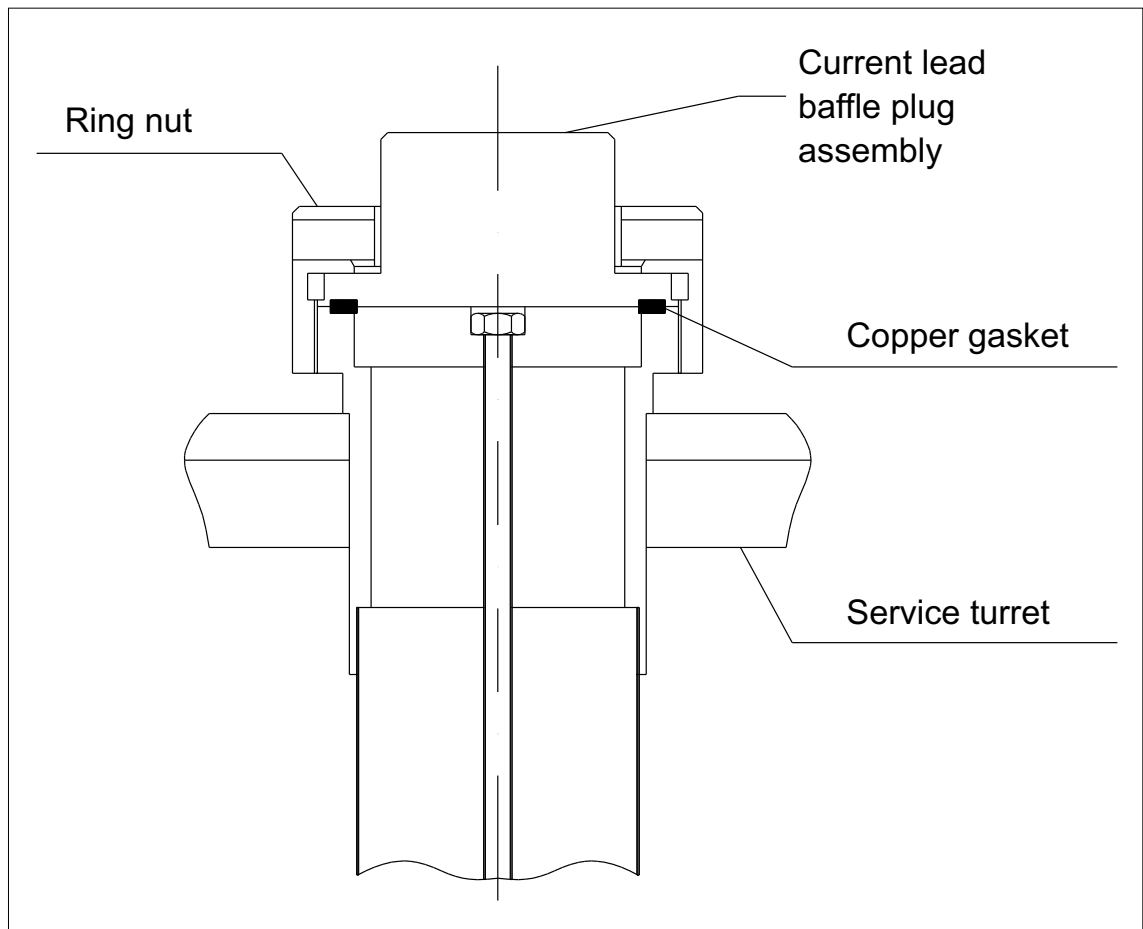
Very cold gas being emitted from the entry port.

If not observed cold burns can occur

- » Wear gloves for protection. Certain operations must be done quickly. Ensure that the system is open for the minimum time possible.

- Prepare the baffle plug with a **new** copper gasket.
Do not use a vacuum grease on the copper gasket. Make sure the recess is clean.
- Immediately close the depressurization valve after opening any turret openings. Lock the handle with the specially fitted bracket.
- Undo the ring nut, while holding the HDCL in place. Pull the HDCL out quickly to prevent air from entering the helium reservoir, which may form ice on the main magnet connector.
- Immediately insert the baffle plug, taking care to locate it properly on the copper gasket.

Fig. 131: Baffle for the current probe port



- Refit the ring nut and tighten it using the 'C' spanner provided without using excessive force.
- Disconnect the HDCL vent line from the 1/3 psi valve.
- Remove the 1/3 psi valve from the quench valve, and blank the open port.
 - » The system will develop some underpressure at first and should normally start to repressurize to approx. 40 mbar (2/3 psi) (a different pressure is available at mobile magnets) after one day. Failure to do so may indicate a leak on the turret. If you suspect a leak on the HDCL port, remove the ring nut and check the condition of the copper seal to ensure that it is located correctly and shows an even ring indent on both sides where the knife edges cut into it. If the knife edges are damaged, contact the hotline; if they are ok, carefully reseal the port using a new copper gasket.
- Disconnect the current leads from the magnet (unlock and release the connections at the upper and lower magnet connectors).

NOTICE

The MPS cables should be connected to the MPS only when ramping the magnet.

- » Never connect the MPS cables during patient operation. All cables must be disconnected and removed from the examination room.

- Warm and dry the HDCL, and store it with the 1/3 psi valve in the transport box.
- Switch off the water pump with circuit breaker F1 of the refrigerator in the RCA.
- Reconnect the body coil to the cooling water and switch on the pump again (refer to (→ Fig. 109 Page 128)).

! CAUTION

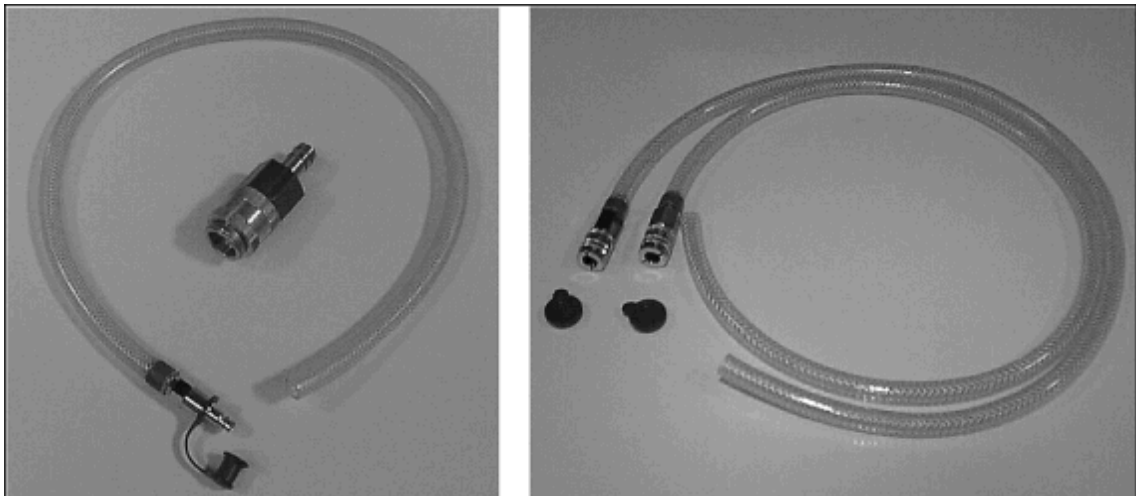
MPS transport at temperature below 0 °C

MPS can be damaged by frozen water.

- » Since the MPS may be transported at temperatures below 0 °C (especially during winter time), the water must always be emptied.

- Use the following plugs and hoses to drain the water hoses and the MPS. Blow out the water from the MPS.

Fig. 132: Connectors and hoses to empty the water



- Install the magnet covers (refer to (→ Fig. 100 Page 120)).
- Pack the MPS and all accessories into the transport boxes

NOTICE**Packing the MPS**

- » To avoid damage on the connectors of the control cable (connection control box to power box) when moving the MPS into the transport box we recommend to disconnect the control cable.



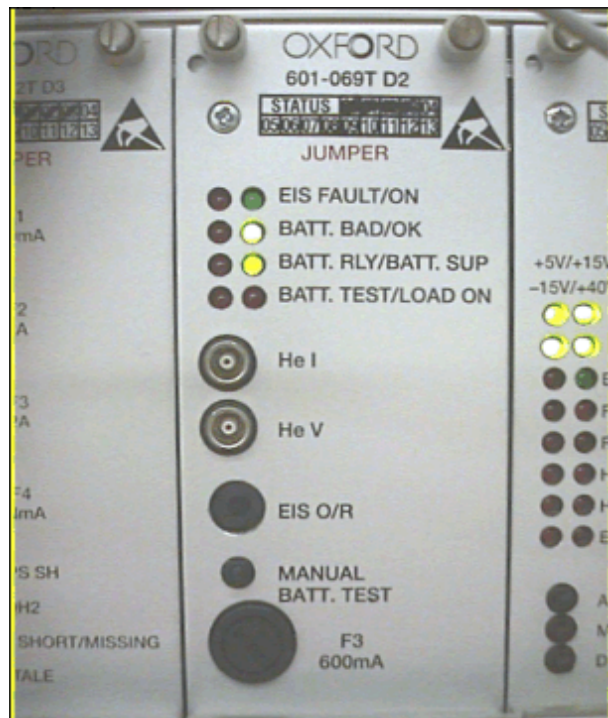
Check image orientation after ramping the magnet.

5.19 Appendix

5.19.1 Energizing the EIS circuit

After opening the turret port(s) for service, the system pressure can be restored more quickly by energizing the EIS circuit. It is essential that slight **positive pressure** exists in the system prior to performing any operations that require you to open any of the turret ports (e.g. insertion/removal of the HDCL).

Fig. 133: EIS circuit of D2



- Insert the shorting plug into D2 to energize the EIS circuit.
- The manual samples may be discontinued when sufficient pressure has been reached. The pressure should measure at least 0.3 psi (on the compound gauge this corresponds to the pointer being in the yellow sector and closer to the green than the red sector). When the system has reached sufficient pressure, it can then be depressurized again for insertion/removal of the HDCL etc., if necessary.

NOTICE

High boil off

- » The shorting plug should be removed when sufficient pressure has been reached.

After removing the shorting plug:

1. LED "EIS FAULT/ON goes out

2. LED "EIS" has to be reseted with the ACKNOWLEDGE button

NOTICE

Pressure has not started to rise after 2 hours

Check accessible seals with a leak detector gun or suitable leak detection solution. There may be a leak in one or more of the turret seals or the quench valve bursting disc/poppet seal may be broken.

- » If no leak is found, replace the bursting disk.

5.19.2 Ramping down the magnet



CAUTION

Cooling of the HDCL

Risk of a quench

- » The 1/3 psi valve indicated with a *, (→ Fig. 111 Page 131), must be completely defrosted prior to ramping. The DCL may be left venting through the 1/3 psi valve, if further ramping is needed. Otherwise remove the HDCL (refer to removing the demountable current lead). The HDCL vent MUST NOT be blanked with the magnet at field.

- The HDCL must be connected (refer to (→ Inserting the hardwired demountable current lead (HDCL) / Page 134)).
- Check the LHe level. It must be higher than 30%. ¹
- Turn the system on.
- Switch on circuit breaker F5 on top of the CCA and power on the MPS.
- Press the **"ON"** button.
 - The lamp in the ON button is illuminated.
 - The lamps in the "Run to Field" and "Run to Zero" buttons flash alternately.
- Set the magnet current (refer to the "Installation Protocol" in the Service Software), if not yet done.
- Press the **"Run to Zero"** button.
 - The lamp in the "Run to Zero" button is illuminated.
 - The Self Test LED > "ON" indicates that the self-test of the magnet electronics is in progress.

1. For more information, refer to the footnote "LHe Level" in the Appendix.



When ramping down the magnet, the magnet electronics runs a self-test and a cable test as well .

- Check the 1/3 psi valves



CAUTION

No sufficient cooling through the HDCL

Risk of a quench

- » If the 1/3 psi vent valves below the Quench valve frost up during self test, the ramp should be stopped. This indicates that there is not sufficient cooling through the HDCL. The 1/3 psi valves may need to be warmed up before continuing.

- After the self-tests have been completed:
 - LED **Self Test** > "OFF"
 - LED **Active** - > "ON"
- After the magnet has been ramped down: (wait approximately 3 min.)
 - LED **Active** - > "OFF"
 - LED **Finish** - > "ON"
- Press the OFF button and switch off the MPS.
- Switch off circuit breaker F5 on top of the CCA.



In the event of a power failure during magnet ramping, follow the procedure (→ Power failure during magnet ramping / Page 162)

5.19.3 Power failure during magnet ramping

A power failure when ramping the magnet up or down will cause the magnet power supply to trip out of RUN mode, closing the output shorting contactor. The magnet current will then run very slowly down through the magnet leads and shorting contactor.



WARNING

Current in leads

If not observed, death or serious physical injury can occur.

- » The MPS must remain connected to the HDCL during and following a power failure. The magnet leads and the shorting contactor are not indicated. Take no action until the power has returned and the following procedure is carried out.



After a power failure, there is NO indication on the power supply until mains power is restored.

When power returns, or after a momentary power failure, the red LED "Warning - Current in leads" will be illuminated as a warning that current is still flowing in the magnet, magnet leads and shorting contactor circuit.

- Press S1 and confirm that mains power has returned (the green lamp in S1 should be illuminated).
- Check that the MPS has powered up in the OFF state by verifying that the LED in the **ON** button is extinguished.
- Check whether the red LED "Warning - Current in leads" is illuminated or not.
 - If it is not illuminated, it is safe to switch off the MPS and circuit breaker F5 and proceed with removal of the HDCL.
- If it is illuminated, press the **ON** button on the MPS.
- Then select **Run to Zero** or **Run to Field** to continue ramping under MPS control.
- Wait until the green LED on the MPS illuminates in **Finish**, then press the **OFF** button.
- Check that the red LED "Warning - Current in leads" is not illuminated before switching off the MPS and circuit breaker F5.
- It is now safe to remove the HDCL and disconnect the magnet leads (follow the procedure in (→ Removing the hardwired demountable current lead (HDCL) / Page 155)).

NOTICE

Removing the HDCL

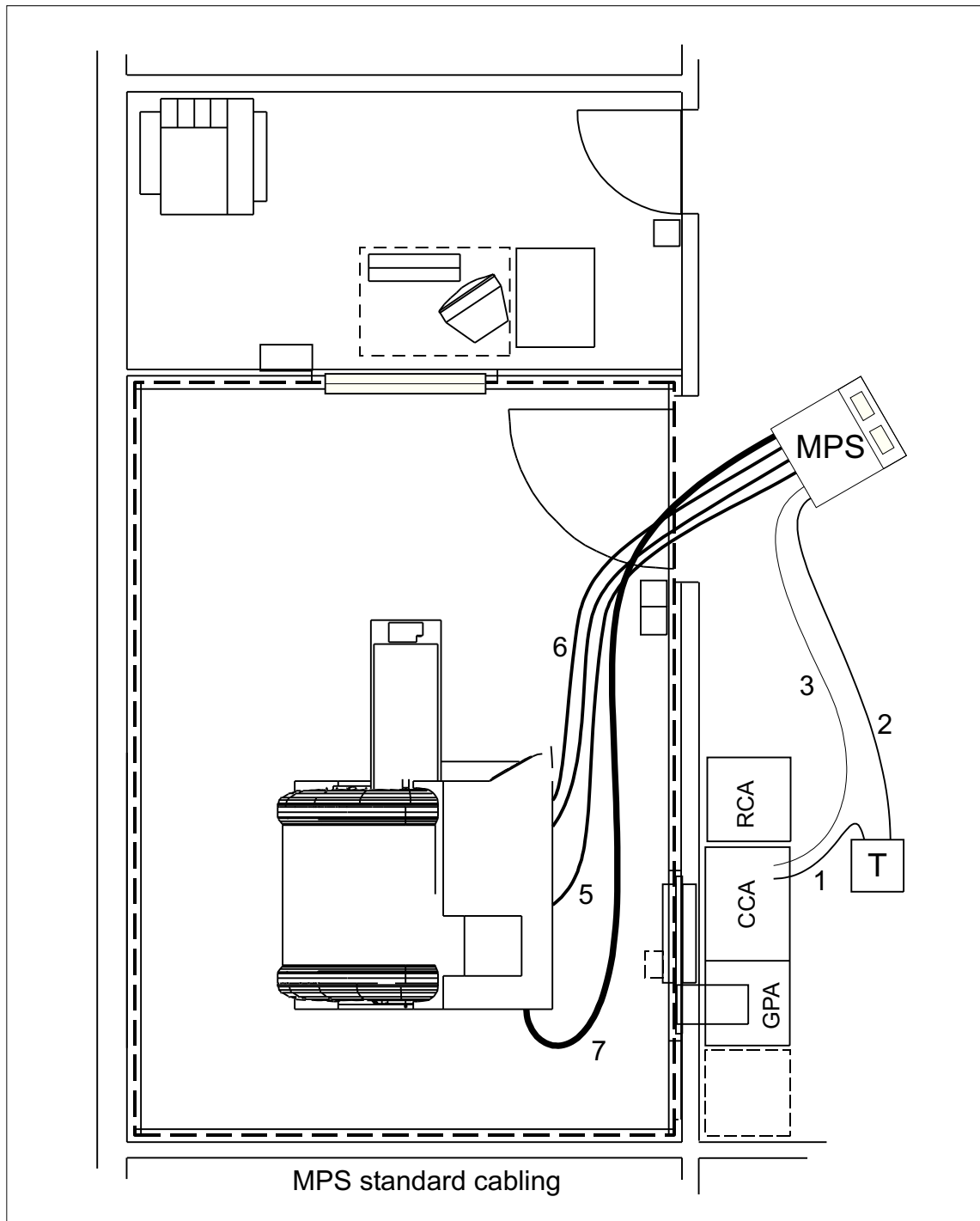
- » The MPS must remain connected until after the HDCL has been removed. The HDCL vent port **MUST NOT** be blanked.

5.19.4 MPS cabling

5.19.4.1 Standard cabling

- If any of the cables or hoses delivered with the MPS are too short, try one of the alternative cabling diagrams as shown on the following pages.

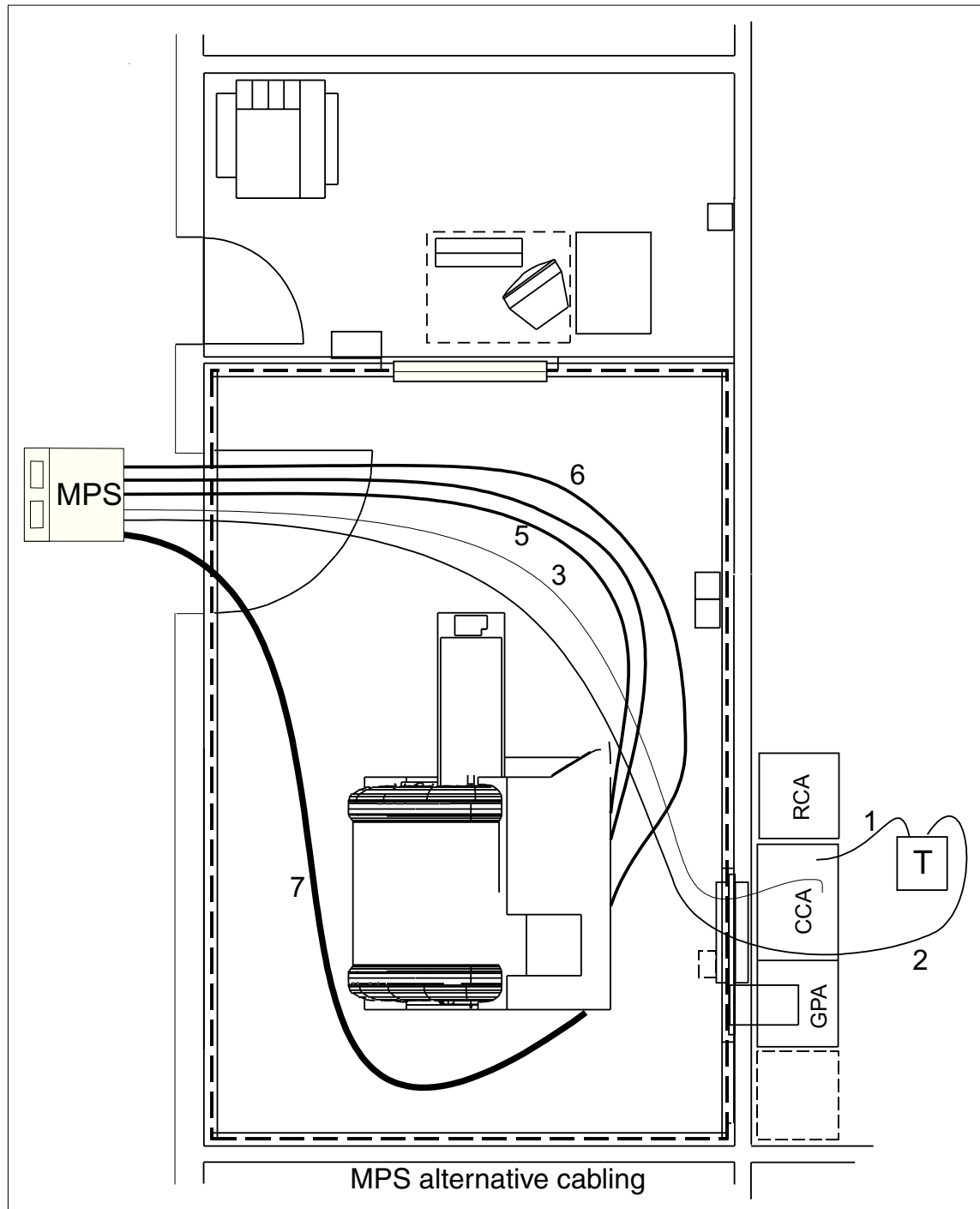
Fig. 134: Standard cabling of the MPS



5.19.4.2 Alternative MPS cabling 1

- Open a blind plate in the filterplate and feed through the cables as shown in (→ Fig. 135 Page 165).

Fig. 135: Alternative cabling of the MPS through the filter-plate



5.19.5 Footnotes

The following footnotes apply to the entire chapter.

LHe Level

The minimum operating LHe level is 30 %.

Normal ramping up the magnet should not be performed below a level of 50%.



CAUTION

Helium level

Magnet can quench

- » Ensure that the helium level is higher than 50% (initially ramping 70%) before ramping up the magnet.

Ramping down the magnet is permitted even at levels below 50%.



After ramping the magnet the reservoir for water of condensation should be drained

5.19.6 Adding Citric Acid to the Secondary Water Circuit

(Does not apply to Allegra and Trio)

After initial shimming procedure (part of start up) has been completed, citric acid must be added to the water circuit (citric acid powder is part of the delivery volume).

The purpose of adding citric acid is to prevent germ build-up in the water circuit.

The citric acid will be delivered from end of august. It is part of the system components.



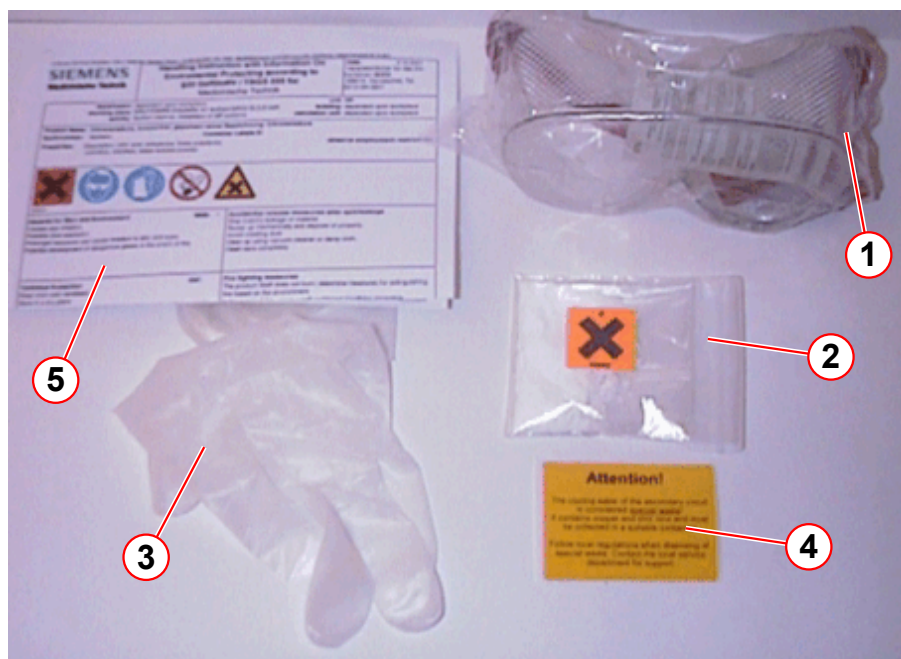
CAUTION

Adding Citric Acid

The citric acid can cause irritation to skin and eyes

- » Avoid contact with eyes and skin Wear the goggles and gloves provided when handling citric acid. Follow the safety instructions in the enclosed "Handling Instruction" (→ 5/Fig. 136 Page 167).

Fig. 136: Citric acid set



- (1) Goggles
- (2) Citric acid powder, 8 grams
- (3) Gloves
- (4) Warning label
- (5) Handling instruction



Two sets of the Citric Acid are included in the delivery.
For single cooler RCA only one set is used.

- Dissolve 8 grams of citric acid in 250 milliliters of distilled water **per cooler**.

Fig. 137: Dissolving the citric acid



- Add the citric acid solution to the cooling circuit.

Fig. 138: Adding the citric acid solution



- Affix the warning label provided to the cooler.

Fig. 139: Warning level



(1) Position for the warning label



CAUTION

Adding citric acid

The citric acid can cause irritation to skin and eyes

» Wash hands thoroughly after handling .



The coolant turns turquoise after adding the citric acid.



Cooling water containing citric acid that was collected while draining the MPS should be refilled into the secondary cooling circuit.



Follow the country-specific guidelines when disposing of cooling water containing citric acid.



The acidity of the citric acid is neutralized after it has been dissolved in water for several days.

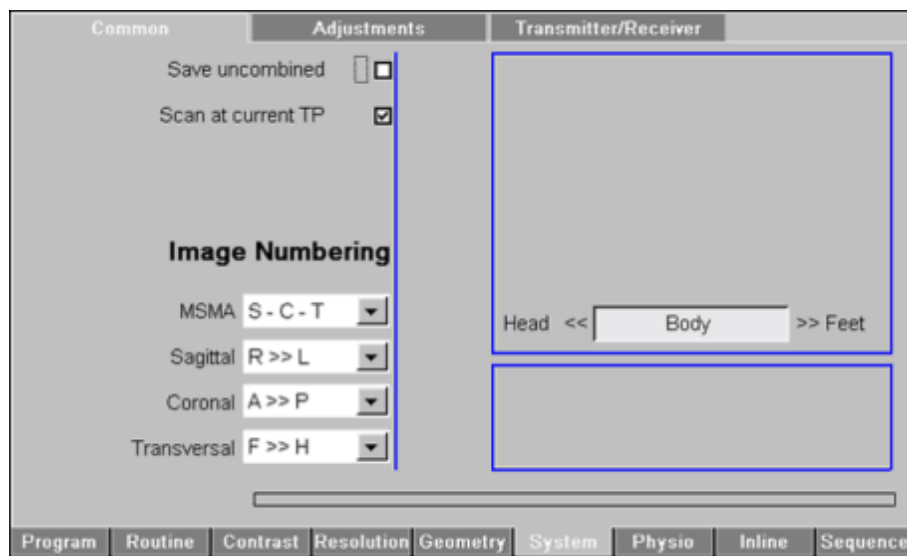
5.19.7 What to do if frequency adjustment failed



This procedure is valid for local coil and the Array shim device!

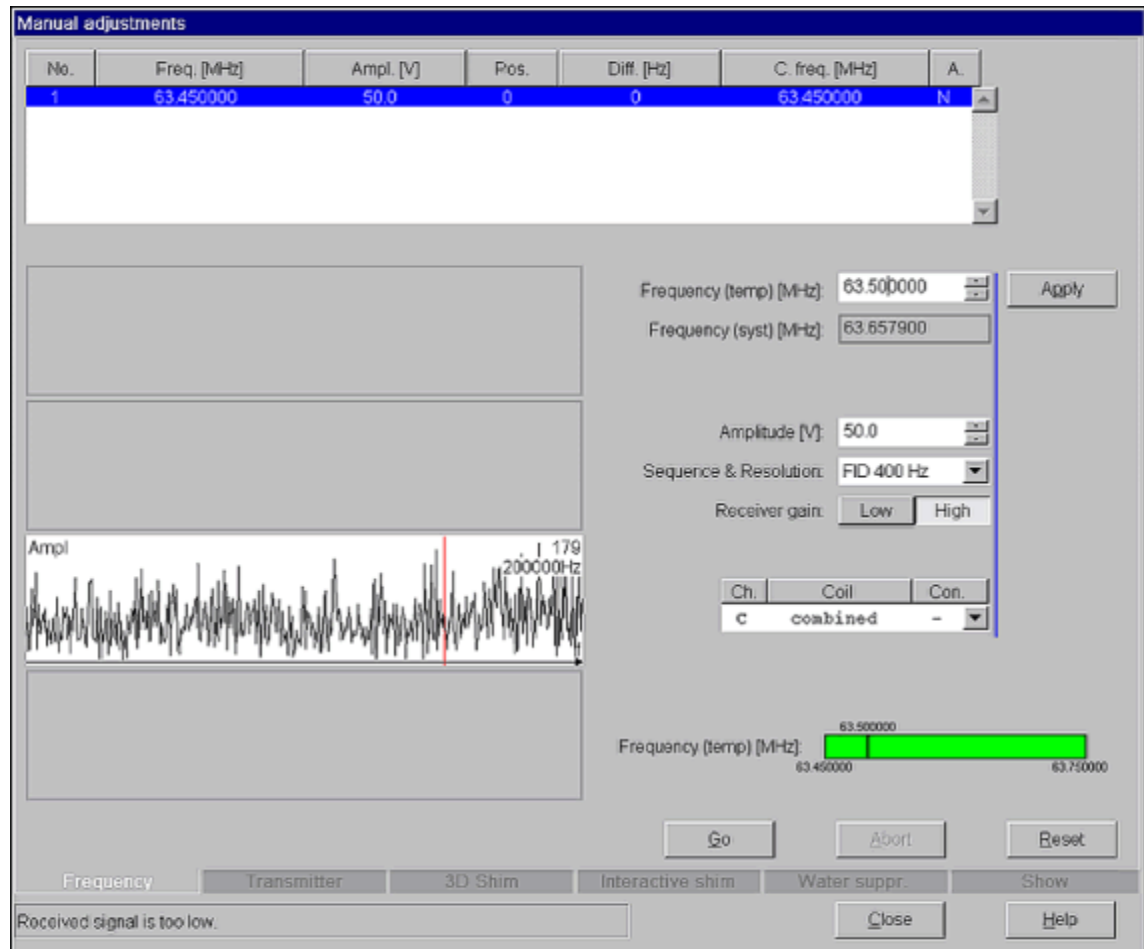
- Register a Patient
- Open the protocol "se_15b130"
- Check if "Scan at current TP" is active

Fig. 140: Manual Adjustments



- Open manual adjustments

Fig. 141: Manual adjustments



- Enter frequency (temp) manually, start from the lowest frequency 63.450 000 MHz
- Select **FID 400Hz** of Sequence and Resolution



The FID 400 Hz uses a frequency bandwidth of +/- 100kHz.

- Click **Go** - button
- Continue in 50 kHz steps from lowest to the highest frequency.
 - » If necessary calculate the new magnet current and ramp the magnet to the required frequency

6.1 Control panel

Two control panels allow you to control the table from either side. However, only the left control panel contains the controls that allow you to adjust the volume for the music.

Fig. 142: Left control panel



No.	Function
1	Patient air (four steps)
2	Tunnel light (four steps)
3	Table STOP
4	Move out table completely
5	Isocenter table
6	Laser light localizer
7	Interrupt measurement
8	Start / Stop measurement
9	Move table up/in or out/down
10	loudspeaker - volume control
11	headphones - volume control

6.2 Converting the table

- » This section describes how to convert a removable table. It is not required when the system is equipped with a standard patient table.

The trolley bracket must be located on the door side of the patient table, so that the trolley can be easily docked. If this is already the case on your system, disregard the following procedure. Check only whether the tabletop and the trolley interconnection is functioning.

- Remove the cover on the foot-end of the table (two M6x12 Allen screws). Pull down and toward the back at the same time to pull the cover off.



Be careful not to bend the light barrier when removing the cover.

Fig. 143: Cover on the foot-end of the table

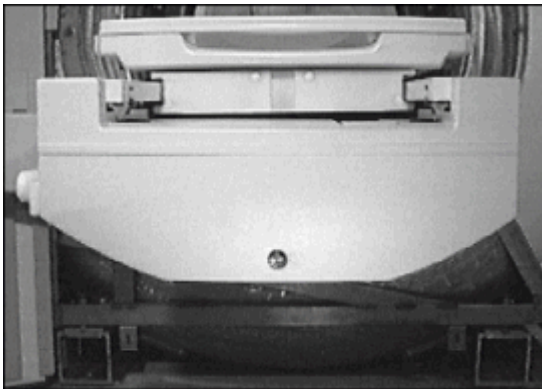
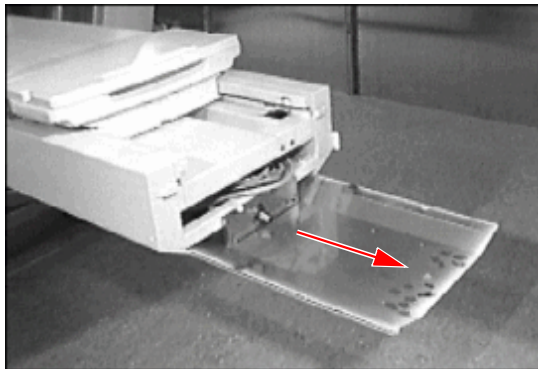
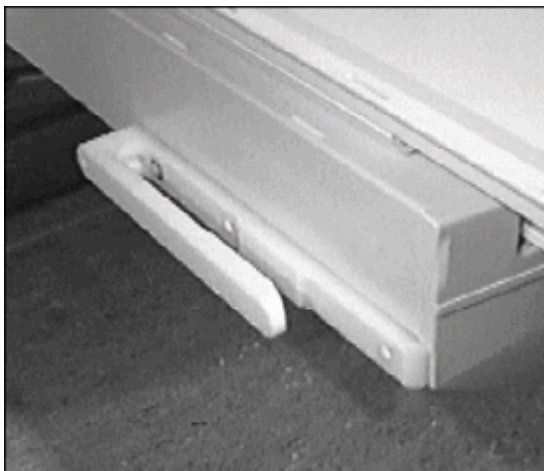


Fig. 144: Lower table panel



- Pull the lower table panel out toward the foot-end.

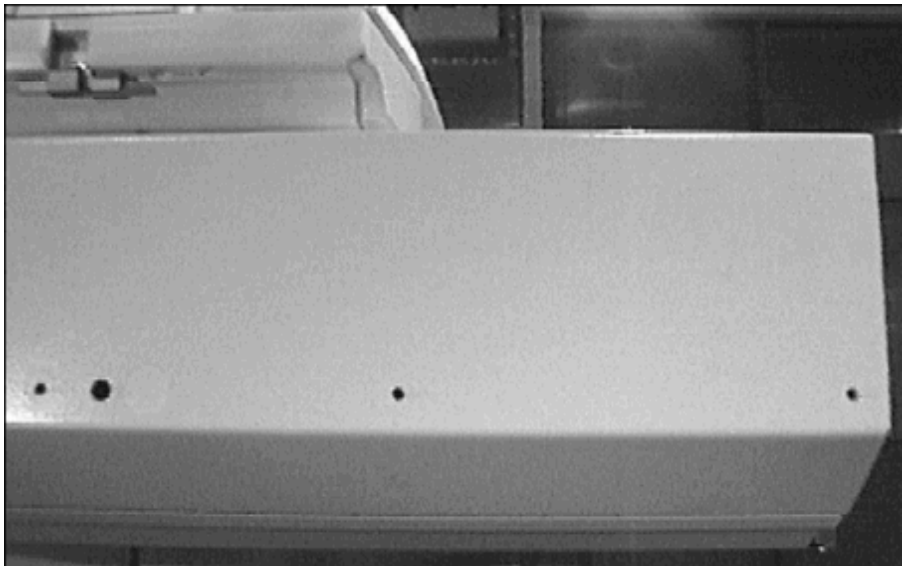
Fig. 145: Bracket (sausage pliers)



- Remove the three plugs and remove the bracket (three M6x20 Allen screws).

- Remove the four plugs on the other side of the table.

Fig. 146: Installation holes for bracket



- Mount the bracket on the other side of the table. Use Loctite to secure the screws.
- Reinstall all seven plugs.
- Remove the lateral light barrier (SZ12) in the table.
- Mount the light barrier on the opposite side of the table.



Center the light barrier to the hole (visually check from the outside) to assure accurate function. The light barrier should detect the trolley when it is inserted at a distance of 3 mm before the mechanical end stop (indicated by the light barrier LED).

Fig. 147: Light barrier

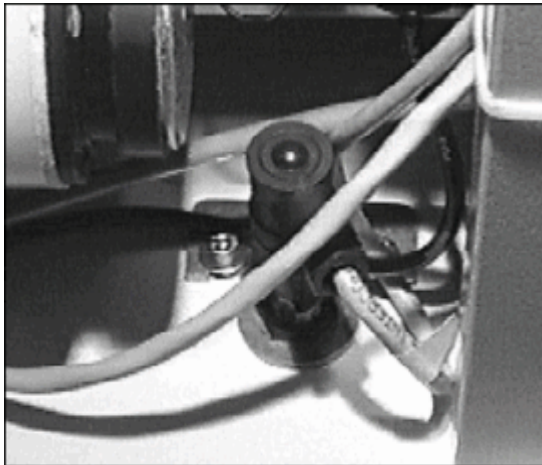
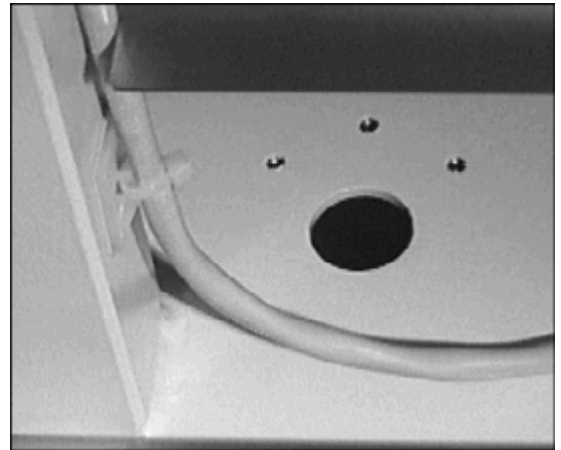


Fig. 148: Mounting holes for the light barrier on the opposite table side



Secure the light barrier cable with cable ties so that it does not contact the tabletop drive belt.

- Push the table panel back in.
- Secure the cover on the foot-end.



Make sure you insert a dummy plug at X28 of the table electronics panel if there is no third operating panel installed.



The intercom cable must be connected at X23 of the table electronics panel so that the table functions can be tested.

6.3 Converting the trolley

- Place the trolley on its side with the railing facing up and carefully remove the plugs with a sharp knife or screwdriver (arrows).

Fig. 149: Remove the white hole plugs

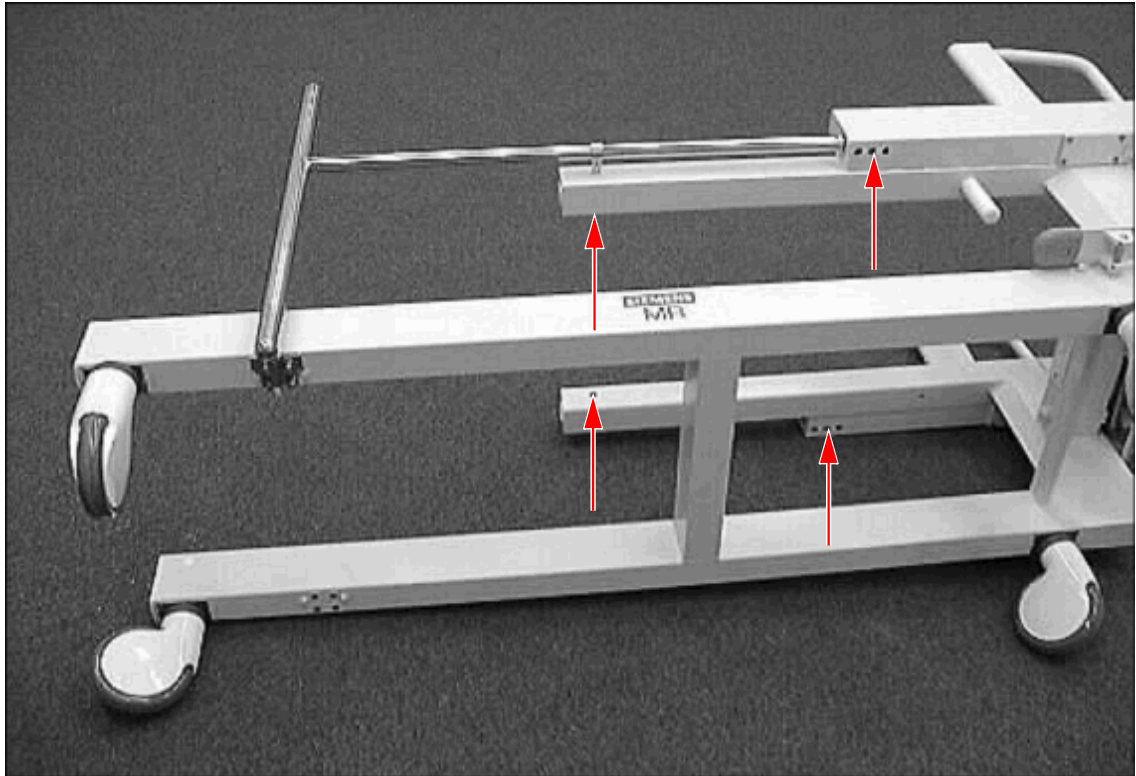
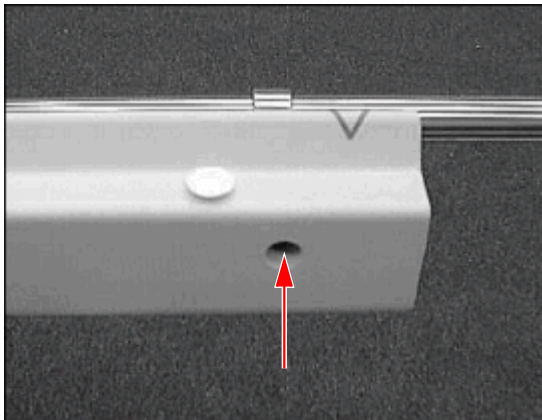
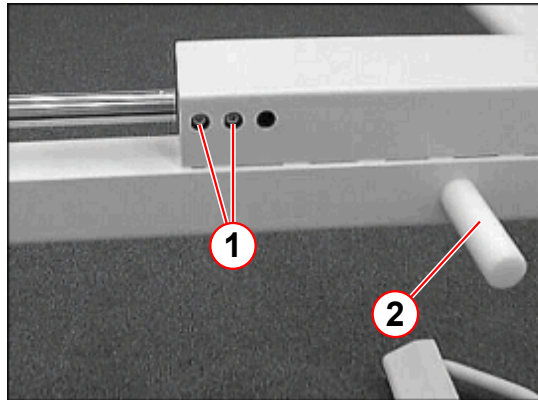


Fig. 150: Allen screw in the center railing



- Remove the center M8x12 Allen screw.

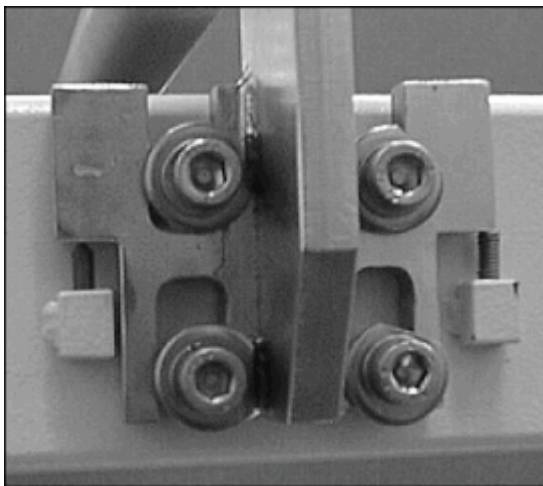
Fig. 151: Plastic lever



- (1) M6 Allen screws
- (2) Plastic lever

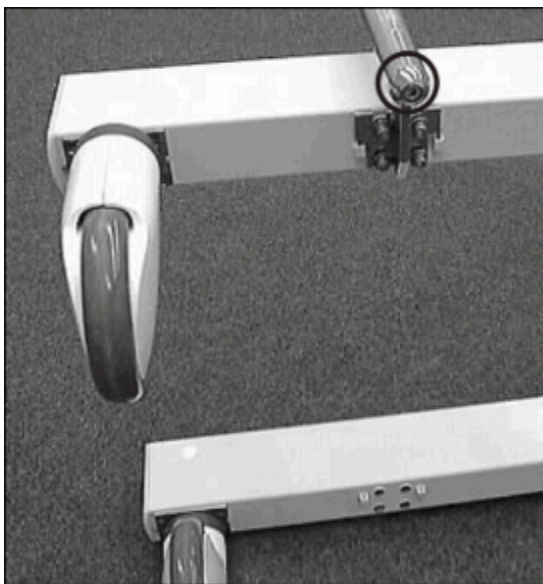
- Remove the two M6 Allen screws (→ 1/Fig. 151 Page 177).
- Unscrew the plastic lever (→ 2/Fig. 151 Page 177) and mount it on the other side of the trolley, securing it with Loctite.

Fig. 152: Lower railing bracket



- Remove the four lower M6x30 Allen screws.

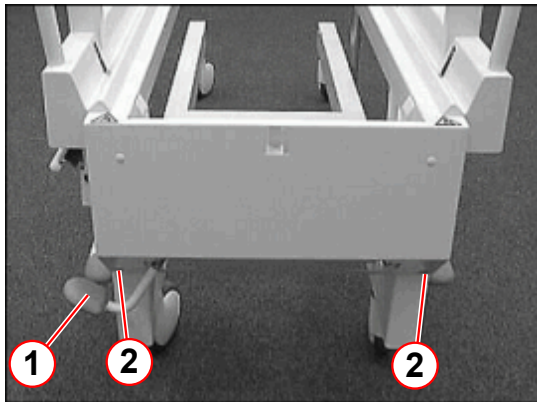
Fig. 153: Rotate the mounting



- Rotate the lower mounting (circled) by 180° using a 6 mm Allen wrench.

- Pull the railing out and insert it on the other side.
- Reinstall all the screws in the reverse order.

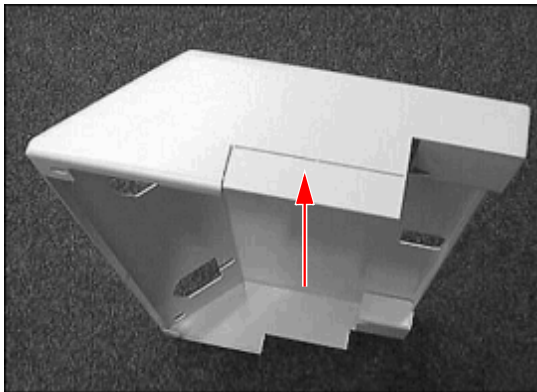
Fig. 154: Disassemble foot lever and cover



- (1) Foot lever
(2) M6x20 Allen screw

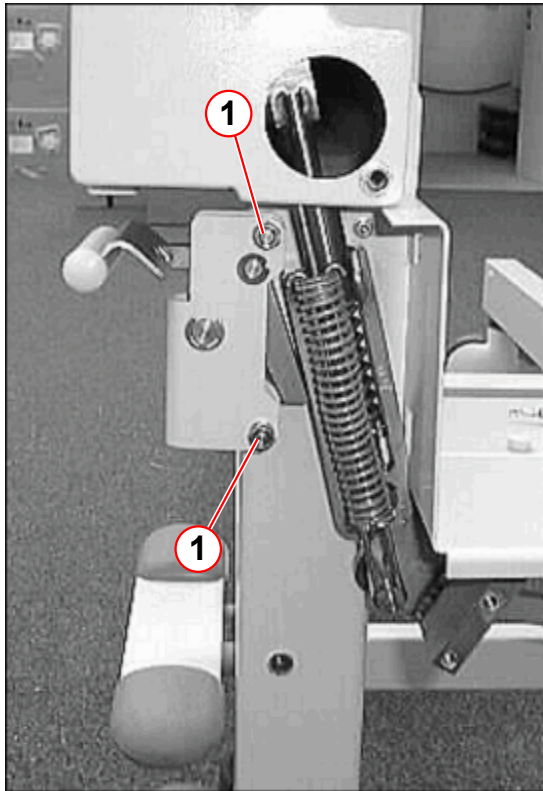
- Disassemble the foot lever (two M6 x 20 Allen screws)
- Remove the foot-end cover. Remove the two M6 x 20 Allen screws (→ 2/Fig. 154 Page 178) and loosen both upper M6 x 20 Allen screws.

Fig. 155: Break-out metal piece at foot-end cover



- Remove the break-out piece and smooth the edges with a file.

Fig. 156: Manual locking lever



(1) M6 nuts and washers

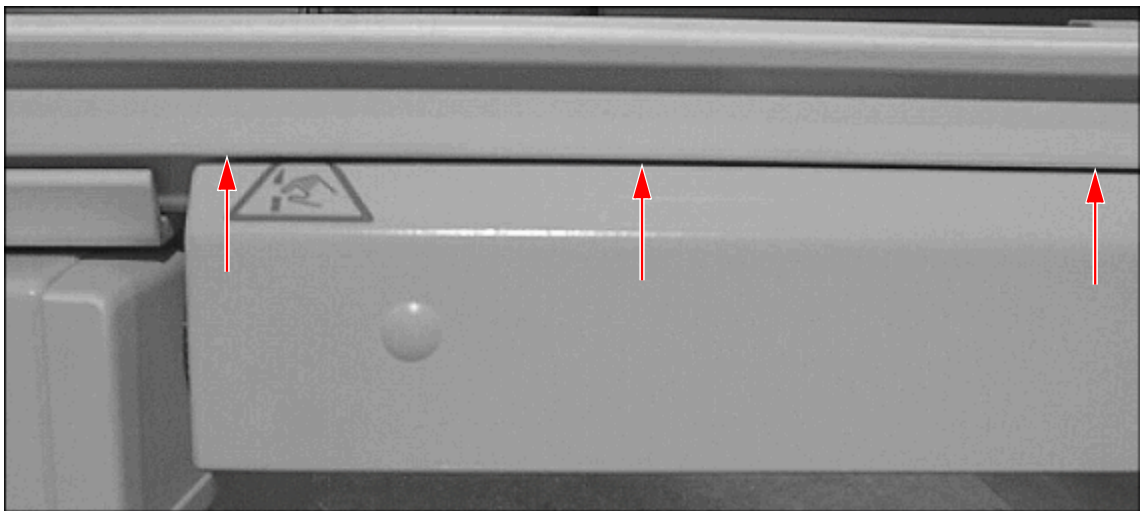
- Remove the two M6 nuts and washers (→ 1/ Fig. 156 Page 179) of the manual locking lever.

- Mount the lever on the other side of the trolley. Use Loctite to secure the threads.
- Install the cover.
- Mount the foot lever on the same side as the manual locking lever. Use Loctite to secure the threads.

6.4 Testing docking

- Move the trolley underneath the table.
- Move the tabletop all the way out of the magnet.
- Place a weight of approximately 80 kg on the table (use all the phantoms or a volunteer).
- Make sure that the tabletop is parallel to the trolley (equal distance - red arrows, (→ Fig. 157 Page 180)) when the table has been lowered.
 - » Otherwise, the docking height must be adjusted as described (→ Adjusting the docking height / Page 181).
 - » If the docking height is correct, the patient table is complete.

Fig. 157: Red arrows indicate equal distance with the table lowered onto the trolley



6.5 Adjusting the docking height

6.5.1 Deinstalling the front cover



If the table is not in the electronic travel range, any attempt to move the table will result in a software block. A software block can only be cancelled by switching the table electronics off and subsequently performing a software reload after the travel range has been reentered by hand crank.

- Remove the trolley.
- Remove the side cover of the magnet.
- Loosen and pull back the PMU display (if applicable).
- Move the table to its uppermost position.
- Using the hand crank, crank the table further up until it reaches the mechanical end stop.
- Remove both guide cover plates from the table support arm slots.
- Remove both retaining plates and the decorative blind (caution - spring is under tension) underneath the support arm and roll up the blind.
- Using the hand crank, crank the table back into the electronic travel range.
- Move the table into its lowest position.
- Using the hand crank, crank the table further down until it reaches the mechanical end stop.
- Remove the three Allen screws on the front cover.
- Remove the hand crank.
- Rotate the adjustment wheel to raise the cover.
- Remove the cover (lift it up off the support arm).
- Crank the table up until it is within the electronic travel range.
- Move the table up with the controls to the uppermost position.

6.5.2 Adjusting the docking height

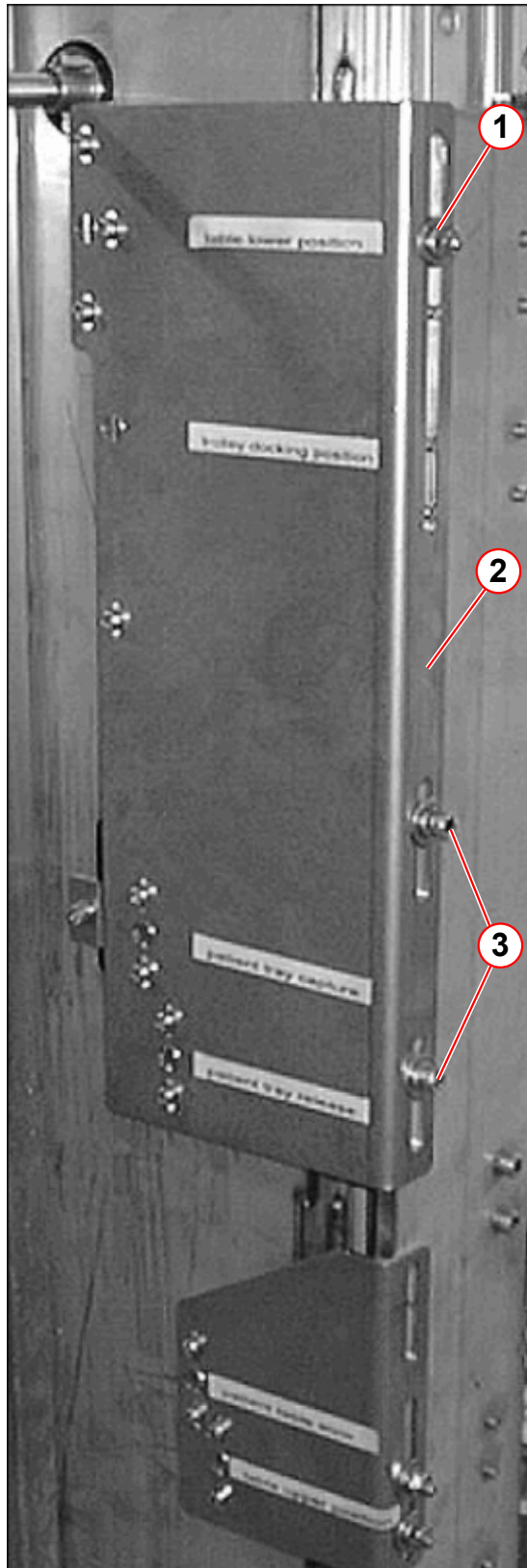
- Position the trolley under the table. Activate the foot brake.
- Place a weight of approximately 80 kg on the table (use all the phantoms or a volunteer).
- Move the table down until it stops above the trolley.
 - » The space between the table and the trolley (→ Fig. 157 Page 180) must be uniform. You must be able to lock in the tabletop using the foot lever from this position (both latches snap in).

If this is not the case, adjust as follows:

- Move the table up.

- Loosen the upper cam plate (refer to (→ 1, 3/ Fig. 158 Page 183). Move the cam plate to its lowest position (this is equivalent to the highest position for the table) and re-tighten it.

Fig. 158: Cam plate for table height adjustment



- (1) M6 nut
- (2) Upper cam plate
- (3) M6 Allen screw

- Move the table back down and measure the distance between the trolley and the tabletop.
- Move the table back up.
- Move the large cam plate upward by the amount measured.
- Move the table back down and check the distance again (refer to (→ 1, 3/ Fig. 157 Page 180)).
 - » The tabletop must now lie flush on the trolley, must be able to be raised and the trolley locking mechanism must be functioning correctly, that is, the table can be moved downward only.

6.5.3 Testing docking after adjustments

- Lock in the tabletop using the foot lever on the trolley.
- Move the table into its lowest position.
- As a test, remove the trolley with the weighted tabletop.
- Move the trolley back to the table again (align it and activate the foot brake).
- Move the table up as far as possible (it should stop automatically at docking height).
- Unlock the trolley.
- Move the tabletop further up and remove the trolley.
 - » As long as the transfer of the tabletop to the trolley and back functions correctly, the adjustment is complete.



If a heavy patient is put on the table, the foot end of the table might be "lifted". The electrical connection on the foot end can be lost. In that case a fine adjustment of the SY4 and SY3 will be necessary.

6.5.4 Fine adjustment of docking height

- To check the proper connection of the foot end connector (ASV=self-aligning plug assembly), if present remove foot end cover.
- Move the table **down** and adjust SY 4 (cam "patient tray capture", located on the upper cam plate) so that the table and trolley are touching and parallel.
- Check the proper connection at the foot end connector (ASV).
- If SY 4 needs to be readjusted, move the table up adjust SY 4 and recheck by moving the table **down**.
- If SY4 adjustment satisfied, keep moving the table down couple centimeters past the previously adjusted position.
- Move the table **up** and adjust SY3 (cam "patient tray release", located on the upper cam plate) so that the table and trolley are touching and parallel.
- If SY 3 needs to be readjusted, move the table down adjust SY 4 and recheck by moving the table up.
- Check the proper connection at the foot end connector (ASV).

For really heavy patients we suggest to readjust the SY4 cam so that the table would stop a couple millimeters earlier (higher) and readjust SY3 cam for the same position.

6.5.5 Reinstalling the front cover

Reinstall the front cover in the reverse order.

- Move the table to its lowest position.
- Using the hand crank, crank the table down until it reaches the mechanical end stop.
- Lift the cover over the table support arm into position.
- Rotate the adjustment wheel to lower the cover. Make sure that the cover is not resting on the kick plate.
- Attach the three Allen screws of the front cover.
- Crank the table up into the electronic travel range.
- Move the table back up into its uppermost position.
- Using the hand crank, crank the table up until it reaches the mechanical end stop.
- Attach both cover plates for the table support arm slots.
- Attach the decorative blind (the spring must be under tension - 40 revolutions) with both retaining plates underneath the table support arm.
- Using the hand crank, crank the table until it reaches the electronic travel range.
- Reattach (if applicable) the PMU display. (Before reattaching, switch off F15 .) Align the display with the front cover.
- Reattach the side cover of the magnet.
- Perform a final function test of the table.

7.1 General



Valid for Germany as well as some EU countries.

If the monitor of an MR-system is used for medical reporting a special acceptance test according to DIN V6868-57 is required. The monitor has to be labeled as such by the installation personnel.

According to the documentation listed below:

1. Acceptance testing for image display devices (→ Acceptance testing for image display devices / MR00-000.820.01).
2. Acceptance test protocol for image display devices. (→ Acceptance test protocol for image display devices / MR00-000.820.02)



The monitor only needs to be checked (and adjusted as necessary) if at least one of the parts listed below is replaced.

- Monitor
- Graphic board
- PC

Refer to (→ Host / Imager / MR-000.841.12)"Replacement of parts"

8.1 Installation



CAUTION

Always switch the MR system off before performing service work on the intercom.

- » The intercom could be damaged if cable connections are connected or disconnected while the intercom is switched on.

The following intercom cable connections must exist to operate the table:

- Cable W140: X25.D9.K2209-X2.D15-4.Filterplate
- Cable W 58: X1.D15-4.Filterplate-X11.ICU (Intercom Central Unit)
- X12.ICU-IUI (Intercom User Interface)
- Carefully turn the two potentiometers marked R1 and R2 located on the front plate of the new central unit (p.n. 4757204) in a clockwise direction until the end position is reached. This sets the microphones in the operating unit (p.n. 4757261) and the RF room to maximum sensitivity.
- Attach the **ICU** (Intercom Central Unit) central unit with the enclosed screws to the back wall of the control console (with threaded inserts) as shown in (→ Fig. 159 Page 187)

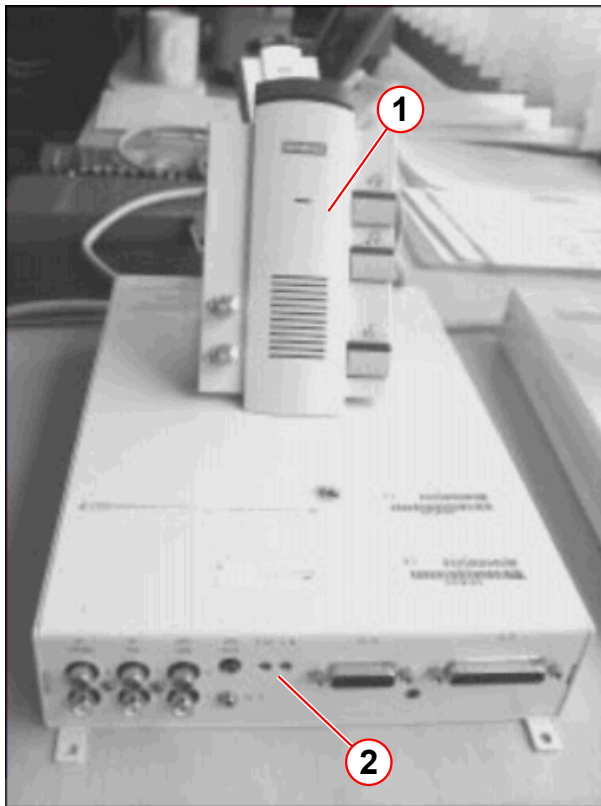
Fig. 159: ICU installed at back of MRC (table optional)



- Place the control unit **IUI** (Intercom User Interface) next to the MRC monitor on the control console.

- Connect the cable to the **IUI** at the 15 pin socket connection X12 of the ICU. Lock the connector with the locking screws.

Fig. 160: Old and new intercom models



- (1) New Operating Unit p.n. 4757261
(2) Two potentiometers on the new Central Unit p.n. 4757204

8.2 Start-up

- Switch on and boot the system.
- Switch the operating unit (p.n. 4757261) to position "CBT". This switch is located on the left side of the operating unit.
- Close the patient application (Patient/ Close Patient) in case patient is active.
- Press <Ctrl> and <ESC> and select Programs/ Accessories/ Advanced User and enter the password: meduser1. Click <OK> to exit.
- Press <Ctrl> and <ESC> and select Programs/ Accessories/ Multimedia/ Volume Control. Set all six regulators in the volume control window to maximum (upper position), as shown in (→ Fig. 161 Page 189). Deactivate the four controls (CD Audio Bal., Synt., AuxB, ILS) on the right side of the window by clicking mute. "Mute" does not have to be activated for the two controls on the left side (Vol. Contr., Wave Output). In case the volume control menu has a different design than the figure below, set the "Volume Control" and wave output to maximum, and set all other controls to "Mute". Select Options/ Properties in the same window and click <Recording>. Press <OK> and set the regulators for the "Recording Control" as shown in (→ Fig. 162 Page 190). Set the recording control volume to approx. 25% and the volume for the other four regulators to approx. 50%. Click <Select> for the microphone. In case the recording control menu has a different design than the figure below, set the volume to approx. 25% and all others to 50%. To select the microphone, click <Select> with all others deselected, as shown in the example below. Refer to online help, if one of the controls for the device is missing. Close the window via Options/Exit.

Fig. 161: Volume Control

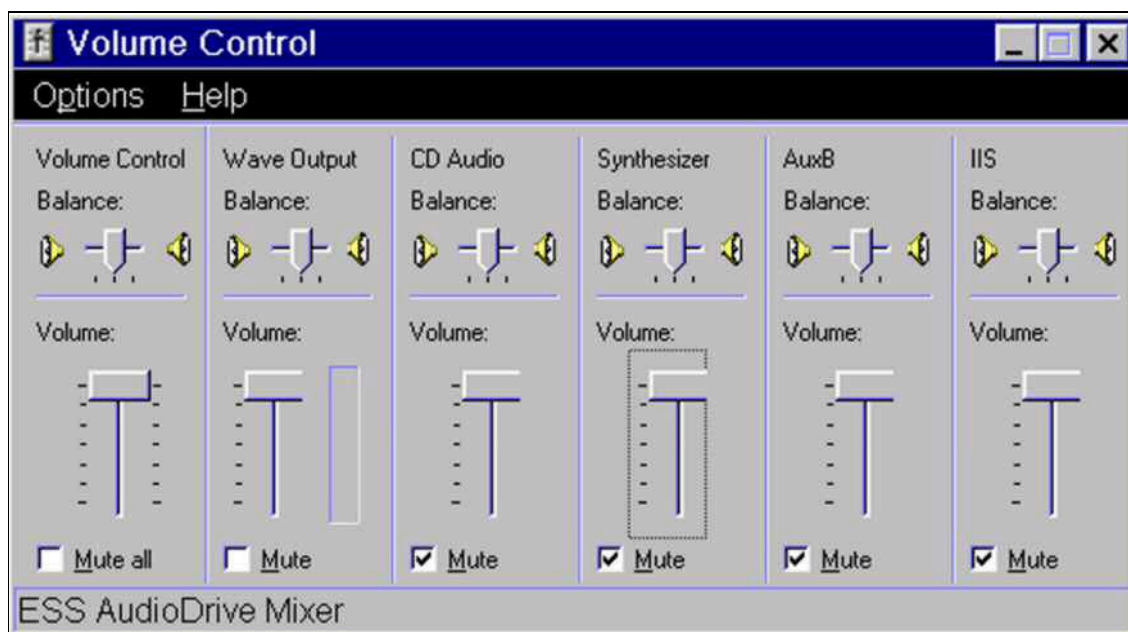
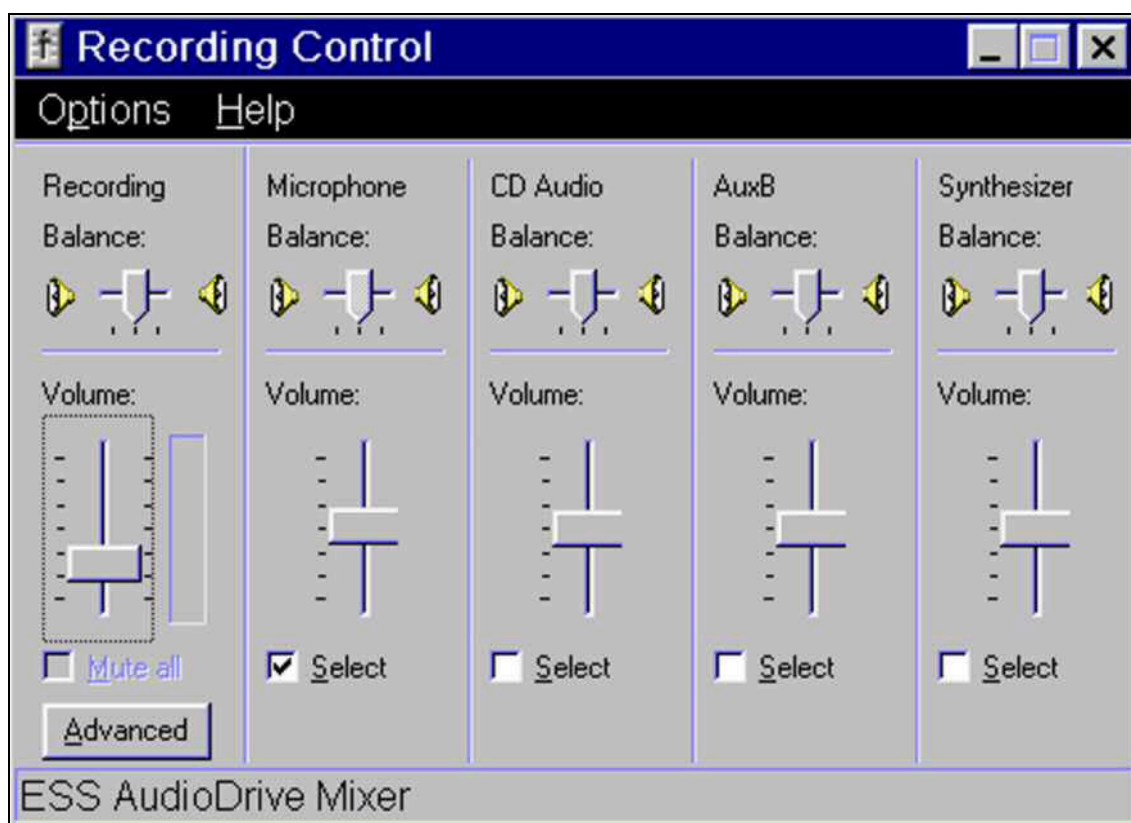


Fig. 162: Recording Control

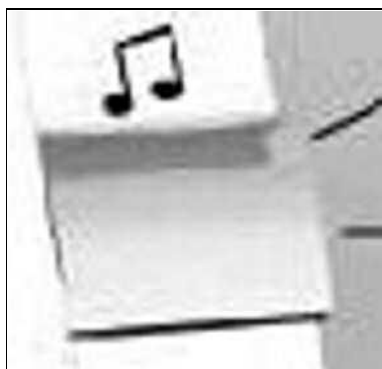


- Setting the computer voice (CV) volume: Select Options/ Configuration/ ExamUI and set the volume at approx. 70% of maximum. To save the settings and exit the windows, select <Apply> and <OK>.
- Setting the music volume, i.e. CD player: Turn both potentiometers on the left patient table control in clockwise direction to maximum. Switch on the music source, i.e. CD player, and adjust the output volume control on the music source to a volume just below the level when the music starts to sound distorted (open the door to the RF room to listen). Keep this setting for the music source and adjust the two potentiometers on the left patient table control panel to a comfortable volume level (depending on the customer) for the speaker and the head phones.

8.3 Final Check

- Perform a final check for the voice output (new option as of VA21A). Close the patient application (Patient/ Close Patient) in case Patient is active. Select Queue/ Voice Output/ Organize... and click <New...>. Enter a name, e.g. "test", and select <OK> and click the ● button. Press the lower right-hand button on the intercom's operating unit and keep it pressed during this procedure. Click the ● button in the record voice window. Speak the command into the operating unit (keep a distance of approx. 40 cm). After finishing the text, click the ■ button in the record voice window and release the lower right-hand button on the operating unit. The voice recording is now finished. To check the recording, switch the operating unit from CBT to CV and click the > button. The recorded test command should be complete. A snapping sound at the beginning of the recording means that the procedure has been performed incorrectly. Compare the volume of your test recording with that of a default recording; both recordings should have just about the same sound level. If there is a difference in levels between the two recordings, readjust the volume of the recording control. Refer to the online help for more information regarding this new option. Close the voice output windows.
- The potentiometers for the operating unit (p.n. 4757261) affect the volume of the computer voice (CV) as well as the voice commands for the operator. The potentiometers on the operating unit do not require a basic volume adjustment, since these are set individually by the user. To select a starting value, turn the two potentiometers on the operating unit to approx. 50% of the volume. The user decides on an individual basis whether to use music or computer voice. The computer voice should not be used at the same time as music, because it may be difficult for the patient to understand the computer voice commands.
- If the customer prefers music: Switch the operating unit to "CBT" and press the play music button located in the center of the right side (music active = switch light on) as shown in (→ Fig. 163 Page 191).
- If the customer prefers the computer voice (CV): Switch the operating unit to "CV" and switch off the play music button in the center of the right side (music deactivated = switch light off). Notice: For computer voice recordings, the operating unit has to be switched back to "CBT". After the recording is complete, switch back to "CV".

Fig. 163: Music switch



8.4 Customer Information

Inform the customer about this new option by explaining how to record automatic voice output and what to keep in mind when using this new option.

9.1 Tune-up



There is no written document for the Tune-up. Use Seso Online Help for detailed information.

- Perform the Tune-up of the system (Local Service / Tune-up). Most Tune-up steps are performed fully automatic. Follow the instructions on the screen. Popup windows will inform you about necessary actions. Please consult SeSo Online Help for detailed information.

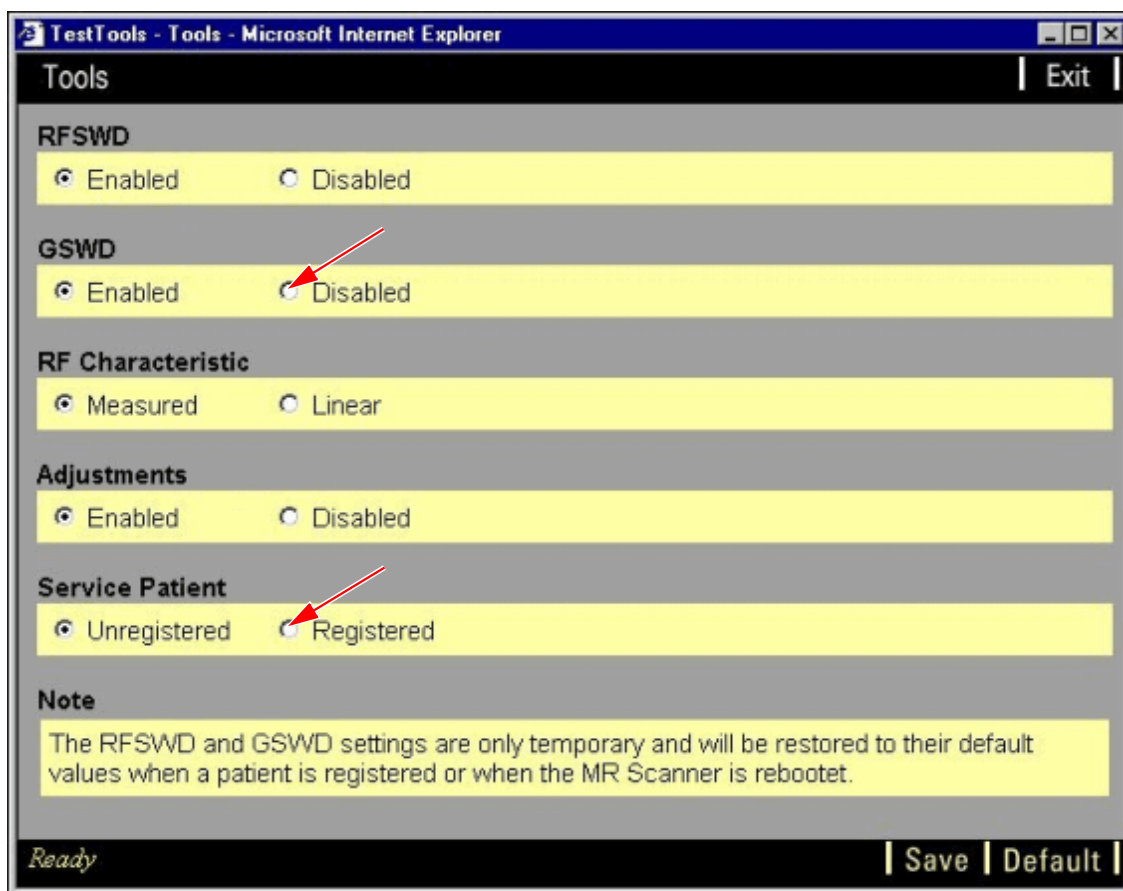
9.2 Gradient Performance Check

The tests described in this section are designed to make sure the gradient final stages of the Cascade Gradient Cabinet K2117 (used in Magnetom Harmony / Symphony, Sonata, Trio) are performing as specified.

9.2.1 Gradient Risetime Check

- Position the Spherical Body Phantom and Loader.
- Open the "Local Service" platform.
- Select "Test Tools / Tools".

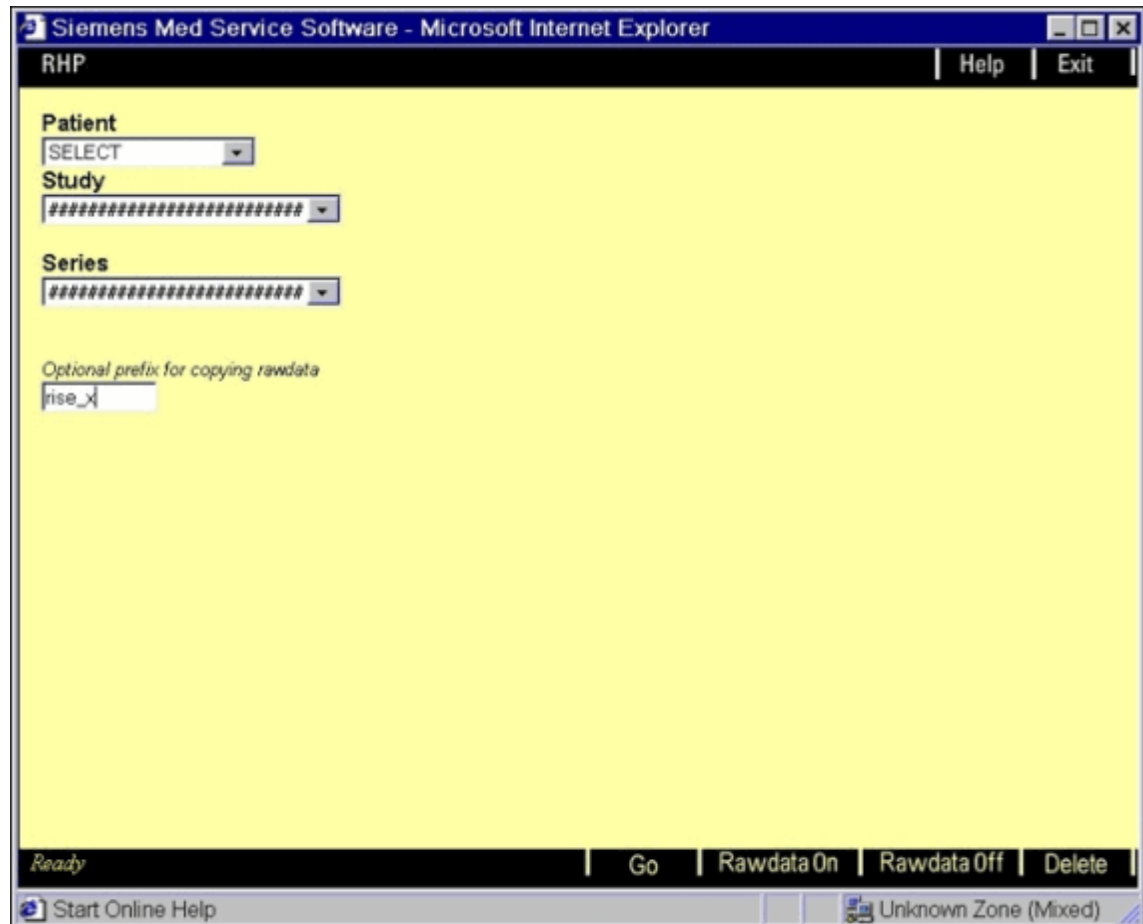
Fig. 164:



- Select "GSWD Disabled" and "Service Patient Registered"
"Save" this selection and "Exit" the platform.

- Select "Test Tools / RHP".

Fig. 165:



- Enter a filename for the rawdata (e. g. rise_x)
- Open the "Examination" task card.
- Open the "Examination Explorer" and select the sequence SIEMENS - Sequence region - Service Sequences - Default Protocols - grad_risetime.
- Copy the sequence into the examination queue (drag and drop).
- Select the orientation in the routine card according to the axis to be measured. The axis to be measured must be set as read out gradient
 - X axis: orientation Transverse, phase encoding normal (A>>P)
 - Y axis: orientation Transverse, phase encoding swapped (R>>L)
 - Z axis: orientation Sagittal, phase encoding normal (A>>P)
- Open the "Contrast Card".
- The parameter TI(2) must be toggled to "1".

TI(2) -> "1"

- Keep the defaults for all other sequence parameters.
- Run the sequence. Click "Apply".

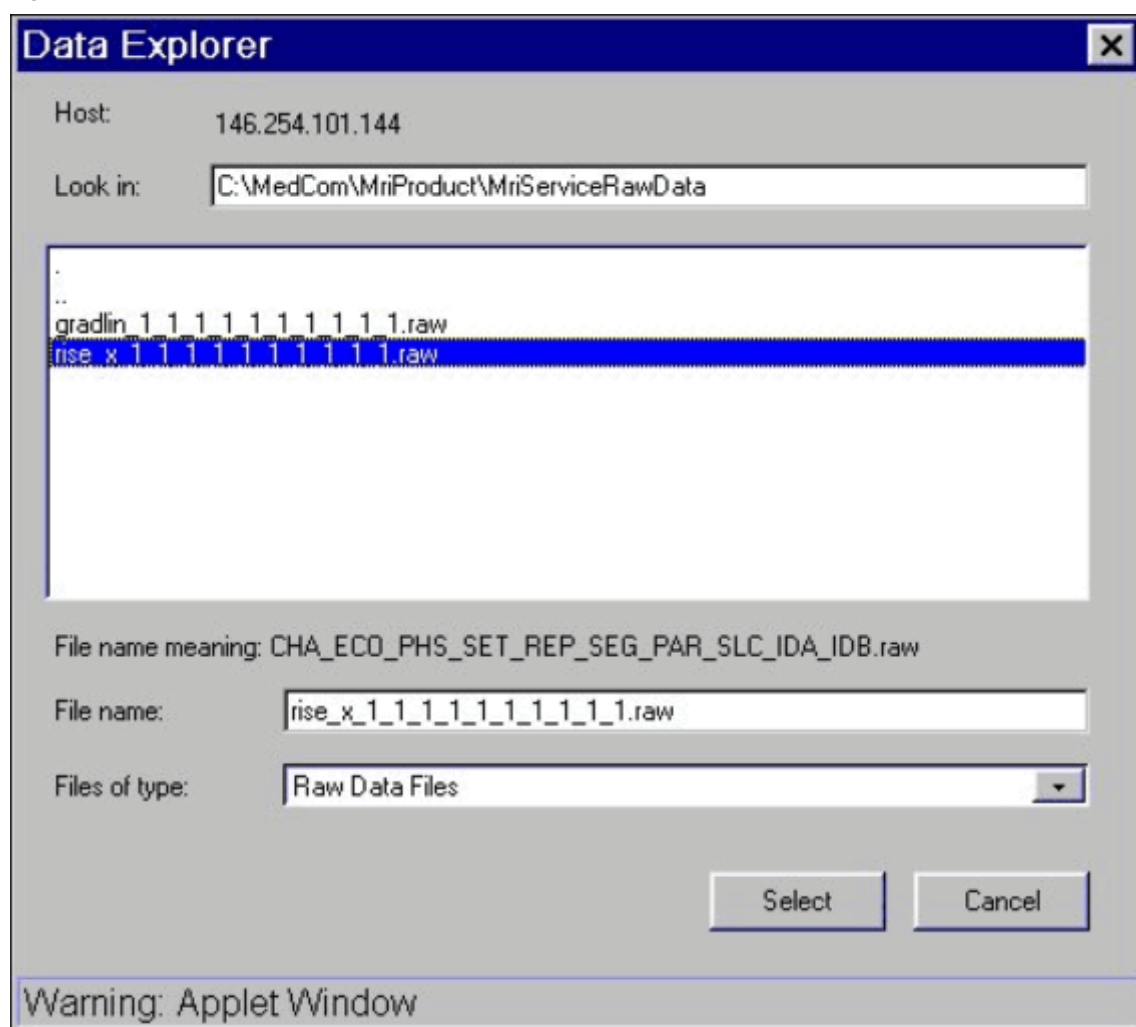
- RHP platform: Select "Go".

Fig. 166:



- RHP main Menu: Select "Evaluate".
- Evaluation platform: Select "Browse".

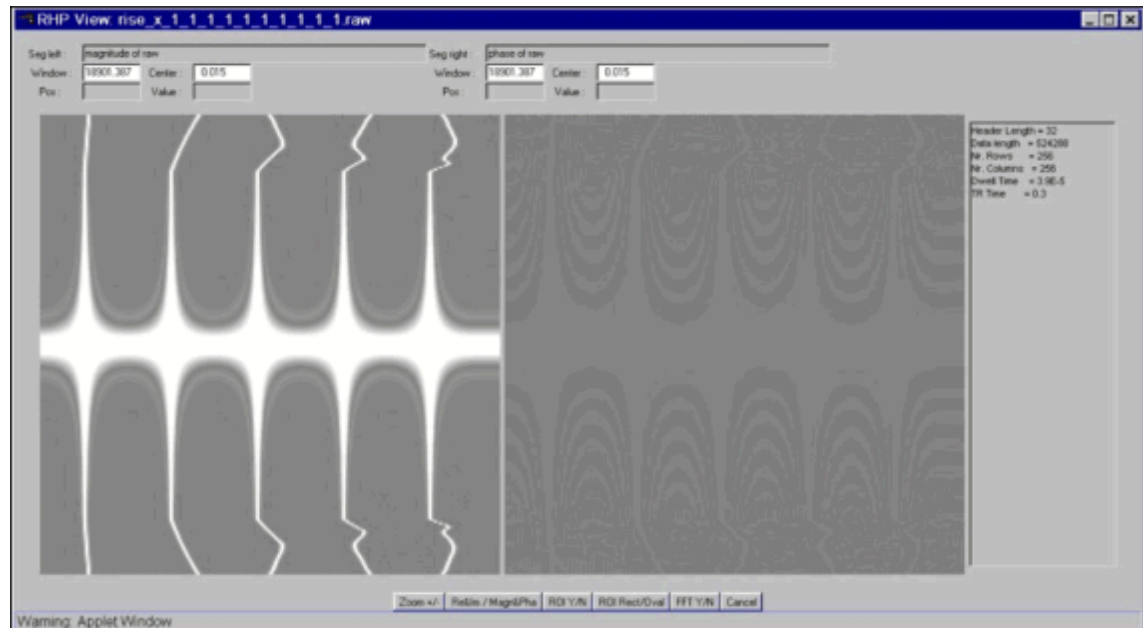
Fig. 167:



- Evaluation platform: Select the desired rawdata file with doubleclick.

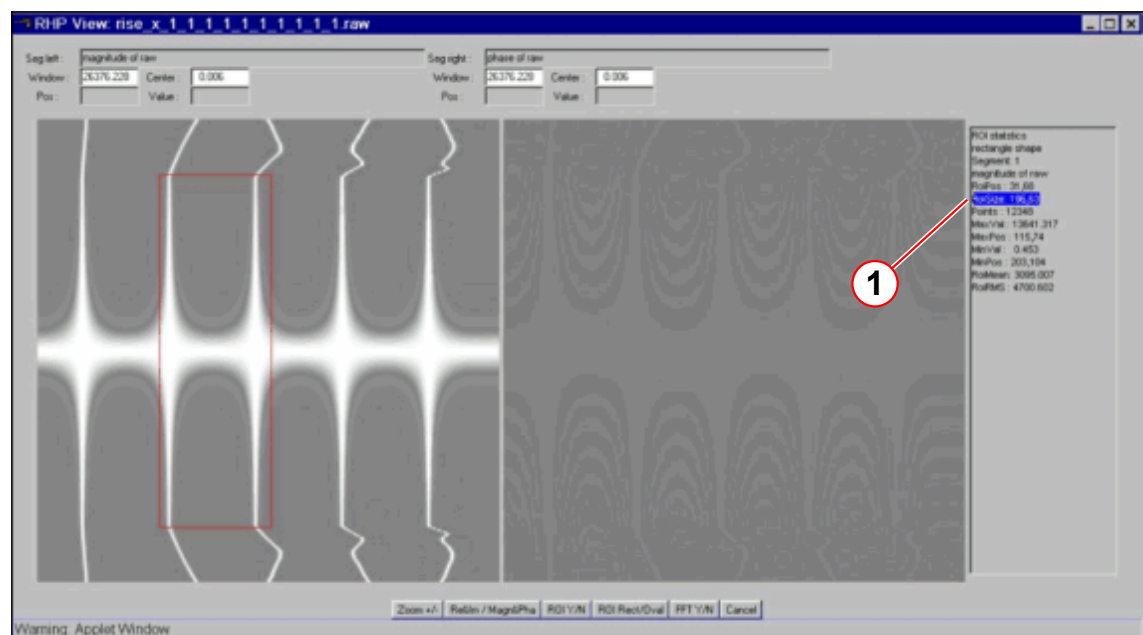
- Evaluation platform: Select "View"

Fig. 168:



- Viewing platform: Select "Zoom +/-", "Re&Im / Magn&Pha" and adjust the window and center to obtain the above representation.

Fig. 169:



(1) ROI Size (height, width)

- Click "ROI Y/N". Draw a rectangular ROI as shown above. The upper and lower boundaries of the ROI must be placed where the graphs of the gradient pulses start to deviate from the straight vertical lines. The width of the ROI is not important.
- Evaluate the value of ROI size (height):

ROI size ≥ 171

- Change the filename in the RHP platform for the measurement of the next axis (rise_y, rise_z).
- Open the grad_risetime protocol and change the orientation of the measurement ($\rightarrow$ Gradient Performance Test Protocol / Page 200).
- Repeat the measurement and evaluate it in the same way for the other orientations.

9.2.2 Possible reasons for this test to fail:

- DC supply voltages for the final stages out of specification.
Measure the incoming line voltage.
Check the jumpering of the gradient transformer T1.
- Line resistance too high. (Refer to Gradient_Burst check.)
- UPS (if present) does not work correctly or switches into bypass mode.
- Hardware problems of GPA (final stages (partly) defective, one of the 5 individual DC supply voltages missing, problems of small signal unit GSSU).

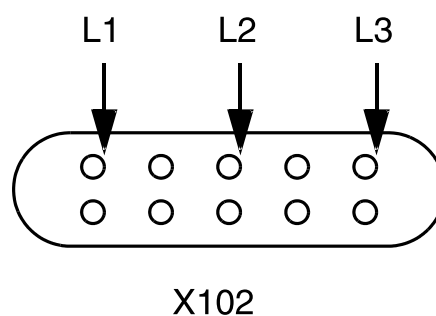
9.2.3 Gradient Burst Check

Fig. 170:



- Connect a DVM at X102 on top of the CCA cabinet.
- Switch on circuit breaker F5 for the MPS.

Fig. 171:



- Record the line voltage between phases for the idle condition

 V_{idle12} , V_{idle13} , V_{idle12}

- Open the "Examination" task card.
- Open the "Examination Explorer" and select the sequence SIEMENS - Sequence region - Service Sequences - Default Protocols - grad_burst.
- Copy the sequence into the examination queue (drag and drop).
- Run the sequence.

- While the sequence is running (duration 52s) record the line voltage for the loaded condition

$$V_{\text{load12}},$$

$$V_{\text{load13}},$$

$$V_{\text{load23}}.$$

- The line voltage must not decrease by more than 4% (Sonata / Trio / Allegra) and 6% for systems with Quantum gradients (all three phases).

$$V_{\text{load}} / V_{\text{idle}} \geq 0,96 \text{ (Sonata / Trio)}$$

$$V_{\text{load}} / V_{\text{idle}} \geq 0,94 \text{ (Quantum Gradients)}$$

- Switch off circuit breaker F5 and disconnect the DVM.



If the voltage drop exceeds the above values, it is very likely the line resistance of the systems power supply is out of spec

>200mOhm (Symphony Ultra, Sprint, Quantum)

>120mOhm (Sonata, Trio).

9.3 Gradient Performance Test Protocol

System serial number:

Customer:

Country/City:

Gradient System: Quantum ☐ Sonata ☐ Trio ☐

The transformers of the CCA and GPA are jumpered to:

360V ☐ 380V ☐ 400V ☐ 420V ☐ 440V ☐ 460V ☐ 480V ☐

Gradient_Burst Check:

	L1 - L2	L1 - L3	L2 - L3
V_{idle}			
V_{load}			
V_{load} / V_{idle}			

$V_{load} / V_{idle} \geq 0,96$ (Sonata / Trio), $V_{load} / V_{idle} \geq 0,94$ (Quantum Gradients)

Gradient_Risetime Check:

	X	Y	Z
ROI size			

ROI size ≥ 171

	yes	no
The gradient_risetime and the gradient_burst measurements are within specification.	☐	☐

Place, date

Name in printed letters

Signature

Please attach a copy of this protocol to the standard Installation Protocol. Distribute the protocol as usual.

9.4 QA (Quality Assurance)



There is no written document for the QA. Use Seso Online Help for detailed information.

- Perform the QA (Quality Assurance) of the system (Local Service / QA). Most QA steps are performed fully automatic. Follow the instructions on the screen. Popup windows will inform you about necessary actions. Please consult Seso Online Help for detailed information.

9.5 Safety Related Tests

Perform the (→ Safety Related Test Instructions / M1-010.833.01). Refer to the (→ syngo MR: Harmony / Symphony / Sonata / M1-010.831.01)"Maintenance Instructions" for details.

9.6 Installation Protocol

Complete the software based "Installation Protocol". Refer to Seso Online Help for details.

9.7 Dynamical Backup

Perform "Backup of dynamical data". Refer to Seso Online Help for details.

9.8 Trolley with removable Tabletop (Option)

- Hand over the Waring letter to the customer
- Choose one label in your language from the warning label set, see (→ Fig. 172 Page 205).

Fig. 172:

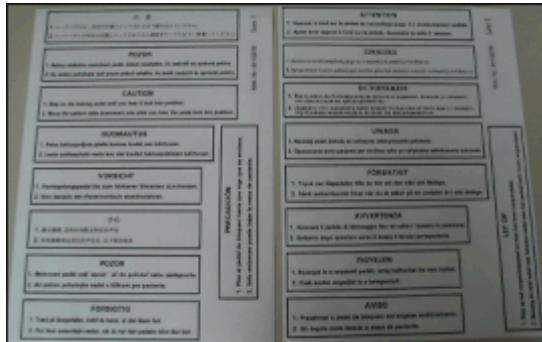
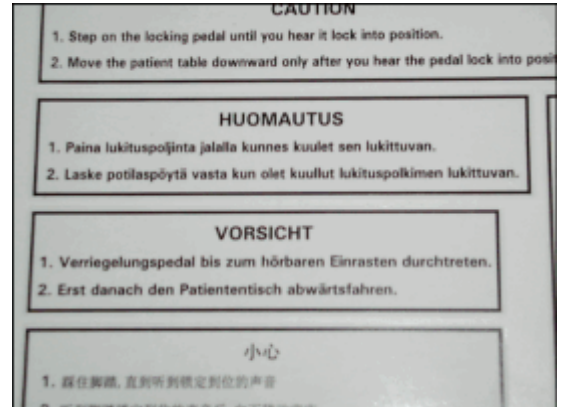


Fig. 173:



- Stick the warning lable on the trolley, see (→ Fig. 173 Page 205).

Fig. 174:



Fig. 175:



9.9 Cleaning up



This completes the start-up of your new MR system. If no other options are being installed in your system, you should now complete the following final steps. If you are installing other options, complete these final steps after installation of the options is finished.

Please consider that your customer has purchased an excellent, high-quality MR system with a modern design. The visual impression of the system is equally important to your customer's satisfaction.

- Assemble the surface coil cart according the installation manual attached.
- Close all magnet covers.
- Close all electronic cabinet doors.
- Touch up any paint chips with matching color paint.
- Clean any fingerprints off the monitor screens.
- Clean any fingerprints off the table display and off the PMU display.
- Clean any fingerprints off the system with a mild cleaner.
- Clean the patient table and the surface coils with a mild cleaner.
- Clean the area of the system, dispose of empty cartons or other packing materials.
- Make sure that the MR system rooms are cleaned by the cleaning crew.

Migration from Frame-Word to CMS.

Chapter 08

Transfer monitor information into the REP.

Chapter 03 is supplemented: "Cleaning the water filter in case of a local move and Refurbished Systems"

Chapter 05:Shimming the magnet is supplemented: Note: shimmed from bare...

There are no Hazard IDs in this document.

All documents may only be used for rendering services on Siemens Healthcare Products. Any document in electronic form may be printed once. Copy and distribution of electronic documents and hardcopies is prohibited. Offenders will be liable for damages. All other rights are reserved.

healthcare.siemens.com/services

Siemens Healthcare Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthcare

Print No.: M1-010.815.01.13.02 | Replaces: M1-010.815.01.12.02
Doc. Gen. Date: 05.10 | Language: English
Part No.: n.a.
© Siemens Healthcare GmbH, 1997

siemens.com/healthcare